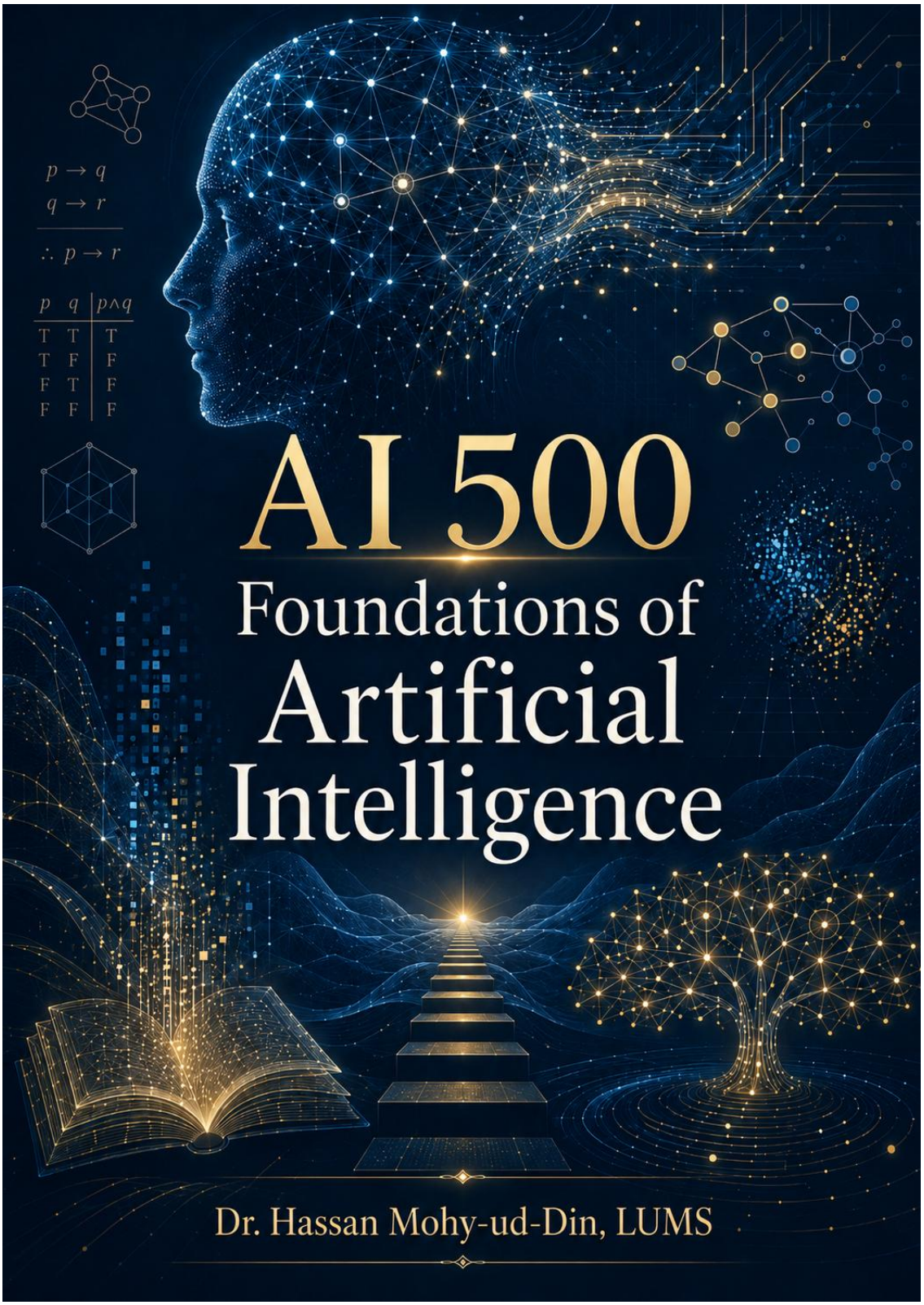

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About the Cover Image

The cover image presents artificial intelligence as a journey from foundational knowledge to purposeful innovation. The luminous human profile, interwoven with neural networks, mathematical symbols, logical structures, and computational pathways, represents the union of human reasoning and machine intelligence. The open book symbolizes scholarship and the enduring intellectual foundations of the field, while the ascending staircase suggests progressive learning – from data representation, probability, optimization, and algorithms to machine learning, evaluation, deployment, and responsible decision-making. The networked tree signifies the growth of knowledge, the interconnectedness of AI disciplines, and the emergence of increasingly sophisticated systems from fundamental principles. Together, these elements reflect the central philosophy of this course is not merely a collection of algorithms, but a complete, evolving discipline grounded in mathematics, data, computation, judgment, ethics, and real-world application.

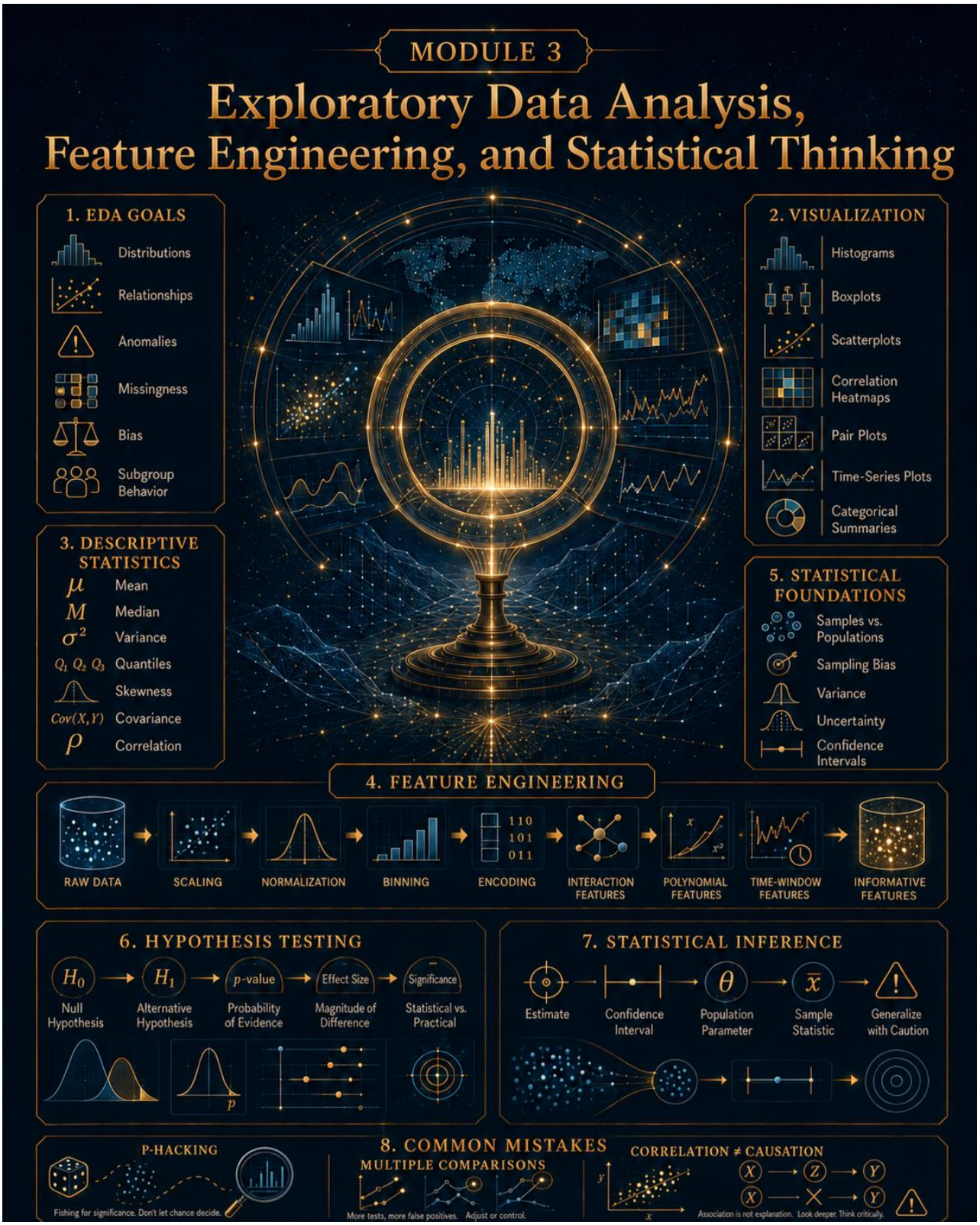
Disclaimer

In the compilation of this document, I have utilized GPT-based tools, including ChatGPT 5.6 Sol, to assist with the generation of cover and module images, schematics, flowcharts, infographics, tables, workflows, and synthetic examples, as well as with the identification and synthesis of published references, historical information, and other relevant background material. These tools were also used to support the organization, refinement, and presentation of the material. All generated content and references were reviewed, edited, and, where appropriate/possible, independently verified by the author. Nevertheless, errors, omissions, or inaccuracies may remain, and I would be grateful to readers for identifying and bringing any such issues to my attention so that they may be corrected in future revisions.

I do not harbor unbridled enthusiasm or fascination of AI. I firmly believe that technology should remain *semi-autonomous* i.e., operating under informed judgment, ethical oversight, and deliberate control of a wise human mind. In this spirit, AI is treated here as a supportive tool rather than a substitute for human reasoning, creativity, or responsibility.

About the Instructor

Dr. Hassan Mohy-ud-Din is the Director of Algorithms in Theory and Practice Lab and an Assistant Professor of Electrical Engineering, Syed Babar Ali School of Science and Engineering, LUMS. He completed his PhD and MSE in Electrical and Computer Engineering and MA in Applied Mathematics and Statistics from Johns Hopkins University (2009 – 2015). He was recipient of PhD Fellowship from Johns Hopkins University. From 2015 – 2017 he was a postdoctoral associate in the Department of Radiology and Biomedical Imaging at the Yale School of Medicine. From 2017 – 2018 he was a Clinical Research Scientist at Shaukat Khanum Memorial Cancer Hospital and Research Centre, Lahore, Pakistan. His research is at the intersection of applied mathematics and clinical imaging – exploiting tools in machine learning, optimization, statistics, and information theory to develop novel algorithms for clinical and translational imaging. He has done extensive research in multimodality imaging including PET/CT, SPECT/CT, PET/MR, Low-dose CT, and multiparametric MRI and developed computational pipelines for brain imaging, cardiac imaging, and abdominal imaging. His work on dynamic cardiac PET imaging won the 2014 SNMMI Bradley-Alavi fellowship and the 2014 SIAM student award. He is also a recipient of the 2019 Charles Wallace Fellowship from the British Council, Pakistan. He also serves as a reviewer on major scientific journals and a member of IEEE and SIAM societies. He carries a university teaching experience of over fifteen years (including five years at Johns Hopkins University) and is deeply interested in the development, implementation, and evaluation of (pragmatic and effective) pedagogical approaches in schools, colleges, and universities.



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Exploratory Data Analysis

Exploratory data analysis (EDA) emerged as a distinct approach to statistical investigation through the work of American statistician **John W. Tukey**, particularly his influential 1977 book *Exploratory Data Analysis*[259]. Tukey argued that data should **not be treated merely as materia**l for confirming hypotheses decided in advance; instead, analysts should **actively explore data** to **uncover structure**, **patterns**, **anomalies**, and **relationships** that *may not have been anticipated*. This perspective helped establish EDA as an important stage of statistical analysis and remains highly relevant in modern data science, where **large and complex datasets** often contain **unexpected features** that **must be understood** before formal modeling.

3.1.1 What is exploratory data analysis?

EDA is the **systematic investigation of a dataset** before formal **statistical modeling**[260], **hypothesis testing**[261], or **confirmatory inference**[262]**.** It provides an initial opportunity to understand **what the data contains**, how the **variables behave**, and whether the **dataset is suitable** for the intended analytical task.

The main purpose of EDA is not simply to produce attractive graphs or calculate descriptive statistics. Rather, it is a **process of asking questions** about the data and **using numerical** and **graphical techniques** to develop a reliable understanding of their structure. Through this process, an analyst can:

- understand the structure and composition of the dataset;
- examine the distributions of individual variables;

[259] **John W. Tukey** (1915-2000) was an American statistician and mathematician best known for his major contributions to statistical analysis and data visualization. He coined the terms *bit* and *software* and developed influential statistical techniques, including the fast **Fourier transform** (FFT). His **1977** book Exploratory Data Analysis played a foundational role in promoting systematic methods for discovering patterns and insights in data.

[260] **Statistical modeling** is the process of **constructing a mathematical or probabilistic representation** of the relationships among variables while accounting for random variation and uncertainty. For example, a researcher may use multiple linear regression to model house prices as a function of house size, number of bedrooms, and location.

[261] **Hypothesis testing** is a **formal statistical procedure** used to **evaluate evidence** in sample data **against a specified null hypothesis**, typically using a test statistic, *p*-value, and significance level. For example, a researcher testing whether a new drug lowers blood pressure might compare the mean blood pressure of patients receiving the drug with that of a control group.

[262] **Confirmatory inference** refers to **statistical inference** conducted to **evaluate hypotheses** or **theoretical claims** that were **specified before examining** the data, using a predetermined study design, statistical model, and decision criterion. For example, in a clinical trial, researchers may pre-specify that the primary objective is to determine whether a treatment reduces blood pressure compared with a placebo and then use the collected data to assess that hypothesis.

- discover patterns, trends, and relationships between variables;
- identify missing, duplicated, inconsistent, or erroneous observations;
- detect outliers and other unusual observations;
- assess whether important statistical assumptions appear reasonable;
- generate plausible hypotheses for further investigation;
- determine whether transformations or other preprocessing steps are necessary;
- identify potentially important variables and relationships;
- evaluate whether the proposed analytical task is feasible; and
- guide the selection of appropriate statistical models and methods.

EDA is described as **exploratory** because the analyst **does not begin** with a single **rigid conclusion** that the data are expected to confirm. Instead, the **analysis develops iteratively**. The analyst examines the data from **different perspectives**, notices something interesting or unusual, asks a new question, and then investigates that question using additional summaries or visualizations.

The **National Institute of Standards and Technology (NIST) Statistical Handbook**[263] describes EDA as an approach to data analysis that seeks to **maximize insight** into a dataset, uncover its **underlying structure**, detect **outliers and anomalies**, test **underlying assumptions**, develop **parsimonious models**, and determine **appropriate factor settings**. In this sense, EDA provides an important link between initial examination of raw data and subsequent statistical modeling, helping analysts understand the structure and characteristics of a dataset before applying more formal analytical methods.

3.1.2 The EDA pipeline

The EDA pipeline is depicted in **Figure 22**. For example, an **analyst** may begin by **asking** whether two variables are related. A **scatter plot** might reveal a relationship but also show several **unusual observations**. **Numerical summaries** can then be used to examine those observations more closely,

[263] N. A. Heckert and J. Filliben, *NIST/SEMATECH e-Handbook of Statistical Methods: Chapter 1, Exploratory Data Analysis*, National Institute of Standards and Technology, 2003.

leading to a new question about whether they **represent genuine** cases, **measurement errors**, or **data-entry problems**. Thus, EDA is not necessarily a linear procedure; it is an iterative process in which each discovery can lead to further investigation.

EDA commonly combines **graphical methods** with **numerical summaries**. **Graphical techniques**[264] such as histograms, box plots, scatter plots, bar charts, density plots, and time-series plots help **reveal patterns** that may be difficult to detect from numbers alone. **Numerical measures**[265] such as the mean, median, standard deviation, range, quartiles, and correlation coefficients provide **quantitative descriptions** that complement those visual observations.

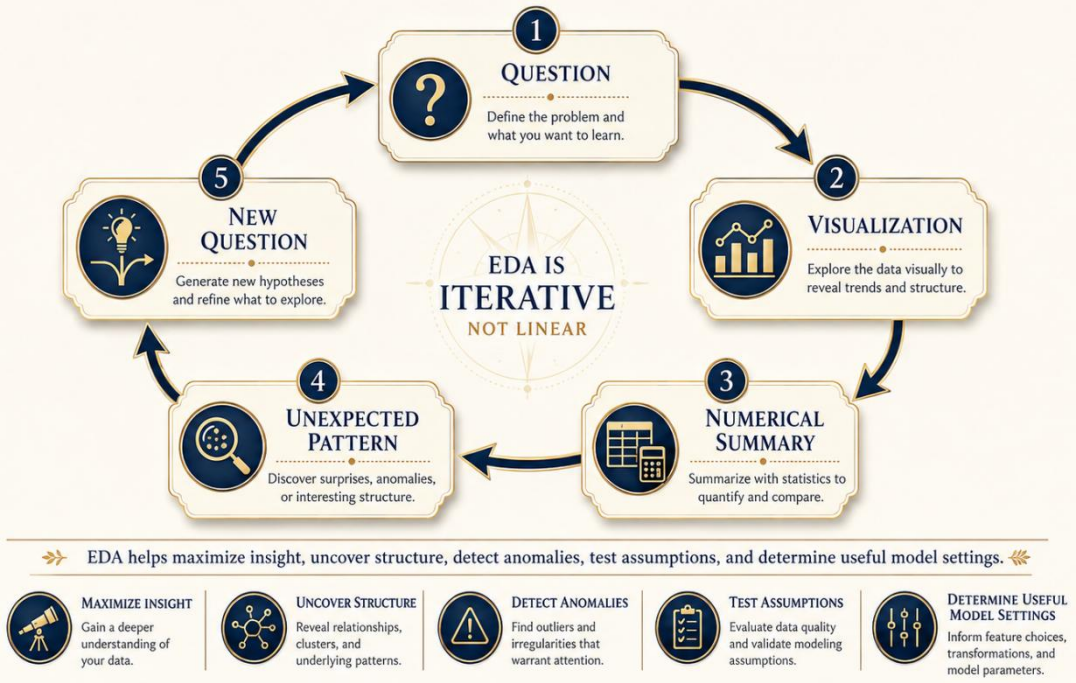

Figure 22: Iterative cycle of exploratory data analysis (EDA). It depicts EDA as a continuous process in which an initial question motivates visualization and numerical summarization,

[264] **Graphical methods** are statistical techniques that use visual representations of data to reveal patterns, distributions, relationships, trends, clusters, and potential anomalies that may not be apparent from numerical calculations alone. Common examples include **histograms** for examining the distribution of a continuous variable, **box plots** for comparing distributions and identifying potential outliers, **scatterplots** for examining relationships between two quantitative variables, and **bar charts** for comparing frequencies or proportions across categories.

[265] **Numerical summaries** are quantitative measures that concisely describe important characteristics of a dataset, such as its center, variability, and distribution. Examples include the **mean**, **median**, **standard deviation**, **variance**, **range**, **interquartile range (IQR)**, and **percentiles**. For example, when exploring students' examination scores, an EDA may use a histogram to visualize the distribution of scores and report the mean, median, and standard deviation to numerically summarize the same data.

unexpected patterns generate new hypotheses, and these in turn lead to refined questions. This iterative cycle supports the core goals of EDA: maximizing insight, uncovering structure, detecting anomalies, testing assumptions, and informing subsequent model specifications.

An important advantage of EDA is that it **can reveal problems** that might otherwise be carried into later stages of analysis. For example, a dataset may contain **substantial missingness**, **extreme outliers**, **highly skewed variables**, **duplicated records**, **incorrect values**, or **unexpected relationships** between variables. Detecting these issues early can prevent inappropriate statistical conclusions and help the analyst decide whether the data requires cleaning, transformation, or additional investigation.

EDA should therefore be viewed as a **foundation for subsequent statistical analysis**, rather than as a replacement for formal inference. **Exploratory findings** can **suggest hypotheses** and **analytical strategies**, while confirmatory methods can subsequently be used to test those hypotheses more rigorously. Keeping these stages conceptually distinct is important because **patterns discovered through exploration** may be the **result of chance**, **selection effects**, or **characteristics** of the particular dataset being examined.

Example # 65: Relationship Between Sleep Duration and Exam Performance

Suppose a researcher examines the relationship between **sleep duration and students' exam performance**. During EDA, they plot the data and notice that students who report sleeping $7-8$ hours tend to have higher exam scores than students who sleep fewer than 5 hours. This exploratory pattern may lead the researcher to hypothesize that **adequate sleep is associated with better academic performance** and to consider statistical methods such as correlation or regression for further investigation.

However, the researcher should not immediately conclude that getting $7-8$ hours of sleep **cause** higher exam scores. The observed pattern could have arisen by chance, or it could be influenced by **selection effects or other characteristics of the sample**. For example, students who sleep longer may also experience less stress, have better study habits, or have more stable schedules. Therefore, the researcher could use a subsequent confirmatory analysis to formally test the hypothesis while controlling for relevant factors.

For example, the researcher could use a regression analysis to test whether sleep duration is significantly associated with exam performance while controlling for factors such as study time and stress level.

Thus, EDA helps researchers **discover interesting patterns** and **formulate hypotheses**, whereas formal **statistical inference** is used to **evaluate** those **hypotheses more rigorously**. Keeping the two stages separate reduces the risk of treating an interesting pattern found in one dataset as if it were already established evidence.

3.1.3 Exploratory analysis versus Confirmatory analysis

Exploratory data analysis (EDA) and confirmatory analysis serve **different but complementary purposes** in statistical investigation. **Exploration** is concerned primarily with **discovering** what the data **may reveal**, whereas **confirmation** is concerned with **evaluating specific** claims or hypotheses using methods whose assumptions and analytical procedures have been defined in advance. Both are important, but they should not be treated as interchangeable stages of analysis. Table 30 provides a comparative summary of EDA vs confirmatory analysis.

Table 30: Comparison of exploratory and confirmatory analysis.

Exploratory analysis	Confirmatory analysis
Searches for patterns, trends, and relationships	Tests prespecified hypotheses or claims
Generates hypotheses for further investigation	Evaluates hypotheses using formal statistical procedures
Uses flexible graphical and numerical summaries	Uses predefined inferential methods
May examine many variables and alternative representations	Typically follows a specified analytical plan
Emphasizes discovery and understanding	Emphasizes evidence, uncertainty, and quantification
Is often iterative and adaptive	Is ideally specified before examining the data used for confirmation
Can inform feature construction and variable selection	Estimates or tests predefined parameters or effects
Can reveal unexpected findings	Determines how strongly the observed data support a particular claim

The distinction is important because **an interesting pattern discovered during exploration is not automatically evidence that has been**

independently confirmed. Exploratory analysis can suggest that a relationship exists, identify potentially important variables, or generate a new hypothesis. However, the same data that were used to discover a pattern should generally not be treated as though they were completely independent evidence for confirming that pattern.

For example, suppose an analyst examines many variables and discovers that one particular predictor appears to be associated with an outcome. The analyst then selects that predictor, chooses a model that fits the observed relationship well, and conducts a statistical significance test on the same data. The resulting ***p*-value**, **confidence interval**, or other **inferential quantity** may not have the interpretation that it would have had if the hypothesis and analysis had been specified independently of the data.

This problem can be understood as a consequence of **data-dependent analytical decisions**. If the same dataset is repeatedly used to search for patterns, select hypotheses, choose variables, choose models, and then test the selected hypothesis, the apparent strength of the final result can be exaggerated. The analysis has effectively used the data both to discover the result and to provide evidence for that result. Consider the following sequence:

Search for pattern → Select a hypothesis → Choose variables → Choose a model → Test the selected hypothesis

When all of these decisions are made using the same data, the final inferential result may fail to account for the selection process that preceded it. This does not make exploratory analysis invalid; rather, it means that **exploratory findings and confirmatory evidence should be interpreted differently**.

Example # 66: Sleep Duration and Exam Performance

Suppose a researcher has a dataset containing **sleep duration**, **exam scores**, **study time**, **stress level**, and **other characteristics** of **500 students**. The researcher begins by exploring the data and notices that students who sleep $7-8$ hours appear to have higher exam scores. The researcher then makes a series of decisions based on the same dataset:

- **Search for a pattern:** The researcher examines several variables and discovers that sleep duration appears to be related to exam scores.

- **Select a hypothesis:** Based on this pattern, the researcher decides to investigate whether sleep duration predicts exam performance.
- **Choose variables:** The researcher selects sleep duration as the main predictor and decides to control for study time and stress, perhaps after examining which variables appear related to exam scores.
- **Choose a model:** The researcher tries several possible approaches and ultimately selects a regression model that produces a particularly strong relationship.
- **Test the selected hypothesis:** The researcher then reports that sleep duration is significantly associated with exam performance.

The problem is that **the same data was used to discover the relationship** and then to **provide evidence for that relationship**. Because the researcher had many opportunities to search for patterns and make analytical choices, the **final result may appear stronger** or more **statistically convincing** than it would have been if the hypothesis had been specified independently beforehand.

For instance, if the researcher initially examined **sleep duration**, **exercise**, **screen time**, **breakfast habits**, **caffeine consumption**, and **study time**, a noteworthy relationship might emerge simply by chance. If the researcher then focuses only on the most promising relationship and tests it using the same data, the usual statistical test does not fully reflect the fact that many possible relationships were explored first.

This **does not** mean that the exploratory analysis was **wrong or unnecessary**. The exploratory analysis was useful because it helped the researcher **identify a potentially interesting hypothesis**. However, the resulting hypothesis should ideally be evaluated using **new data**, a **separate validation sample**, or an **appropriately designed confirmatory analysis**. Hence,

Exploration uses data to discover what might be interesting; confirmation uses independent or appropriately controlled evidence to determine whether that finding is likely to be genuine.

3.1.4 Separating exploration from confirmation

A responsible analytical workflow therefore distinguishes between **exploration** and **final confirmation**. The goal is not necessarily to prevent analysts from learning from their data. On the contrary, **exploration** is often **essential** for **understanding complex datasets**. The important issue is to ensure that **claims of confirmation** are supported by an **appropriate source** of **independent** or properly **controlled evidence**. Several approaches can help maintain this distinction:

- **Separate exploratory and confirmatory datasets:** One dataset can be used to discover patterns and develop hypotheses, while another is reserved for testing those hypotheses.
- **Training, validation, and test partitions:** In predictive modeling, different subsets of the data can be assigned different roles so that model development and final performance assessment are not based on exactly the same observations.
- **Preregistration:** Hypotheses, variables, outcomes, and analytical procedures can be specified before the confirmatory analysis is conducted, reducing the opportunity for results to influence methodological decisions.
- **Replication with new data:** A finding discovered in one dataset can be examined using an independent dataset or a new sample.
- **Transparent reporting:** Analysts should clearly document important analytical decisions, including variable selection, exclusions, transformations, model selection, and other choices that could influence the results.

Exploration helps researchers **discover** what may be important**; confirmation** helps determine how strongly the available **evidence supports** a specific claim**.** Treating the two stages as complementary, while being transparent about which role a particular analysis serves leads to more credible and reproducible statistical conclusions.

3.1.5 The unit of analysis must be defined first

One of the first questions an analyst should answer when working with a dataset is: **What does one row represent?** This question should be resolved before producing visualizations, calculating summary statistics, fitting models, or interpreting sample sizes. The answer defines the **unit of analysis** or **unit of observation** and determines how individual records should be understood statistically.

A row in a dataset **does not necessarily** represent one **independent individual**. Depending on the context, a single row might represent:

- a person;
- a hospital admission;
- an image;
- a financial transaction;
- a sensor reading;
- a household;
- a geographical region;
- a user session;
- a device-day; or
- a student-course combination.

The distinction becomes particularly important when multiple rows are associated with the same underlying entity. For example, suppose a medical imaging dataset contains **50,000 radiological images from 2,000 patients**. Although the dataset contains 50,000 image records, it does not necessarily provide 50,000 independent observations for a patient-level analysis. A single patient may contribute many images, and those images are likely to share characteristics because they come from the same person. Treating all 50,000 images as independent patient-level observations could therefore give a misleading impression of the amount of independent information available.

The same issue occurs in many other settings. A customer may generate hundreds of transactions, a device may produce thousands of sensor readings, or a student may appear in multiple course records. In each case, the number of **rows** may be substantially larger than the number of independent entities represented by those rows.

For this reason, an analyst should explicitly document at least four related concepts:

- **Unit of observation:** the entity or event represented by an individual record.
- **Unit of prediction:** the entity for which a prediction is ultimately required.
- **Unit of sampling:** the entity or group from which observations were selected.
- **Unit of evaluation:** the entity at which model performance or statistical results are assessed.

These units may be identical, but they do not have to be. For example, an image-classification dataset may have an **image** as its unit of observation,

while the desired prediction is ultimately made at the **patient** level. Similarly, a study may collect repeated measurements from individuals, making the measurement the unit of observation while the individual is the relevant unit for sampling or inference.

Example # 67: Applications of Observation, Prediction, Sampling, and Evaluation Units

Here are several other applications where the four units can be clearly distinguished.

Healthcare: Predicting Hospital Readmission
Suppose a hospital wants to predict whether a patient will be readmitted within 30 days.

- *Unit of observation:* A **hospital admission**. A patient who is admitted three times may contribute three records.
- *Unit of prediction:* The **individual patient admission** for which readmission risk is being predicted.
- *Unit of sampling:* The **hospital or hospital department**, if hospitals and/or departments are selected first and patients are then included.
- *Unit of evaluation:* The **individual patient admission**, where predicted and actual readmission outcomes are compared.

Treating multiple admissions from the same patient as independent observations could underestimate uncertainty because those observations are related.

Retail: Predicting Customer Churn
Suppose a telecommunications company wants to predict which customers will cancel their subscription.

- *Unit of observation:* A **customer-month**. Each row represents one customer's information during a particular month.
- *Unit of prediction:* The **individual customer** whose likelihood of cancelling is predicted.
- *Unit of sampling:* The **geographic region or customer account group**, if customers are selected through these groups.

- *Unit of evaluation:* The **individual customer**, by comparing predicted churn with actual churn.

One customer can generate many observations, so the analyst must avoid accidentally allowing information from the same customer to appear in both training and test data.

Agriculture: Predicting Crop Yield
Suppose researchers want to predict the yield of wheat.

- *Unit of observation:* A **field plot** measured during a particular growing season.
- *Unit of prediction:* A **field** for which future crop yield is to be predicted.
- *Unit of sampling:* A **farm**, if farms are randomly selected and several fields are then observed within each farm.
- *Unit of evaluation:* The **field** where predicted yield is compared with actual yield.

Fields belonging to the same farm may experience similar soil, weather, and management practices. Therefore, treating every field as completely independent could produce misleading statistical conclusions.

Education: Predicting Student Dropout
Suppose an education department wants to predict whether students will drop out of university.

- *Unit of observation:* A **student-year record**.
- *Unit of prediction:* The **individual student**.
- *Unit of sampling:* The **university or academic department**, if institutions or departments are sampled first.
- *Unit of evaluation:* The **individual student**, based on whether the predicted dropout outcome matches the actual outcome.

In this application, **the unit of sampling can differ substantially from the unit of prediction**. Students may be sampled through universities, but predictions are ultimately made for individual students.

3.1.6 Why the unit of analysis matters

Failing to distinguish between these units can lead to **serious analytical errors**. In particular, it can produce:

- **overstated sample sizes**, when multiple observations from the same entity are counted as independent;
- **underestimated standard errors**, when dependence between observations is ignored;
- **data leakage between training and testing sets**, when records belonging to the same entity appear in both partitions;
- **misleading subgroup summaries**, when some entities contribute many more observations than others; and
- **inappropriate statistical tests or models**, when the assumed structure of independence does not match the actual data-generating process.

The issue is especially important in machine learning. If multiple records from the same person, patient, household, device, or other entity are randomly distributed across training and test datasets, information about that entity may effectively be present in both sets. A model can then **appear to generalize** well even though it has been evaluated on entities that it has **indirectly encountered during training**.

Consequently, **row count should not automatically be interpreted as sample size**. Before analyzing a dataset, the analyst should identify the entity represented by each row, determine whether observations are independent, and establish which unit is relevant to the intended inference or prediction.

A clear definition of the unit of analysis provides an essential foundation for the rest of the EDA process. It helps determine which observations can legitimately be compared, how summaries should be calculated, how data should be partitioned, and which statistical methods are appropriate. In practical terms, *understanding what one row represents is one of the first requirements for understanding what the dataset actually represents*.

3.1.7 Recommended EDA workflow

Exploratory data analysis is most effective when it is conducted as a **structured**, **iterative workflow** rather than as an unplanned collection of charts and summary statistics (**Figure 23**). The **exact sequence will vary** according to the research question, dataset, and data-generating process, but a systematic workflow helps ensure that important issues are not overlooked. The analysis should generally move from understanding the context in which the data were produced, to inspecting their structure and individual variables,

then examining relationships, time-dependent behavior, and subgroup differences. Throughout the process, findings should be documented so that important discoveries, assumptions, analytical decisions, and unresolved uncertainties remain transparent and can inform subsequent modeling or confirmatory analysis.

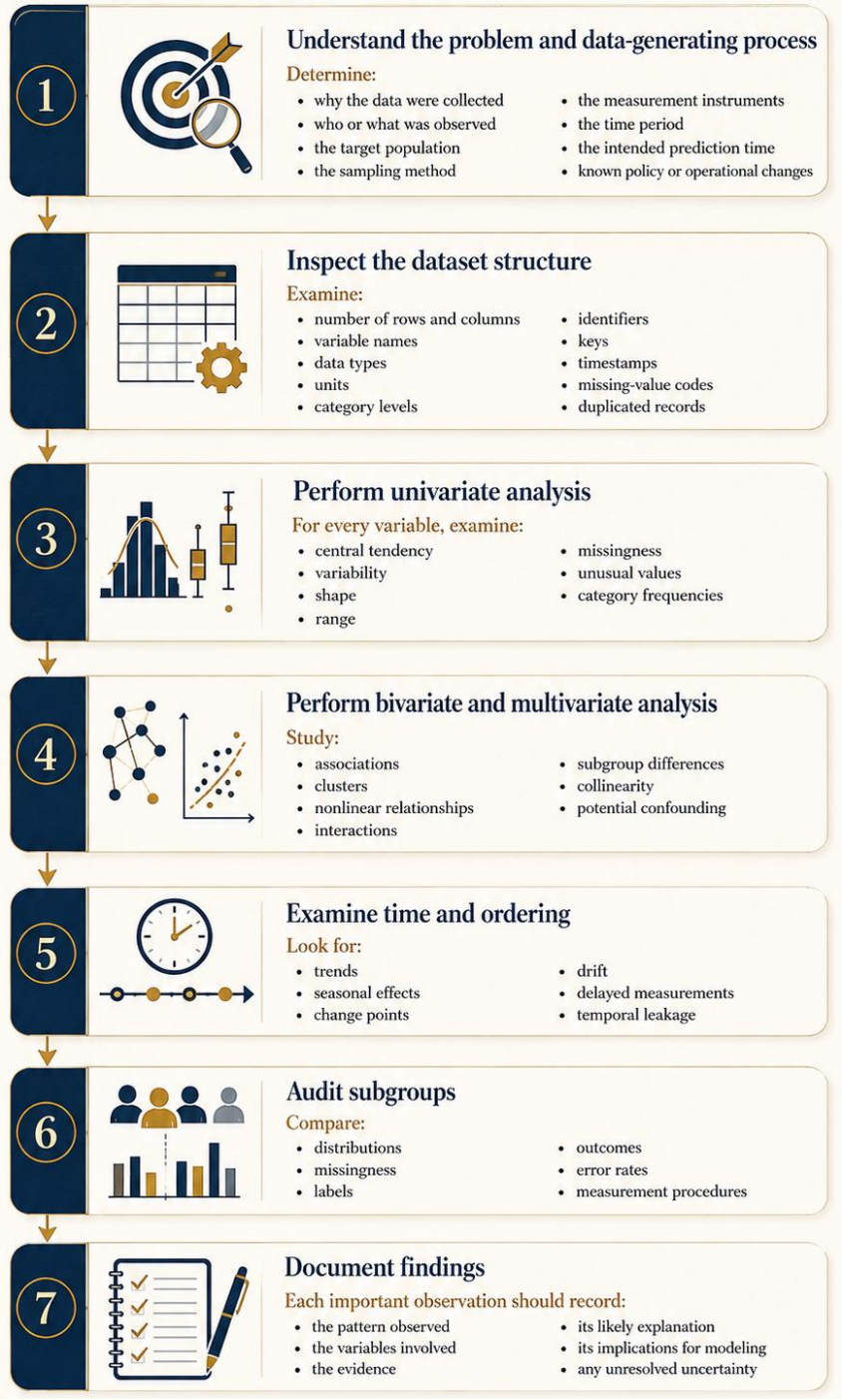

Figure 23: Recommended workflow for exploratory data analysis (EDA).

Core Goals of Exploratory Data Analysis

Exploratory data analysis has several interconnected goals, all centered on developing a **reliable understanding** of what a **dataset contains** and what its **patterns** may mean. Rather than focusing only on producing summary statistics, EDA examines **distributions**, **relationships**, **anomalies**, **missingness**, **potential sources of bias**, and **differences between subgroups**. These investigations help the analyst *distinguish genuine structure from artifacts* of measurement, sampling, data processing, or chance. They also provide the information needed to decide how the data should be cleaned, transformed, modeled, and interpreted. The sections below describe the major objectives of EDA and the questions that should be considered during each stage.

3.2.1 Understanding Distributions

A **distribution** describes how the **observed values** of a variable are **arranged**. Understanding a variable's distribution is one of the most fundamental tasks in EDA because **summary measures** such as the mean or standard deviation **can conceal** important features of the underlying data. For each variable, the analyst should ask:

- Where are the observations concentrated?
- Is the distribution approximately symmetric or skewed?
- Is it unimodal, bimodal, or multimodal?
- Are there gaps or unusually sparse regions?
- Are the values naturally bounded?
- Does the distribution have heavy or light tails?
- Are extreme observations present?
- Does the distribution differ substantially across relevant subgroups?

Distributional characteristics

A numerical distribution can be understood in terms of several complementary characteristics (**Figure 24**):

- **Location** describes where the observations tend to be centered, using measures such as the mean or median.

- **Spread** describes how widely the observations vary, using measures such as the range, variance, standard deviation, or interquartile range.
- **Shape** describes characteristics such as symmetry, skewness, modality, and concentration.
- **Tails** describe the behavior of observations far from the center.
- **Unusual observations** may indicate either legitimate extreme cases or potential data-quality problems.

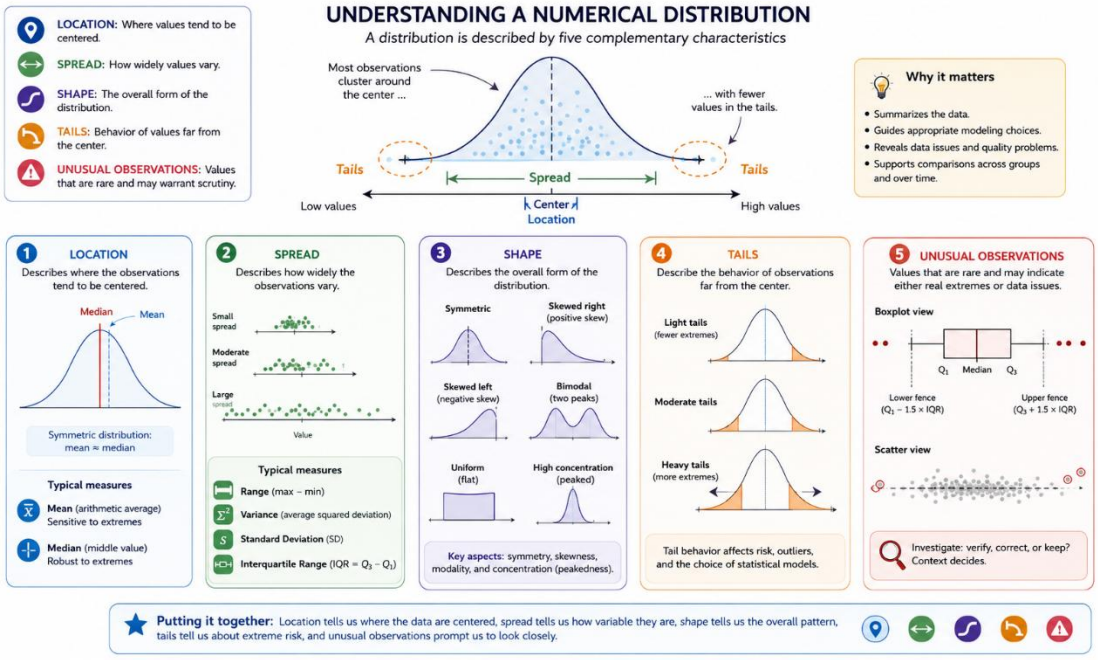

Figure 24: Key characteristics of a numerical distribution. A numerical distribution can be described through **five complementary properties**: **location**, **spread**, **shape**, **tails**, and **unusual observations**. Together, these characteristics provide a structured framework for exploratory data analysis, statistical modeling, and identifying patterns that may warrant further investigation.

Graphical methods such as histograms, density plots, box plots, empirical cumulative distribution functions, and quantile plots are therefore valuable complements to numerical summaries (**Figure 25**).

Pathological examples

Graphical summaries are powerful because they make the structure of a dataset visible, but they **also reveal patterns** that **depart from** the simple distributions commonly assumed in introductory statistical analysis.

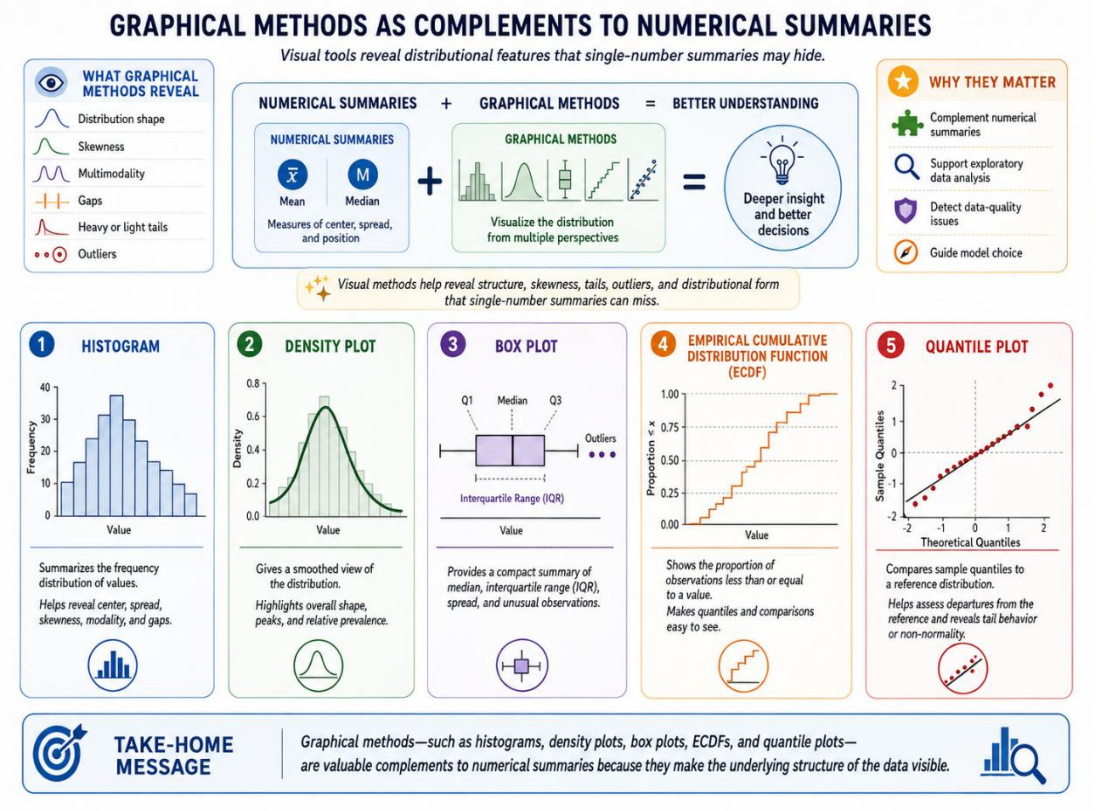

Figure 25: Graphical methods as complements to numerical summaries. Histograms, density plots, box plots, empirical cumulative distribution functions (ECDFs), and quantile plots provide **complementary visual perspectives** on a numerical distribution. Together, they **help reveal features** such as center, spread, skewness, modality, gaps, tail behavior, and unusual observations that may be obscured by summary statistics alone. Used alongside numerical measures, these graphical tools support exploratory data analysis, distributional comparison, data-quality assessment, and informed statistical modeling.

Such **pathological patterns** in numerical data may **reflect genuine structure**, such as skewness, heavy tails, multimodality, or truncation, or arise from data-quality and visualization issues. Examining these patterns across complementary graphical methods helps **distinguish meaningful variation from artifacts** and supports more appropriate statistical analysis and modeling.

- **Pathological distributions:** It exhibits structural characteristics that **differ substantially** from a **simple**, **approximately symmetric**, **unimodal distribution**. Important examples include heavy tails, strong positive or negative skewness, multimodality, zero inflation, truncation, and heterogeneous mixtures (**Figure 26**). Such patterns frequently arise because the data-generating process contains **multiple subpopulations**, **physical** or **administrative limits**, **rare extreme events**, excess zeros, or **nonlinear mechanisms**. Recognizing these characteristics is important

because measures such as the mean and standard deviation may provide an incomplete or misleading summary, while statistical procedures based on normality, constant variance, or a single homogeneous population may perform poorly.

Real-world examples: Income and household wealth are typically strongly right-skewed, with a small number of extremely large values. Insurance claim amounts and financial losses may exhibit heavy tails because rare events can generate very large costs. Hospital length of stay is commonly right-skewed because most patients are discharged relatively quickly while a smaller group remains hospitalized for much longer. Healthcare utilization data may be zero-inflated because many individuals have no hospital visits during a given period. Multimodal distributions can occur when data from distinct demographic groups, disease subtypes, scanners, hospitals, or operating regimes are pooled.

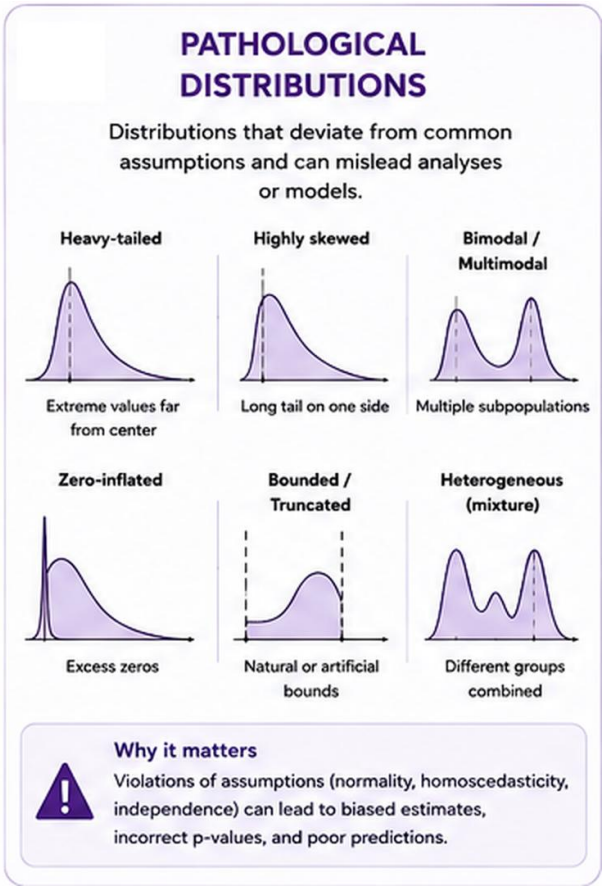

Figure 26: Examples of pathological numerical distributions. Heavy tails, strong skewness, multimodality, zero inflation, truncation, and heterogeneous mixtures may signal important characteristics of the underlying data-generating process. Such patterns can invalidate simplistic distributional assumptions and motivate robust summaries, stratified analysis, transformations, or alternative statistical models.

- **Pathological histograms:** A histogram can reveal **important anomalies** in a dataset, but it can **also create misleading** visual impressions when the bin width, bin boundaries, or plotting scale are poorly chosen. Pathological histogram patterns include apparent gaps, isolated spikes, heaping at particular values, excessive concentration at zero, dominance by extreme observations, and dramatic changes in appearance as the number of bins changes (Figure 27). **Some of these** patterns **reflect genuine structure**, while others result from rounding, digit preference, measurement resolution, censoring, coding conventions, or inappropriate binning. Histograms should therefore be interpreted together with knowledge of how the variables were measured and recorded.

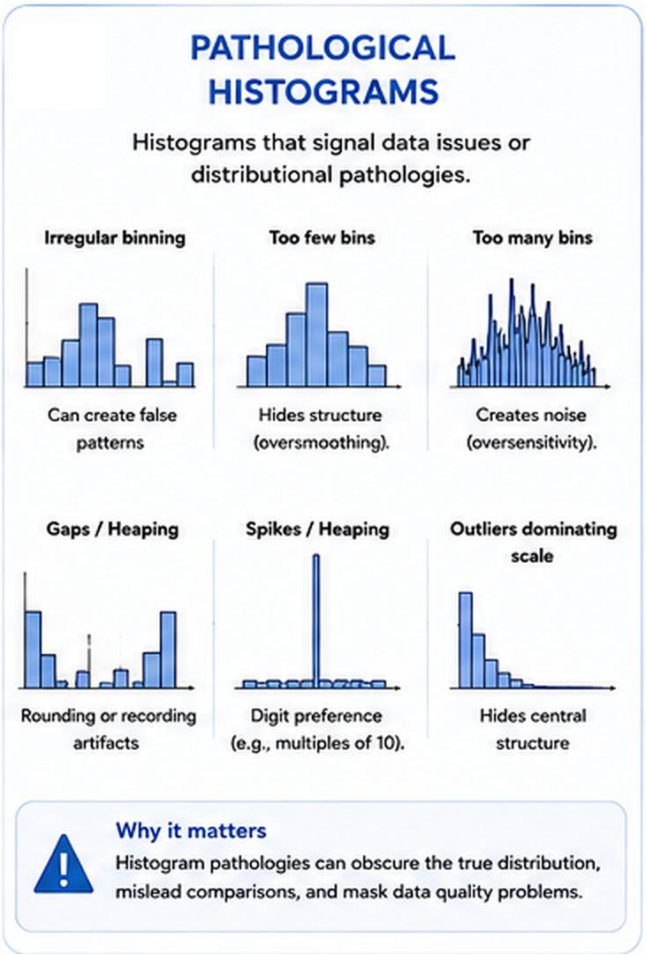

Figure 27: Pathological patterns in histograms. Irregular binning, inappropriate numbers of bins, gaps, spikes, value heaping, and extreme observations can either reveal genuine characteristics of the data or create misleading visual structure. Histogram interpretation should therefore consider both plotting choices and the underlying measurement process.

Real-world examples: Patient ages may show spikes at multiples of five or ten when age is self-reported or rounded. Retail prices often cluster around

values such as 9.99 or 19.99, producing noticeable heaping. Blood-pressure measurements recorded manually may accumulate at multiples of 5 or 10 mmHg. Sensor data may exhibit gaps because a device cannot report values within certain ranges or because measurements were missing during downtime. A histogram of household expenditure may appear compressed when a few extremely high expenditures dominate the horizontal scale.

- **Pathological density plots:** Kernel density plots provide a **smooth representation of a distribution**, but their appearance depends strongly on the smoothing bandwidth and on assumptions made near the boundaries of the variable. **Excessive smoothing** can erase genuine peaks or subpopulations, whereas **insufficient smoothing** can produce spurious oscillations that resemble meaningful modes. **Boundary bias** can distort the estimated density for variables restricted to a finite interval, such as proportions or nonnegative measurements. Other **diagnostically important patterns** include multiple modes, heavy tails, shoulders, abrupt cut-offs, and sharp peaks caused by zero inflation or measurement limits (Figure 28). Consequently, density plots should be viewed as estimates of distributional shape rather than literal representations of the observed data.

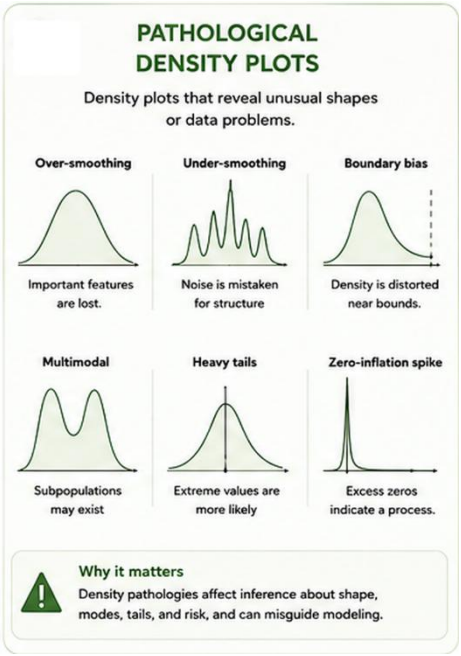

Figure 28: Pathological features in density plots. Over-smoothing, under-smoothing, boundary bias, multimodality, heavy tails, and sharp probability concentrations can substantially alter the apparent shape of a kernel density estimate. Density plots should therefore be interpreted alongside the raw data and with careful attention to bandwidth and variable boundaries.

Real-world examples: A density plot of patient body mass index may show several modes if healthy, overweight, and obese populations are mixed. Household income may exhibit a long right tail. A density estimate of proportions bounded between 0 and 1 can incorrectly assign apparent probability outside the feasible range unless boundary-aware methods are used. Laboratory measurements may show a sharp accumulation at a detection limit when values below that threshold are recorded as the same number. Mixtures of measurements from different hospitals or scanners may create shoulders or secondary peaks.

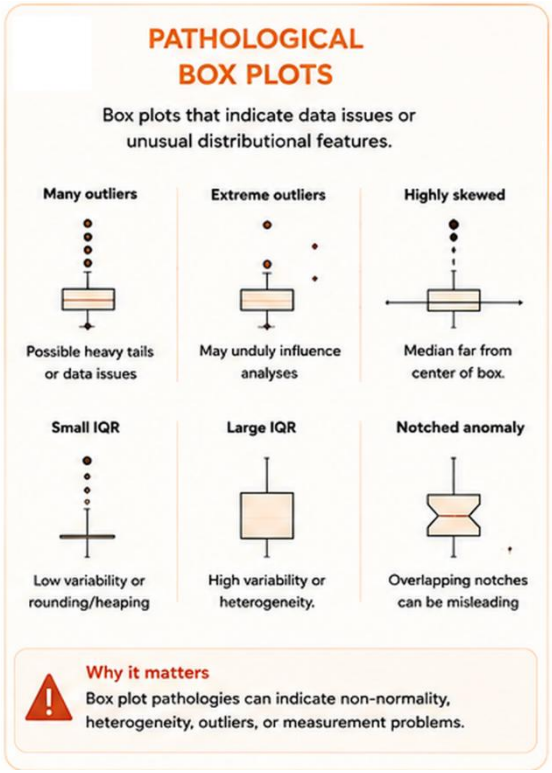

Figure 29: Pathological patterns in box plots. Numerous extreme observations, highly asymmetric whiskers, displaced medians, and unusually small or large interquartile ranges can indicate skewness, heavy tails, heterogeneity, rounding, or data-quality problems. Box-plot outliers should be investigated in context rather than automatically removed.

- **Pathological box plots:** A box plot compactly summarizes the median, interquartile range, and observations lying beyond conventional whisker limits, making it especially useful for detecting asymmetry, extreme observations, and group-level differences in variability. Pathological-looking box plots may contain numerous **flagged points**, extremely **long whiskers**, a **median displaced** toward one side of the box, unusually **small** or **large interquartile ranges**, or **strong differences** between groups (Figure 29). Importantly, points beyond the usual $1.5 \times \mathrm{IQR}$ fences are not

automatically errors; they may represent legitimate observations from skewed or heavy-tailed distributions. The box plot should therefore be treated as a diagnostic device that identifies observations or groups deserving further investigation rather than as an automatic rule for deleting data.

Real-world examples: Hospital charges frequently generate many upper-boxplot outliers because a small number of patients receive exceptionally expensive treatment. Delivery times in logistics may have a long upper whisker because of occasional severe delays. Manufacturing dimensions may show a narrow box but several distant points caused by machine faults or measurement errors. Regional house prices may differ dramatically in both median and IQR because neighborhoods represent heterogeneous markets. Clinical biomarkers with physiological lower bounds may produce strongly asymmetric box plots.

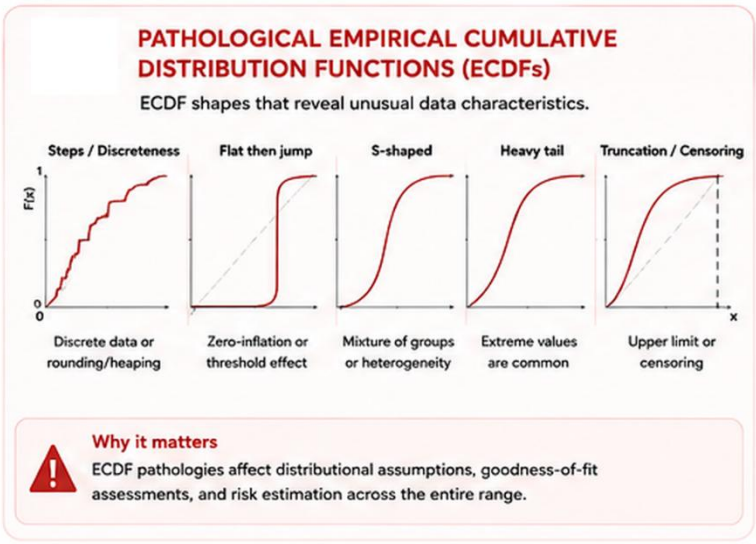

Figure 30: Pathological patterns in empirical cumulative distribution functions. Large steps, flat regions, unusual changes in slope, extended tails, and abrupt truncation can reveal discreteness, heaping, mixture populations, extreme values, censoring, or measurement limits without requiring arbitrary binning or smoothing.

- **Pathological empirical cumulative distribution functions:** The empirical cumulative distribution function, or ECDF, shows the **proportion of observations less than or equal to each observed value**. Unlike a histogram or density plot, it does not require bins or smoothing, making it particularly useful for exposing discreteness, repeated measurements, truncation, censoring, and distributional mixtures. Large vertical steps indicate many observations with the same value, long flat segments indicate ranges containing few or no observations, and abrupt changes in

slope can reveal multiple underlying groups or thresholds. Heavy tails appear as gradual extensions toward extreme values, while truncation or censoring can produce abrupt terminations or accumulations near measurement limits (**Figure 30**).

Real-world examples: Survey scores recorded on a five-point Likert scale produce a strongly stepped ECDF because only a few distinct values are possible. Blood glucose measurements rounded to whole numbers may show repeated jumps. Survival times may exhibit censoring at the end of a study. Device measurements constrained by an upper detection limit can produce a sharp accumulation near that limit. Customer waiting times drawn from two service regimes may create an ECDF with noticeably different slopes across ranges.

- **Pathological quantile plots:** Quantile–quantile, or **Q–Q**, plots compare the **quantiles of observed data** with the **quantiles expected under a reference distribution**, most commonly the normal distribution. A close match produces points approximately along a straight line, whereas systematic departures reveal different forms of distributional mismatch. **Outward or inward departures at the ends** can indicate **heavy or light tails**, respectively; **asymmetric curvature** suggests **skewness**; **changes in slope or clustering** may indicate **mixtures or heterogeneous subpopulations**; and **isolated points far from the reference line** may indicate **outliers or extreme observations** (**Figure 31**). Q–Q plots are therefore especially useful for assessing whether distributional assumptions are plausible and for identifying where and how the observed distribution departs from the reference model.

  Real-world examples: Financial asset returns often deviate from a normal Q–Q line in both tails, reflecting extreme gains and losses occurring more often than a Gaussian model predicts. Biological measurements such as concentrations may show right-skewed departures unless log-transformed. Residuals from a regression model may form an S-shaped Q–Q plot when errors come from multiple subpopulations. Manufacturing measurements may follow the reference distribution except for a handful of distant points corresponding to defective products. Test scores bounded near a maximum may produce lighter upper tails than a normal model would imply.

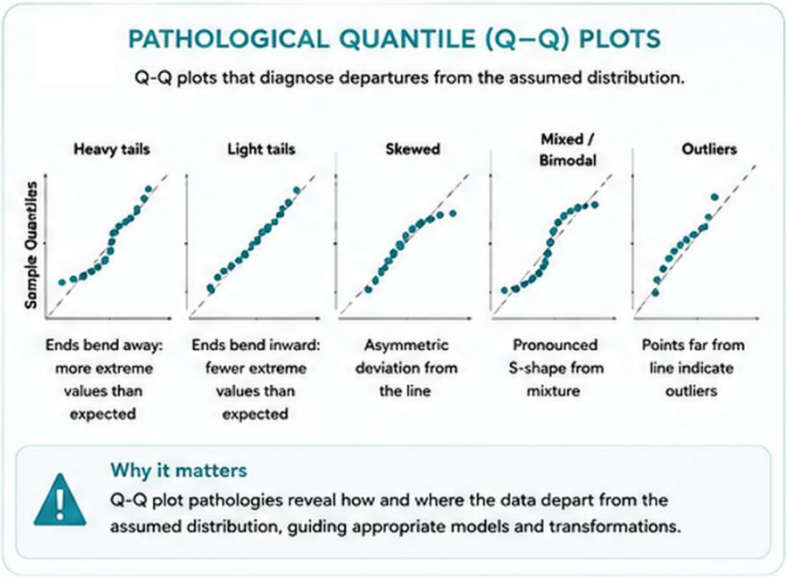

Figure 31: Diagnosing distributional departures using quantile–quantile plots. Systematic deviations from the reference line can reveal **heavy or light tails, skewness, and other distributional departures**, while changes in slope or clustering may suggest **mixtures or heterogeneous subpopulations**, and isolated distant points may indicate **outliers**. Q–Q plots provide a direct visual assessment of where and how an observed distribution departs from an assumed reference distribution.

3.2.2 Understanding relationships

EDA also investigates **how variables vary together**. Relationships between variables can provide important information about the structure of the data and may suggest variables that should be **considered in subsequent statistical** or **predictive models**. Possible forms of association include (Figure 32):

- positive linear relationships;
- negative linear relationships;
- nonlinear relationships;
- monotonic but nonlinear relationships;
- threshold effects;
- interactions between variables;
- clusters or distinct subpopulations;
- apparent absence of association;
- associations driven primarily by outliers; and
- relationships that differ or reverse across subgroups.

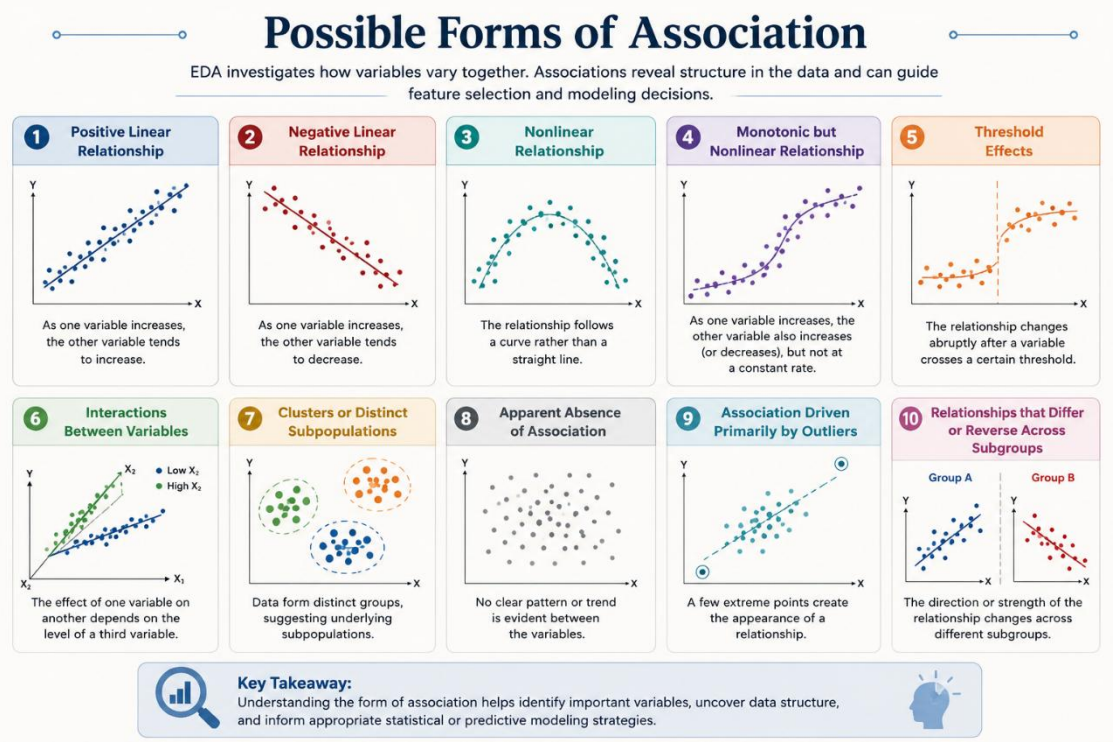

Figure 32: Possible forms of association between variables. Exploratory data analysis can reveal linear, nonlinear, monotonic, threshold-based, interaction-driven, clustered, outlier-driven, or subgroup-dependent relationships between variables, as well as an apparent absence of association. Recognizing these patterns helps uncover underlying data structure and guides subsequent statistical analysis, feature selection, and predictive modeling.

A relationship that appears strong in an aggregate dataset may have a very different interpretation after the data are stratified by another variable. For this reason, relationships should often be examined both overall and within relevant subgroups.

Example # 68: Exercise and blood pressure

Suppose a researcher examines data from **1,000 adults** to investigate the relationship between **weekly exercise time** and **blood pressure**. In the overall dataset, a scatterplot shows a strong negative relationship: people who exercise more tend to have lower blood pressure.

At first glance, the researcher might conclude that **increased exercise is strongly associated with lower blood pressure**. However, the researcher then stratifies the data by **age group**:

- **Younger adults:** the relationship between exercise and blood pressure is weak.
- **Middle-aged adults:** the relationship is moderately negative.
- **Older adults:** the relationship is much stronger.

This suggests that **age may influence the relationship between exercise and blood pressure**. The strong association observed in the aggregate data may partly reflect differences in exercise habits and blood pressure across age groups rather than a uniform relationship within every group.

For example, if older adults generally exercise less and have higher blood pressure, combining them with younger adults could make the overall relationship appear stronger than it is within some age groups.

The key lesson is that an **aggregate relationship** does not necessarily describe the relationship within individual subgroups. **Stratification can reveal** whether an **apparent** association is **consistent**, **weaker**, **stronger**, or even **reversed** across different groups.

Causation

Importantly, **association does not by itself establish causation**[266]. An observed relationship between X and Y may arise for several different reasons. For example:

- X may causally influence Y;
- Y may causally influence X;
- a third variable may influence both X and Y;
- the sampling or selection process may create the association;
- measurement, preprocessing, or data-processing procedures may create the observed pattern; or
- the apparent relationship may have occurred by chance.

[266] **Causation** refers to a relationship in which a **change in one variable** or condition **produces a change in another variable**, such that the outcome would differ if the causal factor were absent or altered. In statistical and scientific contexts, **establishing causation** generally requires **evidence** that the **proposed cause precedes the effect** and that alternative explanations, such as confounding variables or selection bias, have been adequately addressed. For example, a randomized controlled trial may demonstrate that a new medication **causes** a reduction in blood pressure if patients randomly assigned to receive the medication experience a greater reduction than those receiving a placebo. In contrast, observing that people who exercise regularly have lower blood pressure does not, by itself, establish that exercise causes the reduction, because factors such as diet, age, or overall health may also influence blood pressure.

Consequently, EDA can **reveal** and **characterize associations**, but ordinary exploratory analysis generally cannot determine which causal explanation is correct. Establishing **causal claims** requires **additional assumptions**, **research designs**, **interventions**, or **causal models**. Judea Pearl's work on causality, for example, emphasizes the distinction between statistical association and causal relationships and the importance of explicit causal assumptions for causal inference[267]. In a nutshell, *EDA can identify relationships worth investigating; it does not, by itself, establish why those relationships exist*.

Example # 69: Exercise and blood pressure

Suppose an EDA of 2,000 adults reveals that people who **exercise more frequently tend to have lower blood pressure**. A scatterplot shows a clear negative association between weekly exercise hours and blood pressure.

EDA can help the researcher **reveal and characterize** this association; for example, by showing its direction, strength, and whether the relationship is linear or nonlinear. However, the analysis alone cannot establish **why** the relationship exists. Several explanations are possible:

- **Exercise may reduce blood pressure:** Regular physical activity could directly contribute to lower blood pressure.
- **Reverse causation:** People with high blood pressure may exercise less because of their health condition.
- **Confounding:** People who exercise regularly may also have healthier diets, lower stress levels, or better access to healthcare, which could contribute to lower blood pressure.
- **Selection effects:** The people who choose to exercise regularly may differ systematically from those who do not.

Therefore, **observing an association does not tell** us which causal explanation is correct.

[267] Judea Peral, *Causal inference in statistics: An overview*, Statistical Surveys, Volume 3, pp 96-146, 2009.

To make a stronger causal claim, the researcher might conduct a **randomized intervention**[268], where participants are randomly assigned to an exercise program or a control condition and their blood pressure is subsequently compared. Alternatively, researchers could use an appropriate **causal model** together with clearly stated assumptions and observational data.

3.2.3 Detecting anomalies

An **anomaly** is an observation, combination of observations, or broader pattern that **differs substantially** from what would **normally be expected** according to an appropriate reference or data-generating process. Examples include:

- a body temperature recorded as $(350^\circ\ \mathrm{C})$;
- a negative age;
- an account recording thousands of transactions per second;
- a sudden discontinuity in a sensor signal;
- a combination of individually plausible values that is jointly impossible; and
- an unusual error pattern concentrated within a particular subgroup.

An anomaly does not automatically mean that an observation is wrong. It may represent:

- a data-entry error;
- a unit-conversion error;
- an equipment or sensor failure;
- a rare but legitimate observation;
- fraudulent activity;
- an important scientific or operational discovery; or
- evidence that the underlying data distribution has changed.

[268] A **randomized controlled trial (RCT)** is an experimental study in which participants are **randomly assigned** to two or more groups, typically including an intervention or treatment group and a control group, so that the effect of the intervention can be compared with a counterfactual outcome under control conditions. Randomization helps balance both observed and unobserved characteristics between groups, thereby reducing confounding and strengthening the basis for **causal inference**. For example, to determine whether a new drug lowers blood pressure, researchers might randomly assign 500 participants to either receive the new drug or receive a placebo. After a specified treatment period, the researchers compare the groups' average changes in blood pressure. Similarly, an RCT can evaluate an educational program by randomly assigning students or schools to receive either a new teaching intervention or the standard curriculum and then comparing their subsequent test scores.

For this reason, anomalies should generally be **investigated rather than automatically deleted**. Removing extreme observations simply because they appear unusual can eliminate precisely the observations that contain important information. A practical anomaly-investigation workflow is depicted in **Figure 33**:

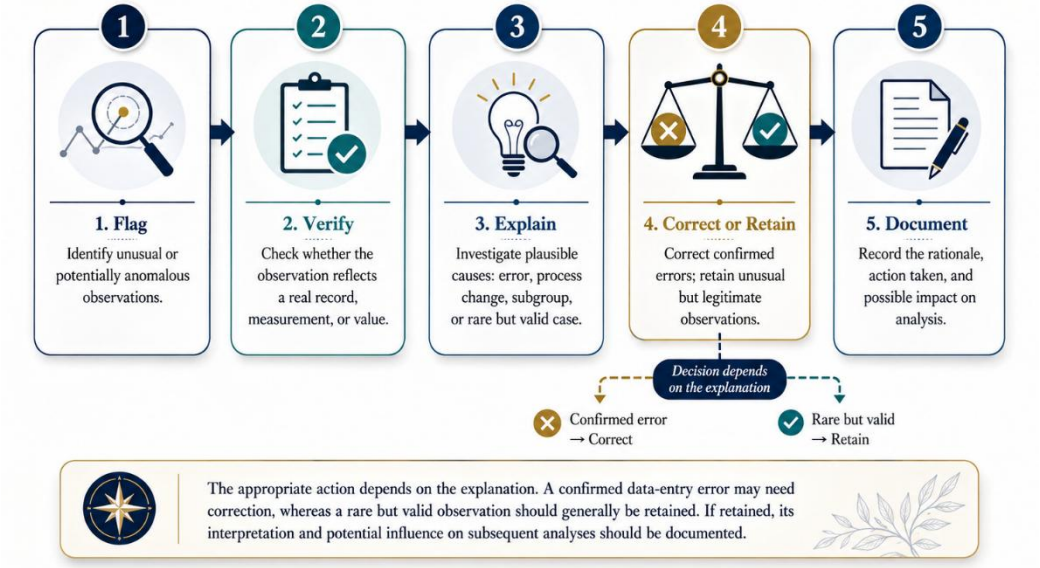

Figure 33: Practical workflow for investigating anomalous observations. Potential anomalies should be handled through a structured sequence of **flagging**, **verification**, **explanation**, **correction** or **retention**, and **documentation**. The appropriate action depends on the underlying cause: confirmed errors may require correction, whereas rare but valid observations should generally be retained. In either case, the rationale for the decision and the potential influence of the observation on subsequent analyses should be documented.

3.2.4 Understanding missingness

Missing data are not merely an inconvenience to be dealt with during preprocessing. The **pattern and mechanism of missingness can itself contain information** about the data-generating process and can affect statistical inference and model performance. Missingness should therefore be examined at several levels.

Variable-level missingness

For feature j, the proportion of missing observations can be expressed as:

$$r_j = \frac{\#\text{ of missing observations in feature } j}{n}$$

where n is the total number of observations (rows/records) in the dataset. This provides an initial assessment of which variables have substantial missingness and may require further investigation.

Record-level missingness

For observation i, the proportion of missing features can be expressed as:

$$r_i = \frac{\#\text{ of missing features in record } i}{p}$$

where p is the total number of features (columns/variables) in the dataset. Examining record-level missingness can reveal observations that are unusually incomplete. Such patterns may indicate specific data-collection processes, system failures, or characteristics of the individuals or entities being observed.

Pattern-level missingness

The analyst should also determine whether particular variables tend to be missing together. For example:

- blood-pressure and heart-rate measurements may both be missing because a monitoring device failed;
- income and employment information may both be missing because respondents declined to provide sensitive information; or
- laboratory measurements may be missing because clinicians did not order the tests for patients who appeared unlikely to have a particular condition.

Subgroup-specific missingness

Missingness should also be compared across relevant groups. For example:

$$P(R_j = 0 | G = g)$$

Where $R_j = 0$ means that feature j is missing and G denotes subgroup membership. Substantial differences in missingness rates across groups can lead to:

- unequal effective sample sizes;
- differential imputation error;
- distorted feature distributions; and
- unequal model reliability or predictive performance.

Consequently, an overall missingness percentage can be misleading if it conceals large differences between groups or time periods.

Example # 70: Understanding missingness in patient data

Suppose a hospital has a dataset of **10,000 patients** containing age, sex, blood pressure, heart rate, income, employment status, and several laboratory measurements. Some values are missing. Rather than simply deleting incomplete records, the analyst examines **how**, **where**, and **why** the **data** are **missing**.

Variable-level missingness: The analyst first calculates the proportion of missing values for each variable. For example:

- Blood pressure: **5% missing**
- Heart rate: **6% missing**
- Income: **25% missing**
- Employment status: **18% missing**
- Laboratory measurement: **3% missing**

This immediately shows that **income has substantially more missing values than most clinical variables**.

Record-level missingness: Next, the analyst examines how many variables are missing for each patient. For example:

- Patient A: 0 missing variables
- Patient B: 1 missing variable
- Patient C: 4 missing variables
- Patient D: 7 missing variables

If a large number of patients have many missing values, simply removing incomplete records could result in a substantial loss of information.

Pattern-level missingness: The analyst then investigates whether variables tend to be missing together. Suppose they discover that **blood pressure and heart rate are frequently missing in the same records**.

Further investigation reveals that both measurements were collected using the same electronic monitoring device, which occasionally failed.

In another pattern, **income and employment status are often missing together** because some patients chose not to answer questions they considered sensitive.

These patterns suggest that missingness is **not necessarily random** and may have identifiable causes.

Subgroup-level missingness: Finally, the analyst examines missingness across patient groups. Suppose laboratory measurements are missing for:

- 2% of **low-risk** patients
- 4% of **moderate-risk** patients
- 15% of **high-risk** patients

This difference is important. If laboratory measurements are disproportionately missing among high-risk patients, the available laboratory data may not represent the overall patient population equally.

This can lead to:

- **unequal effective sample sizes** across groups;
- different levels of **imputation uncertainty**;
- distorted distributions of laboratory variables; and
- **less reliable predictions for groups with greater missingness**.

For example, a model trained primarily on patients with complete laboratory measurements may perform well for low-risk patients but poorly for high-risk patients if laboratory data are frequently unavailable for the latter group.

Example # 71: Subgroup-specific missingness pattern in Income data in a healthcare study

Imagine a hospital network collects patient data to predict diabetes risk. One important feature is **annual household income** (feature j), which is known to correlate with health outcomes. The patient population includes three insurance subgroups (G):

- **Medicare**[269] (primarily elderly, age 65+)

[269] **Medicare** is a U.S. federal health insurance program that primarily provides coverage to people aged **65 and older**, as well as certain younger individuals with qualifying disabilities or specific medical conditions. For example, a 70-year-old patient receiving health coverage through the federal Medicare program is classified as a Medicare beneficiary.

- **Medicaid**[270] (low-income patients)
- **Private Insurance**[271] (working-age, employed patients)

When the data team examines missingness rates, they discover:

- $P(\text{Income missing} \mid \text{Medicare}) = 0.55$ (**55% missing**)
- $P(\text{Income missing} \mid \text{Medicaid}) = 0.10$ (**10% missing**)
- $P(\text{Income missing} \mid \text{Private}) = 0.25$ (**25% missing**)

One might ask, *why does this happen*?

- Medicare patients (elderly) are often retired and may not report income because it feels irrelevant to their medical care, or they may have complex income sources (pensions, Social Security, investments) that are harder to document.
- Medicaid patients must provide income documentation to qualify for the program, so this data is nearly always complete.
- Private insurance patients have moderate missingness because income is sometimes collected during employment verification but not always systematically recorded.

The subgroup-specific missingness may have (four) consequences in practice:

- **Unequal Effective Sample Sizes:**
  - If the dataset has 10,000 patients per insurance group, after removing records with missing income (a common but flawed practice), the team would retain:
    - Medicare: 4,500 patients (45% of original)
    - Medicaid: 9,000 patients (90% of original)
    - Private: 7,500 patients (75% of original)

[270] **Medicaid** is a joint **federal and state** health insurance program that provides coverage to eligible individuals and families with limited income and resources, with eligibility rules varying by state. For example, a low-income adult who meets their state's Medicaid eligibility requirements may receive physician visits, hospital care, and other covered health services through Medicaid.

[271] **Private insurance** refers to health insurance coverage provided by **non-governmental insurance companies**, typically obtained through an employer, purchased individually, or provided through other private arrangements. *Example:* An employee whose employer provides a health plan administered by a private insurer is considered to have private insurance. Although private insurance is common among working-age adults, it is **not limited to employed or working-age patients**; people of various ages and employment statuses may have private coverage.

- The model is now **dominated by Medicaid patients** for learning the income-diabetes relationship, **even though** the **original population** was **balanced**.

- **Differential Imputation Error:**
  - If the team uses imputation[272] to fill in missing income values, the imputation model will be **most accurate for Private insurance** (moderate missingness, good data quality) and **least accurate for Medicare** (high missingness, complex patterns).
  - The imputed income values for elderly Medicare patients may be systematically biased (e.g., underestimating retirement income), introducing **differential measurement error** across groups.

- **Distorted Feature Distributions:**
  - The observed income distribution for Medicare patients (based on only 45% of the population who reported) may be **unrepresentative**. Perhaps only Medicare patients with higher incomes or more complex financial situations bothered to report, creating a **selection bias**.
  - The model learns that $\textbf{Medicare} + \textbf{high income} = \textbf{low diabetes risk}$, but this **relationship is based on a non-representative subset**, not the true Medicare population.

- **Unequal Model Reliability/Predictive Performance:**
  - When deployed, the model's diabetes risk predictions will be **most reliable for Medicaid patients** (income data is real, not imputed) and **least reliable for Medicare patients** (income data is mostly imputed with high uncertainty).
  - However, the model's overall accuracy metric (e.g., 85% AUC) masks this disparity. Clinicians might trust the predictions equally for all

[272] **Data imputation** is a statistical process of **replacing missing** or **incomplete values** in a dataset with **plausible estimated values** based on observed information and a **specified imputation method**. The imputed value **is an estimate** rather than an actual recorded observation, and the purpose is to **reduce information loss** and allow statistical analyses to be conducted when data are incomplete. For example, if patients' ages are recorded as 45, 52, **missing**, 61, and 55, the missing age could be replaced with the mean or median age of an appropriate group. In **regression imputation**, the missing value might instead be predicted using other variables, such as income, education, or occupation. **Multiple imputation** provides another approach in which several plausible values are generated for each missing observation, allowing the analysis to incorporate uncertainty associated with the missing data.

patients, not realizing that Medicare patients' risk scores are based on shaky, imputed data.

- This could lead to **misallocation of preventive care resources**: a high-risk Medicare patient might be under-prioritized because their imputed income artificially lowered their predicted risk.

What is the solution? The data team should:

- **Report missingness rates by subgroup** before modeling (in the Datasheet for Datasets).
- **Use missingness indicators** (binary flags for income was missing) as features, allowing the model to learn that missing income for Medicare patients carries different meaning than missing income for Medicaid patients.
- **Evaluate model performance separately for each subgroup**, not just overall (and document in the model card).
- **Consider whether income should be used at all** for Medicare patients if missingness is too high and imputation too unreliable.

3.2.5 Identifying bias

EDA can provide evidence of several forms of bias that may **affect** the **validity**[273], **fairness**[274], or **generalizability**[275] of an analysis. Bias does not necessarily mean that an analyst has made an error; in many cases, it arises from the way data were collected, measured, selected, labeled, or inherited from an existing system. Important forms include:

Sampling bias

[273] **Validity** refers to the extent to which a statistical analysis, measurement, or study **accurately represents or supports** the **conclusions** it is intended to produce. For example, if a study aims to estimate the effect of a treatment on blood pressure but systematically excludes patients with certain medical characteristics, its estimated treatment effect may have reduced validity.

[274] **Fairness** refers to the extent to which a statistical method, decision, or predictive system produces **equitable outcomes across relevant groups** and **does not systematically disadvantage** individuals because of protected or otherwise sensitive characteristics. For example, a university admissions model may raise fairness concerns if its predictions systematically disadvantage qualified applicants from a particular demographic group.

[275] **Generalizability** refers to the extent to which **findings** obtained from a particular sample, setting, or population can **reasonably be applied to other** populations, settings, or contexts beyond those studied. For example, a medical study conducted only among young adults may have limited generalizability to older adults.

The observed sample **differs systematically** from the target population. As a result, relationships or estimates found in the sample may not generalize to the population of interest.

Example # 72: Examples of sampling bias

- **Online healthcare survey**
  - *Target population:* All adults in a country.
  - *Sample:* People who voluntarily respond to an online survey.
  - *Problem:* People without reliable internet access or those less likely to participate online may be underrepresented.
  - *Consequence:* The estimated level of healthcare satisfaction may not represent the entire population.
- **Student study habits**
  - *Target population:* All university students.
  - *Sample:* Students attending an optional study-skills workshop.
  - *Problem:* Workshop participants may be more motivated or academically concerned than other students.
  - *Consequence:* Their average study time may be higher than that of the overall student population.
- **Customer satisfaction survey**
  - *Target population:* All customers.
  - *Sample:* Customers who recently made a purchase.
  - *Problem:* Customers who stopped purchasing because they were dissatisfied are excluded.
  - *Consequence:* Customer satisfaction may be systematically overestimated.
- **Physical activity study**
  - *Target population:* All adults in a city.
  - *Sample:* People recruited from gyms and sports clubs.
  - *Problem:* Gym members are likely to be more physically active than the general population.
  - *Consequence:* Average physical activity may be overestimated.

Measurement bias

A variable is **measured differently** across groups, locations, devices, or time periods. **Differences in measurement procedures** can create apparent differences that are not entirely attributable to the underlying phenomenon.

Example # 73: Examples of measurement bias

- **Blood pressure measurement**
  - *Situation:* A study compares blood pressure between two hospitals.
  - *Problem:* Hospital A uses a properly calibrated automated device, while Hospital B uses an older device that consistently records slightly higher values.
  - *Consequence:* Patients in Hospital B may appear to have higher blood pressure even if their true blood pressure is similar.
  - *Lesson:* The observed difference may partly reflect **differences in measurement procedures**, not differences in the underlying condition.
- **Exam performance across schools**
  - *Situation:* Researchers compare mathematics performance between two schools.
  - *Problem:* One school administers a standardized examination under strict conditions, while the other uses a teacher-designed test that is easier.
  - *Consequence:* Students from the first school may appear to perform worse, even if their actual mathematical ability is similar.
  - *Lesson:* Differences in the **measurement instrument** can create apparent group differences.
- **Temperature measurements from different devices**
  - *Situation:* A researcher monitors environmental temperature at two locations.
  - *Problem:* One location uses a calibrated digital sensor, while the other uses an older sensor that tends to record temperatures $2°\,\mathrm{C}$ higher.
  - *Consequence:* The second location may appear consistently warmer.
  - *Lesson:* Apparent differences may result from **systematic measurement error**.
- **Income reported through different methods**

- *Situation:* A researcher compares household income between two groups.
- *Problem:* One group reports income using detailed tax records, while the other provides self-reported estimates.
- *Consequence:* Differences in reporting accuracy may create an apparent income difference that does not fully reflect the true difference.
- *Lesson:* The **method of measurement** can influence the observed values.

- **Health measurements over time**
  - *Situation:* A hospital tracks patients' weight over several years.
  - *Problem:* Earlier measurements were taken using one type of scale, while later measurements use a different scale with a systematic offset.
  - *Consequence:* The data may suggest a change in patients' average weight when part of the change is actually caused by the new measurement system.
  - *Lesson:* Changes in **measurement procedures over time** can be mistaken for genuine changes in the underlying phenomenon.

Label bias

The recorded outcome **does not perfectly represent** the **underlying construct** of interest. In many applications, labels are based on human judgments, institutional decisions, administrative records, or imperfect proxies.

Example # 74: Examples of label bias

- **Student dropout prediction**
  - *Underlying construct:* Whether a student is genuinely disengaged from education and likely to discontinue their studies.
  - *Recorded label:* Whether the university officially records the student as a dropout.
  - *Problem:* A student who transfers to another institution may be recorded as a dropout even though they continue their education.

- *Consequence:* A model trained on the administrative label may learn to predict **institutional dropout records** rather than genuine educational disengagement.
- *Lesson:* The recorded label is an **imperfect proxy** for the construct of interest.

- **Medical diagnosis**
  - *Underlying construct:* Whether a patient actually has a particular disease.
  - *Recorded label:* Whether a physician diagnosed the patient with that disease.
  - *Problem:* Diagnosis may depend on the physician's experience, available tests, clinical resources, or subjective judgment.
  - *Consequence:* Patients with similar underlying conditions may receive different labels.
  - *Lesson:* Human or institutional decisions can introduce systematic differences into labels.

- **Credit-risk prediction**
  - *Underlying construct:* A person's underlying ability and willingness to repay a loan.
  - *Recorded label:* Whether the person previously defaulted on a loan.
  - *Problem:* A person may never have defaulted simply because they were never given access to credit. Conversely, a temporary financial shock may result in default even when the person's longer-term repayment ability is good.
  - *Consequence:* The recorded default label may not perfectly represent the underlying creditworthiness of individuals.
  - *Lesson:* Historical outcomes can be **imperfect proxies** for the construct researchers actually want to measure.

- **Employee performance**
  - *Underlying construct:* An employee's actual productivity and quality of work.
  - *Recorded label:* Annual performance rating assigned by a manager.
  - *Problem:* Ratings may be influenced by managerial judgment, personal perceptions, visibility, or differences in evaluation standards.

- *Consequence:* Two employees with similar productivity may receive different labels.
- *Lesson:* Human-assigned labels can contain **subjectivity and inconsistency**.

- **Customer satisfaction**
  - *Underlying construct:* A customer's overall satisfaction with a service.
  - *Recorded label:* Whether the customer submits a complaint.
  - *Problem:* Some dissatisfied customers complain, while others simply leave without reporting their dissatisfaction.
  - *Consequence:* No complaint does not necessarily mean satisfied.
  - *Lesson:* Administrative records often capture only a **specific observable behavior**, rather than the complete underlying construct.

Survivorship bias

The dataset contains **only units that have survived** or **remained observable** through a particular process. Consequently, observations that disappeared, failed, withdrew, or were otherwise excluded may be systematically absent.

Example # 75: Examples of survivorship bias

- **Business success**
  - *Situation:* A researcher studies companies that have been operating successfully for at least 10 years.
  - *Problem:* Companies that failed, went bankrupt, or closed are absent from the dataset.
  - *Consequence:* The researcher may conclude that certain strategies are common among successful companies without realizing that failed companies may have used the same strategies.
  - *Lesson:* Studying only surviving companies can make successful practices appear more effective than they actually are.
- **Investment performance**
  - *Situation:* An analysis examines the historical returns of investment funds that currently exist.
  - *Problem:* Funds that performed poorly may have been closed or merged and therefore disappear from the dataset.

- *Consequence:* The average historical performance of the remaining funds may appear artificially high.
- *Lesson:* Ignoring funds that disappeared can **overestimate investment performance**.

- **University student performance**
  - *Situation:* A university analyzes the academic performance of students who graduate.
  - *Problem:* Students who dropped out or transferred are excluded.
  - *Consequence:* The analysis may suggest that students generally achieve high grades and successfully complete their programs, while overlooking students who struggled and left.
  - *Lesson:* Focusing only on graduates can produce an overly positive picture of student outcomes.
- **Employee retention**
  - *Situation:* A company studies the characteristics of employees who have worked there for more than five years.
  - *Problem:* Employees who resigned or were dismissed earlier are not included.
  - *Consequence:* The company may conclude that long-term employees are highly satisfied and productive, without considering whether similar employees left because of dissatisfaction.
  - *Lesson:* The dataset represents **surviving employees**, not necessarily the entire workforce.
- **Medical treatment**
  - *Situation:* Researchers examine the long-term health outcomes of patients who completed a five-year treatment program.
  - *Problem:* Patients who died, discontinued treatment, or were lost to follow-up are missing from the analysis.
  - *Consequence:* The treatment may appear more successful than it actually was because patients with poor outcomes are systematically underrepresented.
  - *Lesson:* Excluding patients who disappear from observation can **bias estimates of treatment effectiveness**.

Selection bias

Inclusion in the dataset depends on **characteristics related** to the **variables** or **outcomes being studied**. This can produce associations that differ from those in the target population.

Example # 76: Examples of selection bias

- **Hospital study of risk factors**
  - *Situation:* Researchers study the relationship between smoking and heart disease using only patients admitted to a hospital.
  - *Problem:* Hospital admission depends on health conditions, and both smoking and other risk factors may influence the likelihood of hospitalization.
  - *Consequence:* The relationship observed among hospitalized patients may differ from the relationship in the general population.
  - *Lesson:* Restricting the analysis to a selected population can distort associations.
- **University admission and student success**
  - *Situation:* A researcher examines whether high-school grades predict university performance using only students who were admitted to a highly selective university.
  - *Problem:* Admission itself depends partly on previous academic performance and other characteristics.
  - *Consequence:* The relationship between high-school grades and university performance may appear weaker or different because students with lower grades were excluded before the analysis began.
  - *Lesson:* Selection into the dataset can change the relationship between variables.
- **Employment and health**
  - *Situation:* A researcher investigates whether physical health is related to income using only employed individuals.
  - *Problem:* Employment depends partly on health. People with poorer health may be less likely to be employed and therefore less likely to appear in the dataset.

- *Consequence:* The relationship between health and income among employed people may not represent the relationship in the entire working-age population.
- *Lesson:* Conditioning on employment can create or distort associations.

- **Customer satisfaction**
  - *Situation:* A company studies whether customer satisfaction predicts continued purchasing, using only customers who made at least one purchase during the past year.
  - *Problem:* Customers who stopped purchasing are excluded, even though dissatisfaction may have contributed to their decision to leave.
  - *Consequence:* The relationship between satisfaction and continued purchasing may be distorted because inclusion already depends on purchasing behavior.
  - *Lesson:* Selecting observations based on an outcome-related characteristic can bias the observed relationship.

- **Clinical research**
  - *Situation:* Researchers study whether a treatment improves recovery using data only from patients who completed the entire treatment program.
  - *Problem:* Completion may depend on health status, treatment response, side effects, or other factors related to recovery.
  - *Consequence:* The treatment may appear more effective among completers than it would be in the full group of patients who initially received the treatment.
  - *Lesson:* When inclusion depends on characteristics related to the outcome, the observed association may not generalize to the target population.

Historical bias

The data **accurately reflect** an **existing historical system** or **social process** that itself contains **inequalities** or **systematic differences**. A dataset can therefore be carefully collected and internally consistent while **still reproducing historical patterns** that are **inappropriate** for a **particular application**.

Example # 77: Examples of historical bias

- **Hiring and promotion**
  - *Situation:* A company builds a model to predict which applicants are likely to succeed using 20 years of historical hiring and promotion records.
  - *Problem:* Historically, certain groups may have received fewer opportunities because of unequal hiring or promotion practices.
  - *Consequence:* Even if the historical records are accurate, a model trained on them may learn patterns that reflect **past unequal opportunities** rather than genuine differences in ability or potential.
  - *Lesson:* Accurate historical data can reproduce historical inequalities.
- **University admissions**
  - *Situation:* A university uses past admission decisions to develop a model for identifying strong applicants.
  - *Problem:* Historical admission decisions may have been influenced by differences in access to high-quality schools, extracurricular opportunities, or other forms of unequal opportunity.
  - *Consequence:* The model may learn that applicants with certain backgrounds were historically more likely to be admitted and treat this historical pattern as evidence of future suitability.
  - *Lesson:* Historical outcomes may reflect **social and institutional inequalities**, not just merit.
- **Healthcare treatment**
  - *Situation:* Researchers use historical medical records to predict which patients received a particular treatment.
  - *Problem:* In the past, treatment decisions may have differed systematically across patient groups because of unequal access to healthcare or differences in clinical practice.
  - *Consequence:* A model may learn historical treatment patterns and reproduce them, even if those patterns are no longer considered appropriate.
  - *Lesson:* Historical clinical decisions may contain **systematic inequalities** that should not automatically be treated as desirable targets.

- **Credit lending**
  - *Situation:* A bank develops a credit-risk model using decades of historical lending decisions and repayment records.
  - *Problem:* Historical lending practices may have restricted access to credit for certain communities.
  - *Consequence:* The resulting dataset may contain fewer observations of successful borrowing from those communities, making the historical data appear to support continued differences in credit access.
  - *Lesson:* Historical lack of opportunity can become embedded in later models.
- **Policing data**
  - *Situation:* Researchers use historical arrest records to predict where crime is likely to occur.
  - *Problem:* Arrest records reflect not only underlying criminal activity but also **where police historically concentrated their efforts**.
  - *Consequence:* Areas that received more police attention may have more recorded arrests, which can make them appear to have higher crime levels and potentially justify continued concentration of police resources.
  - *Lesson:* Administrative data can accurately record a historical process while still reflecting the **systematic patterns of that process**.

"Just collect more data", Isn't it?

A large dataset does not automatically eliminate these problems. **Sample size primarily addresses random uncertainty; it does not, by itself, remove systematic bias.** Meng's analysis of the *Big Data Paradox*[276,277] demonstrates mathematically that even very large datasets can produce poor estimates of population quantities when the mechanism determining inclusion in the dataset is sufficiently related to the quantity being estimated.

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[276] The **Big Data Paradox**, introduced by statistician **Xiao-Li Meng (2018)**, refers to the phenomenon in which a **very large dataset can still produce biased** or **misleading estimates** when the process by which observations enter the dataset is systematically related to the outcome being studied. Large sample sizes reduce **random uncertainty**, but they do not eliminate **systematic bias** caused by nonrepresentative sampling, selection effects, or measurement problems. For example, millions of social-media users may provide data about political preferences, but if those users differ systematically from the overall population, the resulting estimate may be precise yet inaccurate. Meng demonstrated that even extremely large datasets can have substantial errors when their inclusion mechanisms are sufficiently biased.

[277] Xiao-Li Meng, *Statistical Paradises and Paradoxes in Big Data (I): Law of Large Populations, Big Data Paradox, and the 2016 US Presidential Election*. Annals of Applied Statistics, Volume12(2), pp: 685-726, 2018.

The Big Data Paradox

The **Big Data Paradox** refers to the statistical phenomenon in which an extremely large dataset can produce **misleading or highly overconfident population estimates when the data are affected by systematic selection bias or other data-quality problems**. The key point is that increasing sample size reduces **random sampling uncertainty**, but it does not necessarily eliminate **systematic bias**; indeed, when a very large but nonrepresentative dataset is analyzed as though it were a random sample, the resulting confidence intervals may become deceptively narrow while the estimate remains substantially biased.

For example, Meng examined self-reported voting data from the 2016 U.S. presidential election and showed that even a dataset covering approximately **1% of eligible voters (about 2.3 million people)** could have a mean squared error comparable to that of a genuine simple random sample of only about **400 people** when a small correlation existed between inclusion in the dataset and the outcome of interest.

Similarly, a health-data platform might contain millions of patient records but disproportionately represent younger, wealthier, or more frequently treated patients; analyzing these records without accounting for the selection mechanism could yield a highly precise estimate that is nevertheless poorly representative of the target population. Thus, the Big Data Paradox emphasizes that **data quality and representativeness are not guaranteed by data quantity**.

Meng's Mathematical Argument

Xiao-Li Meng, a prominent statistician at Harvard, proved in non-probability samples[278] (which is what most *Big Data* actually is), **increasing the sample size does not necessarily decrease the error; in fact, it can make the error worse**.

The core equation: Meng proved that the error between the sample estimate ($\bar{X}_S$) and the true population value ($\bar{X}_U$) can be decomposed into three components:

[278] **Non-probability samples** are samples in which individuals or observations are selected through a process that **does not give every member of the target population a known, nonzero probability of selection**. Because the selection probabilities are unknown or unequal in an uncontrolled way, the resulting sample may not be representative of the population, and conventional sampling-error estimates may not fully capture the resulting uncertainty. Examples include convenience samples, voluntary online surveys, social-media datasets, and records from users who choose to participate in a platform. For instance, if millions of people voluntarily respond to an online political poll, the sample may be very large but still systematically differ from the population because people with stronger political opinions may be more likely to participate.

$$\text{Error} = \bar{X}_S - \bar{X}_U = \underbrace{\rho_{R,X}}_{\text{data quality}} \times \underbrace{\sqrt{\left(\frac{N-n}{n}\right)}}_{\text{data quantity}} \times \underbrace{\sigma_X}_{\text{problem difficulty}}$$

where N is the total number of units in the target population (size of the entire population); n is the number of observations included in the dataset/sample (size of sample); $\bar{X}_S$ is the mean of X among the sampled/recorded units (i.e., sample average); $\bar{X}_U$ is the mean of X of across the entire population (i.e., true population average); X is the variable or characteristic being studied (measurement); σ_X is the population standard deviation of X (population variability), R is the **inclusion/recording indicator**: $R_j = 1$, if person j is included and $R_j = 0$ otherwise; and $\rho_{R,X}$ is the correlation between R and X (i.e., how strongly inclusion in the dataset is related to the value being measured; **selection bias**). If inclusion is unrelated to the outcome, then $\rho_{R,X}$ is close to zero. If people with certain values of X are more likely to be included, $\rho_{R,X}$ can be nonzero, creating systematic error[279].

Given the aforementioned setup, the sample mean is given by $\bar{X}_S = \frac{\sum_{j=1}^{N} R_j X_j}{\sum_{j=1}^{N} R_j}$. Because exactly n people are included, $\sum_{j=1}^{N} R_j = n$. So $\bar{X}_S = \frac{1}{n}\sum_{j=1}^{N} R_j X_j$. The population mean is $\bar{X}_U = \frac{1}{N}\sum_{j=1}^{N} X_j$.

Meng then treats a randomly selected population member as the unit of analysis. Under this formulation, $\mathbb{E}(R_J) = \frac{n}{N} = f$ is the sampling fraction. The difference between the sample and population means can then be written as:

$$\bar{X}_S - \bar{X}_U = \frac{\sum_{j=1}^{N} R_j X_j}{\sum_{j=1}^{N} R_j} - \frac{1}{N}\sum_{j=1}^{N} X_j = \frac{\mathbb{E}_J(R_J X_J)}{\mathbb{E}_J(R_J)} - \mathbb{E}_j(X_J) = \frac{\text{Cov}_J(R_J, X_J)}{\mathbb{E}_J(R_J)}$$

Correlation is defined as the normalized covariance, which implies:

$$\rho_{R,X} = \frac{\text{Cov}(R,X)}{\sigma_R \sigma_X} \Rightarrow \text{Cov}(R,X) = \rho_{R,X}\sigma_R\sigma_X$$

Substituting gives $\bar{X}_S - \bar{X}_U = \frac{\rho_{R,X}\sigma_R\sigma_X}{\mathbb{E}_J(R_J)}$. Since R is a binary indicator, we have: $P(R=1) = f$ and $P(R=0) = 1-f$. $\text{Var}(R) = f(1-f)$ and $\sigma_R = \sqrt{f(1-f)}$. This implies:

[279] Xiao-Li Meng, *Statistical Paradises and Paradoxes in Big Data (I): Law of Large Populations, Big Data Paradox, and the 2016 US Presidential Election*. Annals of Applied Statistics, Volume12(2), pp: 685-726, 2018.

$$\bar{X}_S - \bar{X}_U = \rho_{R,X} \frac{\sqrt{f(1-f)}}{f} \sigma_X = \rho_{R,X} \sqrt{\frac{1-f}{f}} \sigma_X$$

We know $\frac{1-f}{f} = \frac{1-\left(\frac{n}{N}\right)}{\frac{n}{N}} = \frac{N-n}{n}$. This yields the following:

$$\bar{X}_S - \bar{X}_U = \rho_{R,X} \sqrt{\frac{N-n}{n}} \sigma_X$$

Implication # 1

The equation makes the paradox especially clear. **Increasing** n reduces the data-quantity term $\sqrt{\frac{N-n}{n}}$ but it does **not necessarily make** $\rho_{R,X}$ **smaller**. If the mechanism determining who enters the dataset is related to the variable being measured, selection bias can remain even when is enormous.

For example, imagine a population of **10 million people** and a dataset containing **1 million** of them. If people with higher incomes are substantially more likely to appear in the dataset, then R and income X are correlated. The dataset may therefore have a very n large but its mean income can still differ systematically from the population mean.

Implication # 2

In a true random sample, the inclusion mechanism has nothing to do with the variable being measured. Therefore, $\rho_{R,X} = 0$. Because the first term is zero, the **systematic bias is zero**. The only error left is **random sampling error**, which **shrinks predictably** as you increase your sample size n.

In Big Data, (e.g., Twitter users, voluntary health app users, web-scraped data), the inclusion is *not* random. People choose to be in the dataset based on their traits. Therefore, $\rho_{R,X} \neq 0$.

Implication # 3

σ_X is the **population standard deviation** of the variable. It represents the amount of **natural variation in the characteristic** being estimated across the entire population.

$$\sigma_X = \sqrt{\frac{1}{N} \sum_{j=1}^{N} \left(X_j - \bar{X}_U\right)^2}$$

Think of σ_X as the ***difficulty* of estimating the population mean**. If everyone in the population has nearly the same value of X, then σ_X is small and even a somewhat imperfect sample may estimate the population mean reasonably well. If X varies greatly across the population, then σ_X is large, and the same degree of sampling bias can produce a much larger error.

σ_X is **not the source of the bias**. The key source of systematic error is $\rho_{R,X}$, the relationship between **who** gets **included** and **what** their **value of** X is. σ_X determines how strongly that relationship translates into an error in the estimated mean.

"Devastating" mathematical realities

- **The Law of Large Populations** (**Bigger Population = Worse Estimate**)**:** If we keep our sample size n fixed, but look at a larger population N, the **error grows** with the $\sqrt{N}$. A $100{,}000$ biased sample of a 10 million population will yield a much worse estimate than a $100{,}000$ biased sample of a $100{,}000$ population.
- **The Illusion of Big Data (Bigger Sample = Slowly Shrinking Error)**: If we increase our sample size n from $100{,}000$ to $1{,}000{,}000$ (a $10\times$ increase in data), the error only shrinks by a factor of $\sqrt{10} \approx 3.16$. We collected 10 times more data, but our accuracy only improved by $3\times$. Meanwhile, the **computational cost** and **complexity skyrocketed**.

3.2.6 Subgroup behavior

Aggregate statistics can **conceal substantial differences** between groups. A dataset may appear well behaved overall while containing important **subgroup-specific patterns**, missingness, anomalies, or performance differences. For groups $g = 1, \cdots, G$, calculate:

$$\bar{x}_g;\ \ s_g;\ \ P(Y = 1|G = g);\ \ P(X \text{ missing}|G = g)$$

Relevant subgroups may be defined by:

- age;
- sex or gender;
- geography;
- socioeconomic status;
- language;
- device type;
- institution;

- time period;
- disease severity; or
- customer segment.

Subgroup analysis can **reveal differences in distributions**, **outcomes**, **data quality**, **measurement practices**, and **model behavior** that disappear when all observations are pooled together.

In some applications, it is also important to examine **intersections of characteristics** rather than considering each characteristic independently. For example, a subgroup might be defined as:

$$G = (\text{gender}, \text{age}, \text{group}, \text{region})$$

Such intersectional analysis can **reveal patterns** that are **not visible when each variable is examined separately**. However, increasingly fine-grained subgroup analysis also creates **methodological challenges**. **Small groups** may produce **unstable estimates**, and examining a very large number of arbitrary subgroups increases the **probability of finding apparently interesting patterns** purely **by chance**.

Subgroups should therefore be selected using **domain knowledge**, **substantive relevance**, and **adequate sample sizes**. Where many subgroup comparisons are performed, the analyst should also consider appropriate statistical adjustments and clearly distinguish **exploratory subgroup findings** from **prespecified confirmatory analyses**.

The objective is not to search every possible subgroup until an interesting result appears. Rather, subgroup analysis should be **used deliberately to determine** whether important relationships, data-quality problems, biases, or model behaviors differ across populations that are meaningful for the research or operational context.

Example # 78: Deliberate subgroup analysis

- **Clinical treatment effectiveness**
  - *Research question:* Does a new treatment reduce blood pressure?

- *Meaningful subgroups:* Age groups, such as younger, middle-aged, and older adults, may be selected because age could plausibly influence treatment response.
- *Problem to avoid:* The researcher should not test dozens of arbitrary subgroups; such as people born in different months, different shoe sizes, or different favorite foods, simply to find one statistically significant result.
- *Interpretation:* If the age-specific analysis was planned in advance and each group has sufficient observations, differences in treatment effects can be investigated more credibly.
- *Lesson:* Subgroups should be chosen because they are **scientifically or operationally meaningful**, not because they produce interesting results.

- **Student performance**
  - *Research question:* Does study time predict examination performance?
  - *Meaningful subgroups:* The researcher may examine undergraduate versus postgraduate students because their study patterns and academic environments differ substantially.
  - *Problem to avoid:* After seeing the results, the researcher could divide students into many groups; by age, residence, preferred study location, device type, and dozens of other characteristics, and report only the groups showing strong relationships.
  - *Interpretation:* If the undergraduate/postgraduate distinction was scientifically justified, subgroup analysis can reveal whether the relationship differs between these populations.
  - Lesson: A small number of **predefined, relevant subgroups** is preferable to extensive post-hoc searching.

- **Predictive model for hospital readmission**
  - *Research question:* How well does a model predict 30-day hospital readmission?
  - *Meaningful subgroups:* Model performance could be evaluated separately for age groups, major clinical categories, or hospital departments if these differences are operationally important.

- *Problem to avoid:* Evaluating performance across hundreds of small subgroups may produce apparently large differences simply through random variation.
- *Interpretation:* A consistently lower model performance in an important patient group may indicate a genuine **model reliability or data-quality problem** that requires investigation.
- *Lesson:* Subgroup analysis can be used to identify **unequal model performance**, not merely statistically significant relationships.

- **Customer churn model**
  - *Research question:* Can a company predict which customers are likely to cancel their subscriptions?
  - *Meaningful subgroups:* The company might evaluate performance separately for new customers, long-term customers, and customers using different service plans.
  - *Interpretation:* If the model performs well for long-term customers but poorly for new customers, this may indicate that the available predictors contain insufficient information about newer customers.
  - *Lesson:* Subgroup analysis can reveal **where a model works well and where it needs improvement**.

Take-home message:

- Select subgroups based on **domain knowledge and substantive relevance**.
- Ensure that each subgroup has an **adequate sample size**.
- When many comparisons are conducted, consider appropriate **multiple-comparison adjustments**.
- Distinguish **prespecified confirmatory subgroup analyses** from exploratory findings discovered after examining the data.
- Do not repeatedly divide the data until an interesting or significant result appears.

Data Visualization

Data visualization is one of the **central tools of exploratory data analysis** because many important properties of a dataset are **difficult to recognize from numerical summaries** alone. Means, medians, variances, correlations, and other statistics necessarily **compress information** into a small number of values. Although this compression is useful, it **can conceal important features** such as **nonlinearity**, **clusters**, **gaps**, **outliers**, changing **variability**, **multimodality**, and **temporal patterns**. Effective visualization restores some of the structure that numerical summaries remove, allowing analysts to see how observations are distributed and how variables behave together. Visualization should therefore be treated not merely as a way of presenting results, but as an **analytical instrument for asking questions**, **detecting problems**, and **developing hypotheses**.

3.3.1 Why visualizations are necessary

Numerical summaries provide **compact descriptions** of data, but a compact description is **not necessarily a complete** description. Two datasets can have similar or even identical summary statistics while exhibiting very different structures.

This principle is famously illustrated by **Anscombe's quartet**, introduced by statistician **Francis J. Anscombe** in **1973**[280]. The four datasets have nearly identical means, variances, correlations, and fitted regression lines, yet their **graphical patterns are dramatically different**. The example demonstrates why statistical summaries should be complemented by appropriate graphical examination. Visualization can reveal features such as:

- nonlinearity;
- clusters;
- gaps;
- outliers;
- heteroscedasticity;
- multimodality;
- temporal change;

[280] F. J. Anscombe, *Graphs in statistical analysis*, The American Statistician, Volume 27(1), pp 17-21, 1973.

- data-entry or measurement errors; and
- subgroup-specific patterns.

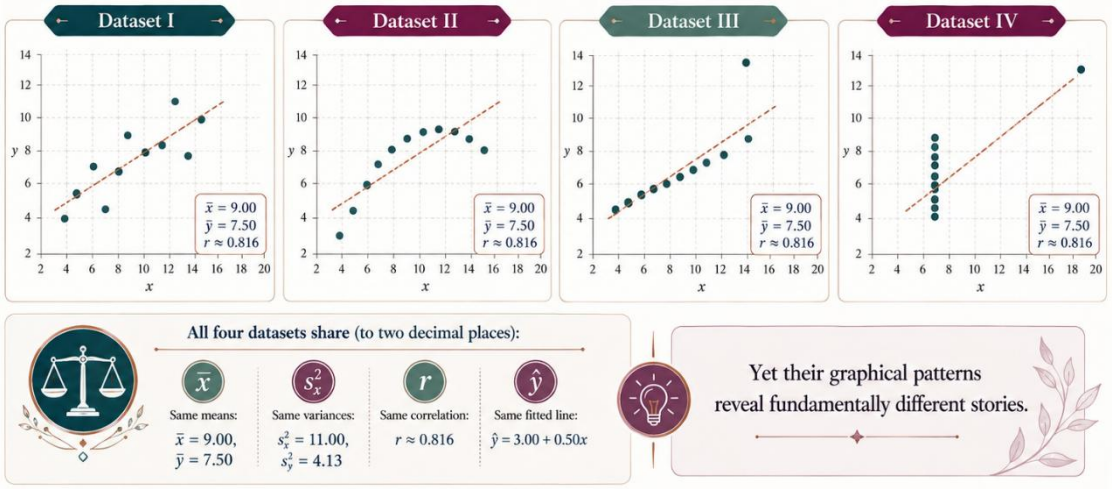

Figure 34: Anscombe's quartet: identical summaries, contrasting data structures. The four datasets exhibit nearly identical means, variances, correlations, and fitted linear regression lines, yet reveal markedly different graphical patterns when visualized. The figure illustrates the central lesson of Anscombe's quartet: numerical summaries alone can conceal important structure, nonlinearity, outliers, and leverage effects, and should therefore be complemented by appropriate graphical examination.

A visualization should therefore begin with a question, rather than with a desire to produce a particular type of chart. The choice of visualization should depend on:

- the types of variables involved;
- the analytical objective;
- the sample size;
- the number of groups;
- whether observations have a meaningful temporal or other ordering; and
- the characteristics and needs of the intended audience.

The objective is not to produce the largest number of charts. It is to select **visual representations** that make **relevant structure** easier to **see** and **interpret**.

3.3.2 Histograms

A **histogram** divides the range of a numerical variable into intervals, or bins, and displays the number or proportion of observations falling within each interval. For bins $\{B_k\}$, the count in bin B_k can be written as:

$$h_k = \#\{x_i : x_i \in B_k\}$$

Histograms are particularly useful for examining the overall shape of a numerical distribution. They can reveal:

- symmetry or skewness;
- multimodality;
- gaps;
- extreme values;
- the overall range; and
- regions where observations are concentrated.

Bin choice matters

The appearance of a histogram depends partly on the **choice of bin width** and **bin boundaries**. Using **too few bins** can **conceal meaningful structure**, whereas using **too many** can make **random fluctuations appear more important** than they are.

Consequently, the analyst should not treat one arbitrary binning scheme as definitive. Examining the distribution using **more than one reasonable bin width** can help determine whether an apparent feature is robust or merely an artifact of the chosen bins.

Appropriate use

Histograms are particularly useful for:

- continuous numerical variables;
- count variables with sufficiently many distinct values; and
- examining distributional shape in moderate or large samples.

When comparing groups, it is generally preferable to use:

- common bin boundaries;
- common axis scales;
- normalized proportions when group sizes differ; and

- separate or faceted displays when overlapping histograms become difficult to interpret.

Example # 79: Infographic of histogram shapes

A histogram summarizes a numerical distribution by grouping values into bins, helping reveal shape, skewness, multimodality, gaps, extremes, and concentration (**Figure 35**). Its interpretation depends on sensible binning: too few bins may hide structure, while too many may exaggerate noise. For group comparisons, use common bins and scales, normalized proportions when needed, and separate or faceted displays when overlap becomes unclear.

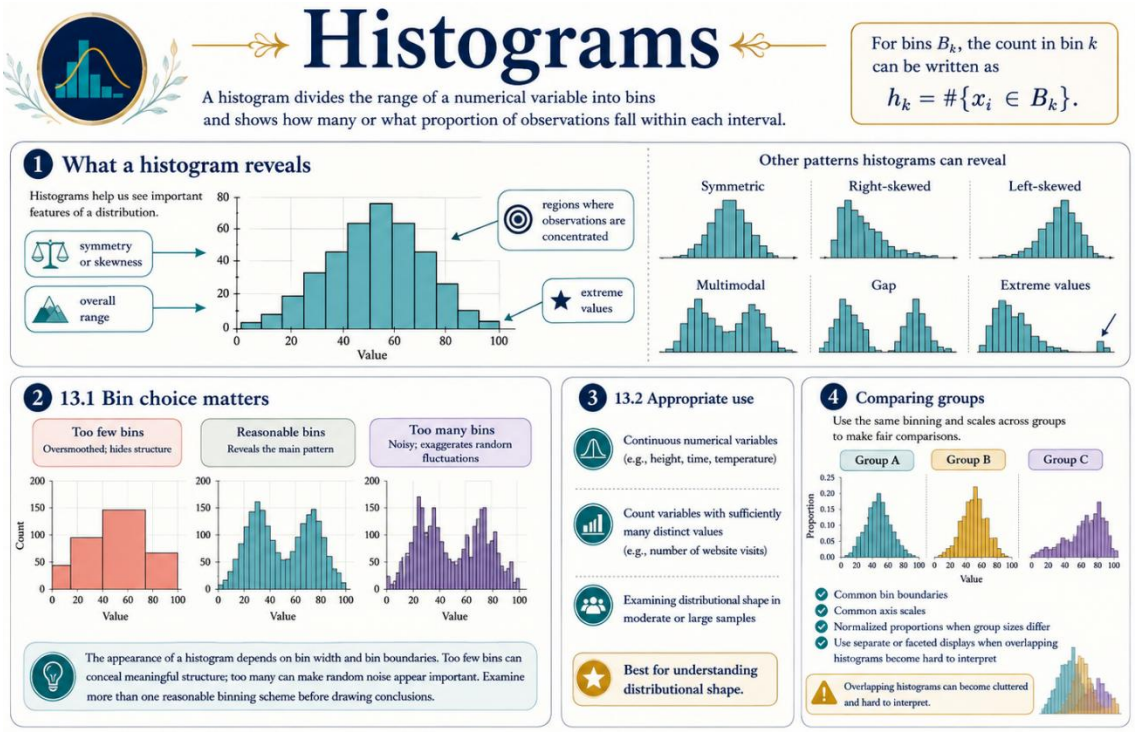

Figure 35: Histograms for exploring the shape of numerical distributions.

3.3.3 Boxplots

A **boxplot** provides a compact summary of a numerical distribution using its quartiles and selected extreme observations. The main components typically include:

- the median;
- the first quartile, (Q_1);
- the third quartile, (Q_3);
- the interquartile range (IQR);

- whiskers[281]; and
- individually displayed observations beyond the whiskers.

The IQR is defined as: $\text{IQR} = Q_3 - Q_1$. A commonly used convention defines the lower and upper fences as:

$$[Q_1 - 1.5 \times \text{IQR}, Q_1 + 1.5 \times \text{IQR}]$$

Observations beyond these conventional limits are often displayed individually.

Important warning

Points beyond the whiskers are **not automatically errors**. They are **observations** that lie **beyond a conventional distributional boundary**. Whether such observations should be retained, investigated, transformed, or removed depends on their context and provenance.

Strengths

Boxplots are particularly effective for comparing groups with respect to:

- central location;
- spread;
- asymmetry;
- extreme observations; and
- differences across multiple categories.

Limitations

A boxplot compresses the distribution considerably and may conceal:

- multimodality;
- gaps;
- small-sample behavior;
- clusters; and
- detailed distributional shape.

[281] **Whiskers** are the lines extending from the boxplot's box to the most extreme observations that are **not considered outliers** under the chosen rule. In the most common boxplot definition: The **lower whisker** extends down to the smallest observation that is at least $Q_1 - 1.5 \times \text{IQR}$. The **upper whisker** extends up to the largest observation that is at most $Q_3 + 1.5 \times \text{IQR}$. Observations beyond these limits are usually plotted **individually as potential outliers**.

For this reason, boxplots are **often strengthened by** displaying individual observations or by using **complementary density** or **violin plots**[282].

Example # 80: Infographic of boxplots

A boxplot summarizes a distribution using the median, first and third quartiles, the interquartile range, whiskers, and individually displayed observations beyond conventional fences (**Figure 36**). It is especially useful for comparing groups with respect to central location, spread, asymmetry, and extreme observations, but it can conceal multimodality, gaps, clusters, small-sample behavior, and detailed distributional shape. Accordingly, boxplots are often **most informative** when **interpreted in context** and **supplemented with displays of individual observations** or complementary plots such as density or violin plots.

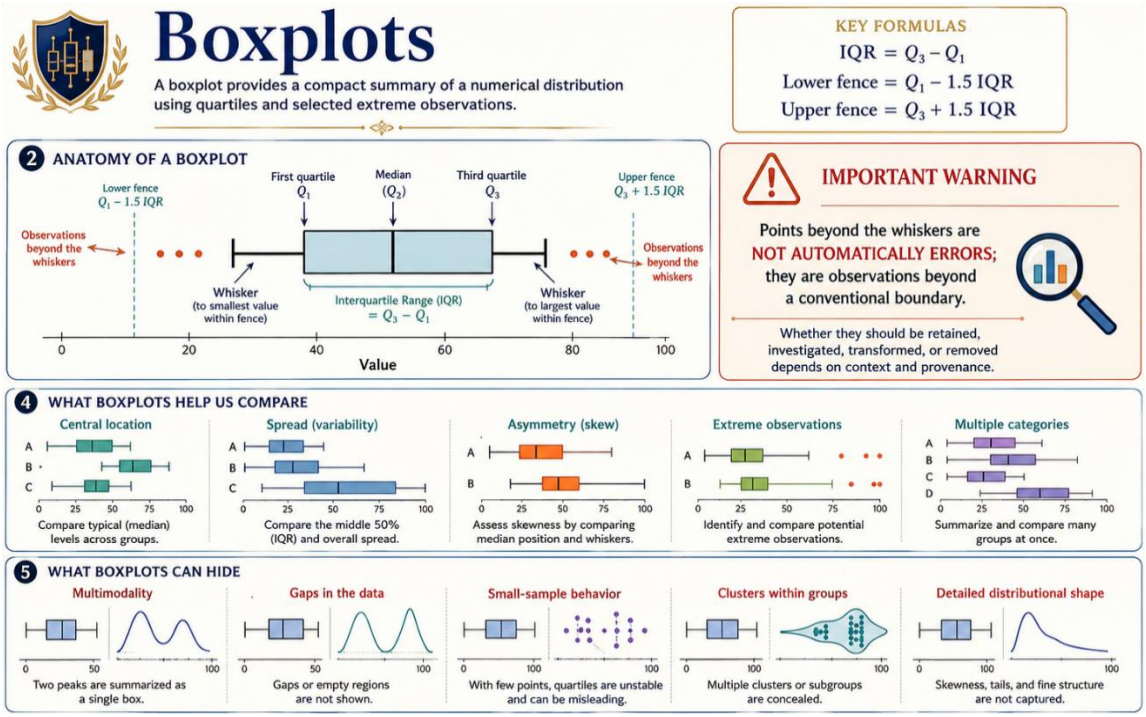

Figure 36: Boxplots as compact summaries of numerical distributions.

[282] A **violin plot** is a statistical graphics method for displaying and comparing the **distribution of a numerical variable across one or more groups**. It combines the summary-statistical information of a **box plot** with a **kernel density estimate**, so that the shape and width of the violin indicate where observations are more or less concentrated, while an embedded box plot may show the median and interquartile range. Violin plots are particularly useful for identifying **skewness**, **multimodality**, **spread**, and **differences in distributional shape** between groups. The method was introduced by **Jerry L. Hintze** and **Ray D. Nelson** in their article, *Violin Plots: A Box Plot-Density Trace Synergism*, published in *The American Statistician*, Volume 52(2), pp 181-184, 1988.

3.3.4 Scatterplots

A **scatterplot** displays paired numerical observations: $(x_i, y_i);\ i = 1, \cdots, n$. Scatterplots are among the most useful tools for **examining relationships** between numerical variables. They can reveal:

- the direction of association;
- the apparent strength of association;
- linearity or nonlinearity;
- clusters;
- outliers;
- potentially influential observations;
- changing variance; and
- subgroup structure.

A scatterplot should generally be examined **before relying solely** on a correlation coefficient or fitted regression line because the **graphical representation can reveal patterns** that a single numerical measure cannot.

Heteroscedasticity

If the conditional variance changes with X, $\mathrm{Var}(Y|X = x)$ is not constant. A funnel-shaped or otherwise changing vertical spread in a scatterplot may provide evidence of **heteroscedasticity**. This can be important because some statistical procedures assume approximately constant variance.

Overplotting

When a dataset contains many observations, points may overlap so heavily that the underlying pattern becomes difficult to see. Possible solutions include:

- transparency;
- jittering;
- hexagonal binning;
- density contours;
- random subsampling; and
- aggregation.

The appropriate method depends on whether individual observations or overall density is the primary analytical interest.

Color and shape

Additional variables can be represented through:

- color;
- marker shape;
- point size; and
- faceting.

These **encodings** can reveal **subgroup structure**, but they should be **used selectively.** Adding **too many visual dimensions** can make a plot harder to interpret rather than more informative.

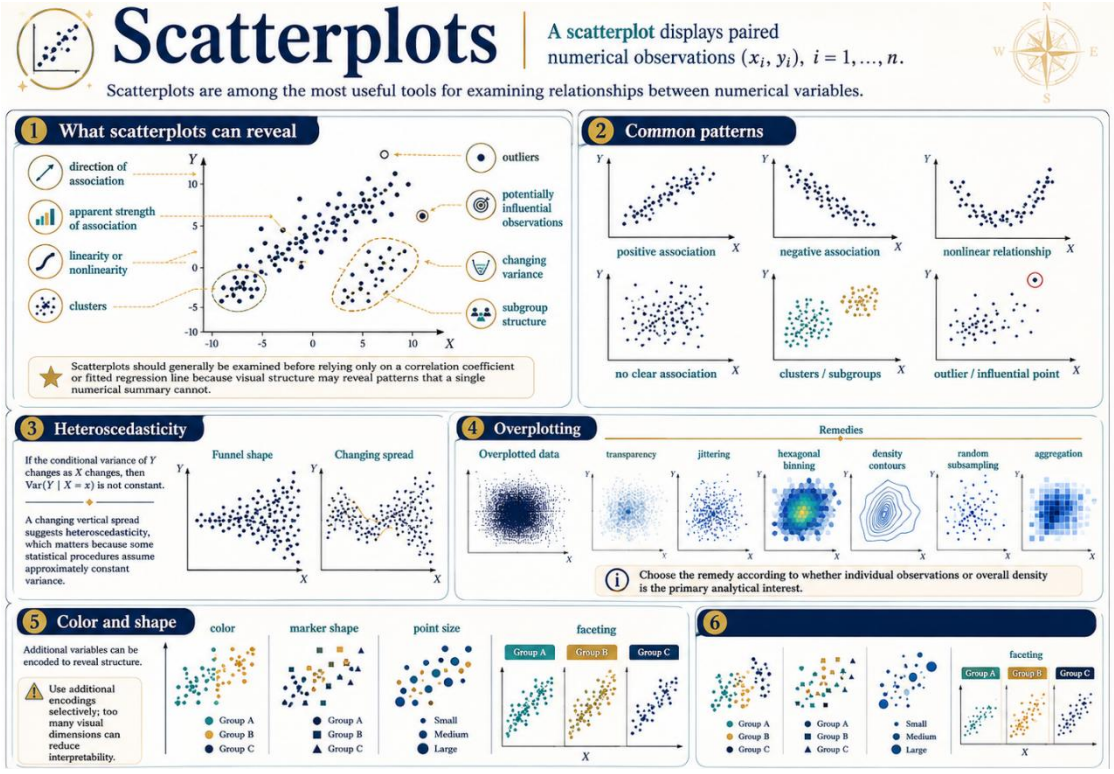

Figure 37: Scatterplots for examining relationships between numerical variables.

Example # 81: Infographic of scatterplots

Scatterplots **display paired observations** and can reveal the direction, strength, and form of association, as well as clusters, outliers, influential observations, changing variance, and subgroup structure (**Figure 37**). Funnel-shaped patterns may indicate heteroscedasticity, while dense datasets may

require transparency, jittering, hexagonal binning, density contours, subsampling, or aggregation to reduce overplotting. Additional variables can be encoded through color, shape, size, or faceting, but these visual dimensions should be used selectively to preserve interpretability.

3.3.5 Correlation heatmaps

A **correlation matrix** summarizes pairwise relationships between numerical variables. Its elements can be written as:

$$R_{jk} = \mathrm{Corr}(X_j, X_k)$$

A correlation heatmap represents these values visually, usually through color intensity or hue.

Uses of correlation heatmaps

Correlation heatmaps can help identify:

- highly related features;
- groups of variables with similar behavior;
- possible multicollinearity;
- redundant measurements; and
- unexpected pairwise associations.

They are particularly useful as an initial screening tool when a dataset contains a moderate number of numerical variables.

Limitations

Correlation heatmaps can be misleading or incomplete when:

- relationships are nonlinear;
- outliers strongly influence correlations;
- variables are categorical or ordinal but are treated as numerical without justification;
- pairwise correlations are calculated from different effective sample sizes because of missing data;
- temporal dependence is ignored;
- a very large number of correlations are examined without considering uncertainty; or
- the color scale exaggerates small differences.

A correlation coefficient also does not establish causation. A heatmap should therefore be regarded as a **screening and exploration tool**, not as evidence that one variable causes another.

Example # 82: Infographic of correlation heatmaps

A correlation heatmap visualizes the entries of a correlation matrix, helping identify highly related features, groups of variables with similar behavior, possible multicollinearity, redundant measurements, and unexpected pairwise associations (Figure 38). Although useful as an initial exploratory tool, correlation heatmaps can be misleading when relationships are nonlinear, outliers strongly affect correlations, missing data alter effective sample sizes, temporal dependence is ignored, or color scaling exaggerates small differences. Accordingly, they should be interpreted as screening devices and complemented with contextual knowledge and more detailed graphical or statistical analysis; in particular, correlation does not imply causation.

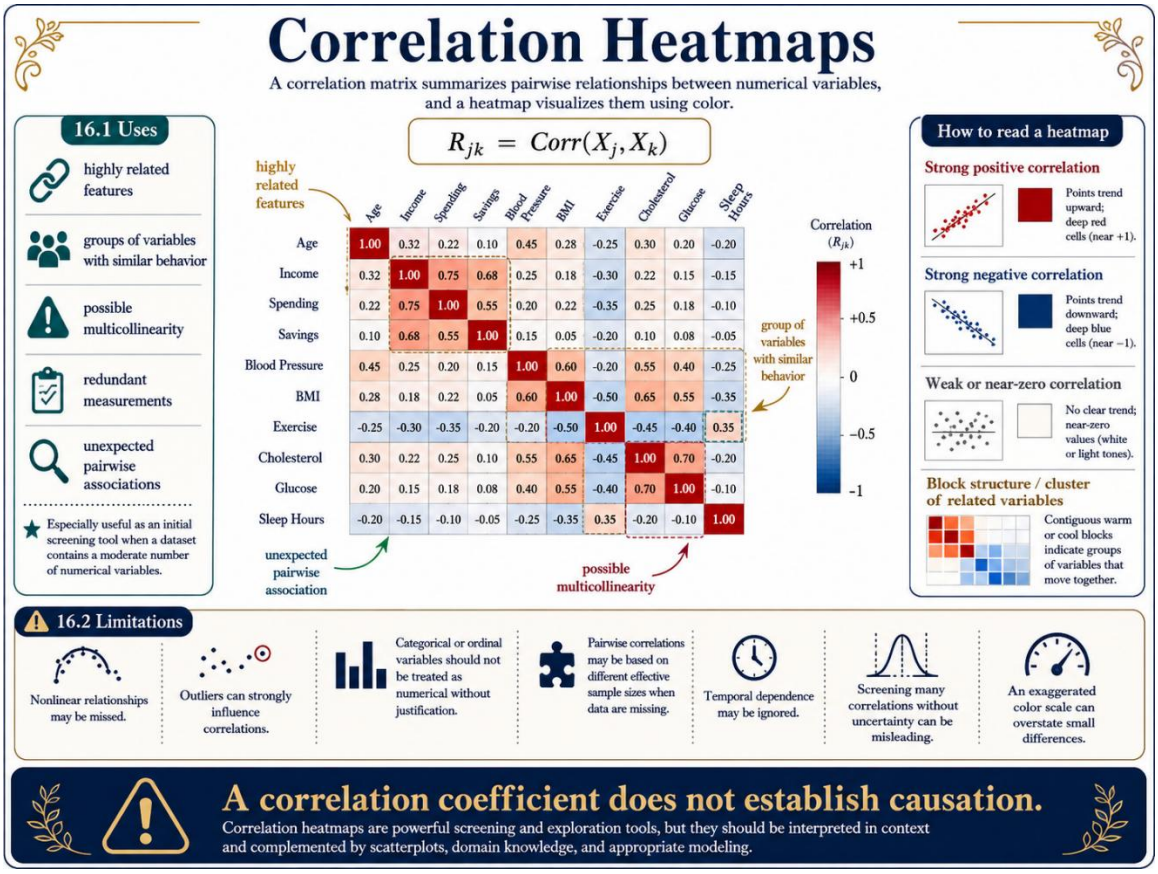

Figure 38: Correlation heatmaps for screening pairwise relationships among numerical variables.

Example # 83: Limitations of correlation heatmaps (**Figure 39**)

- **Nonlinear relationship**
  - *Situation:* Suppose age and blood pressure have a U-shaped relationship: both very young and very old patients tend to have higher blood pressure, while middle-aged patients have lower values.
  - *Problem:* Pearson correlation measures a linear relationship. Because the relationship curves upward at both ends, the correlation may be close to zero.
  - *Consequence:* A heatmap may suggest that age and blood pressure are weakly related even though a strong **nonlinear relationship** exists.
  - *Lesson:* A weak correlation does not necessarily mean that two variables are unrelated.
  - *Solution:* Always visualize data with scatterplots. For monotonic (consistently increasing or decreasing) non-linear relationships, use Spearman's rank correlation[283].

- **Influence of outliers**
  - *Situation:* Suppose a dataset contains 100 employees with salaries between $30,000 and $100,000 and one executive earning $5 million.
  - *Problem:* The extreme salary can substantially change the correlation between salary and another variable.
  - *Consequence:* The heatmap may show a strong correlation that is largely driven by a single observation.
  - *Lesson:* Important correlations should be examined using **scatterplots and outlier diagnostics**, rather than relying only on the heatmap.
  - *Solution:* Use robust correlation measures (like Spearman's rank or biweight midcorrelation[284]) and always inspect scatterplots for outliers.

---

[283] **Spearman's rank correlation** (denoted by ρ_s) is a **nonparametric measure of the strength and direction of a monotonic association between two variables**. It is calculated by converting the observations of each variable into ranks and then measuring the correlation between those ranks. Unlike Pearson's correlation, Spearman's correlation does not require the variables to have a linear relationship and is less sensitive to extreme values. Its value ranges from −1 to +1, where indicates a perfectly increasing monotonic relationship, indicates a perfectly decreasing monotonic relationship, and indicates no monotonic association. The formula is: $\rho_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}$ where n is the number of paired observations and d_i is the difference between the ranks of the two variables for observation i. This simplified formula assumes **no tied ranks**. When ties occur, Spearman's correlation is generally calculated as the **Pearson correlation between the ranked variables**.

[284] **Biweight midcorrelation (bicor)** is a **robust measure of association between two continuous variables** that estimates the strength and direction of their relationship while reducing the influence of extreme observations (outliers). Unlike **Pearson's correlation**, which gives every observation substantial influence, biweight midcorrelation assigns progressively smaller weights to observations that are far from the **median** of their respective variables and can effectively exclude observations beyond a specified cutoff. It is particularly useful for data that contain outliers or are not well described by a normal distribution. The method is based on the **biweight midvariance**, a robust statistical estimator developed by **Beaton and Tukey (1974)** and subsequently adapted for measuring correlation.

- **Categorical variables treated as numerical**
  - *Situation:* A researcher codes education level as $1 =$ high school, $2 =$ bachelor's, $3 =$ master's, and $4 =$ doctorate and calculates ordinary Pearson correlations.
  - *Problem:* The numerical codes imply equal distances between categories, which may not be justified.
  - *Consequence:* The resulting correlation may have an interpretation that the categorical variable does not actually support.
  - *Lesson:* The statistical method should match the **measurement scale and structure of the variable**.
  - *Solution:* Use Spearman's rank correlation for ordinal data[285] or use techniques like polychoric correlation[286].

- **Different sample sizes due to missing data**
  - *Situation:* A dataset contains income, age, education, and health measurements, but income is missing for many participants.
  - *Problem:* The correlation between age and education may be based on 5,000 observations, while the correlation between income and health may be based on only 2,000.
  - *Consequence:* The heatmap displays the correlations together even though they are estimated with very different levels of precision.

The **biweight midcorrelation** is commonly defined using robust, median-centered weights. For paired observations (x_i, y_i), first calculate the medians M_x and M_y, and define the median absolute deviations: $\text{MAD}x = \text{median}(|x_i - M_x|)$ and $\text{MAD}y = \text{median}(|y_i - M_y|)$. Then define the standardized deviations: $u_i = \frac{x_i - M_x}{c\text{MAD}_x}$ and $v_i = \frac{y_i - M_y}{c\text{MAD}_y}$ where is a tuning constant, commonly 9. The biweight weights are: $w_i^{(x)} = \begin{cases} (1-u_i^2)^2, & |u_i| < 1 \\ 0, & |u_i| \geq 1 \end{cases}$ and $w_i^{(y)} = \begin{cases} (1-v_i^2)^2, & |v_i| < 1 \\ 0, & |v_i| \geq 1 \end{cases}$. The biweight midcorrelation is then: $r_{\text{bicor}} = \frac{\sum_{i=1}^{n}(x_i - M_x)(y_i - M_y)\left(w_i^{(x)} w_i^{(y)}\right)}{\sqrt{\sum_{i=1}^{n}(x_i - M_x)^2\left(w_i^{(x)}\right)^2}\sqrt{\sum_{i=1}^{n}(y_i - M_y)^2\left(w_i^{(y)}\right)^2}}$. The **important idea** is that observations close to the median receive high weights, while observations far from the median receive progressively smaller weights; observations beyond the cutoff receive weight zero.

[285] **Ordinal data** are a type of **categorical data in which observations can be classified into distinct categories that possess a meaningful, inherent order or ranking, while the numerical distances between adjacent categories are not assumed to be equal or known**. Ordinal data therefore provide information about the **relative position** of observations but do not necessarily permit meaningful arithmetic operations such as calculating differences between categories. Examples include patient pain severity (none→mild→moderate→severe), customer satisfaction (very dissatisfied→dissatisfied→neutral→satisfied→very satisfied), education level (high school→bachelor's→master's→doctorate), and disease stage (Stage I→Stage II→Stage III→Stage IV). For example, if satisfaction is coded from 1 to 5, the value 5 indicates greater satisfaction than 4, but it cannot be assumed that the increase from 1 to 2 represents the same amount of change as the increase from 4 to 5.

[286] **Polychoric correlation** is a statistical measure of association between **two ordinal variables** that estimates the correlation between the underlying **continuous latent variables** assumed to generate the observed ordinal categories. Rather than treating ordinal categories as equally spaced numerical values, polychoric correlation assumes that each observed ordinal variable results from dividing an unobserved continuous variable into ordered intervals using one or more **thresholds (cut-points)**. The method then estimates the latent correlation that would most plausibly produce the observed cross-tabulation of ordinal responses. The resulting coefficient ranges from −1 to +1, with positive values indicating that higher categories of one variable tend to be associated with higher categories of the other.

- *Lesson:* Correlations should be interpreted alongside their **effective sample sizes and uncertainty**.
- *Solution:* Use *listwise deletion* (dropping any row with a missing value) or multiple imputation to ensure a consistent sample size (n) across the entire matrix.

- **Temporal dependence**
  - *Situation:* A company examines daily advertising expenditure and daily sales over two years.
  - *Problem:* Observations from consecutive days are not necessarily independent. Both variables may increase during seasonal periods or exhibit common time trends.
  - *Consequence:* A strong correlation may partly reflect shared **time trends or seasonality** rather than a direct relationship between advertising and sales.
  - *Lesson:* For time-dependent data, correlations should be examined with appropriate **temporal analysis**.
  - *Solution:* Detrend[287] the data (look at changes from month to month rather than raw totals) or use time-series specific methods like cross-correlation.

- **Large number of correlations**
  - *Situation:* A dataset contains 100 variables. A correlation heatmap displays approximately 4,950 pairwise correlations.
  - *Problem:* With so many comparisons, some strong-looking correlations can occur simply by chance.
  - *Consequence:* The analyst may identify an apparently important relationship that does not replicate in new data.
  - *Lesson:* When many relationships are examined, the analyst should consider **statistical uncertainty**, **multiple comparisons**, and **validation**.

[287] **Detrending data** is the statistical process of **removing a systematic long-term trend from a time series** so that the remaining values primarily represent short-term fluctuations, deviations, or other temporal patterns. A trend may be removed by subtracting a fitted trend line or model from the observed values, by differencing consecutive observations, or through other time-series decomposition methods. For example, if a company's monthly sales increase steadily over several years, correlating the **raw sales totals** of two products may produce a strong correlation simply because both products are growing over time. Detrending can address this by examining month-to-month changes, such as $\Delta X_t = X_t - X_{t-1}$, rather than the raw sales levels. Similarly, if hospital admissions and population size both increase over time, detrending can help determine whether admissions are actually changing in relation to population size rather than merely reflecting their shared upward trend.

- *Solution:* Apply corrections for multiple comparisons (like the Bonferroni correction[288] or False Discovery Rate[289]) and never trust a single *strong* correlation in a massive matrix without a reason for it.

- **Misleading color scale**
  - *Situation:* A heatmap uses a narrow color range where correlations of 0.05, 0.10, and 0.15 appear dramatically different.
  - *Problem:* The visual presentation makes small numerical differences appear substantial.
  - *Consequence:* Users may overinterpret visually striking colors as evidence of strong relationships.
  - *Lesson:* The **color scale and visualization design** can influence interpretation.
  - *Solution:* Always use a symmetric color scale from -1 to 1 for correlation matrices so the visual intensity matches the mathematical reality.

- **Correlation does not imply causation**
  - *Situation:* A heatmap shows a strong positive correlation between ice-cream sales and drowning incidents.
  - *Problem:* It would be incorrect to conclude that buying ice cream causes drowning.
  - *Explanation:* Both variables increase during **hot weather and summer** months, which acts as a third factor.

---

[288] The **Bonferroni correction** is a multiple-comparison adjustment used to control the **family-wise error rate (FWER)**, or the probability of making **at least one Type I error** when conducting multiple statistical hypothesis tests. The method is attributed to **Carlo Emilio Bonferroni (1936)** and is based on the Bonferroni inequality. If hypotheses are tested with an overall significance level of α, the correction sets the significance level for each individual test to $\frac{\alpha}{m}$, or equivalently adjusts each p-value by multiplying it by m: $\alpha_{\text{adjusted}} = \frac{\alpha}{m}$ and $p_{\text{adjusted}} = \min(mp, 1)$. For example, if a researcher conducts 10 statistical tests and wants an overall significance level of 0.05, each test must meet $p < 0.005$ ather than $p < 0.05$. Similarly, if 20 comparisons are made in a clinical study, the Bonferroni-adjusted threshold is $\frac{0.05}{20} = 0.0025$. The correction reduces the likelihood of false-positive findings across multiple tests, although it can be conservative and consequently reduce statistical power. C. E. Bonferroni, *Teoria statistica delle classi e calcolo delle probabilità*, 1936.

[289] The **false discovery rate (FDR)** is a multiple-testing error criterion that controls the **expected proportion of false-positive findings among all hypotheses declared statistically significant**. Unlike the Bonferroni correction, which controls the probability of making **at least one Type I error**, FDR allows some false discoveries but limits their expected proportion among the rejected hypotheses. The concept was formally introduced by **Yoav Benjamini** and **Yosef Hochberg (1995)** through the **Benjamini–Hochberg procedure**. For example, if a researcher tests 1,000 genes for differential expression and identifies 100 as statistically significant while controlling the FDR at 5%, the procedure aims to ensure that, on average, approximately **5% of the discoveries declared significant – about 5 of the 100 – are false discoveries**. Another example is a study examining hundreds of possible associations between dietary factors and diseases; rather than requiring every individual test to meet an extremely strict significance threshold, an FDR procedure ranks the p-values and determines which findings can be considered significant while controlling the expected proportion of false discoveries. Y. Benjamini, and Y. Hochberg, *Controlling the false discovery rate: A practical and powerful approach to multiple testing*, Journal of the Royal Statistical Society: Series B, Volume 57(1), pp 289-300, 1995.

- *Lesson:* A correlation heatmap can identify an association, but it cannot by itself establish a **causal relationship**.
- *Solution:* Use controlled experiments (A/B testing[290]) or causal inference methods (like DAGs[291] or instrumental variables[292]).

3.3.6 Pair plots

A **pair plot** displays multiple pairwise relationships within a collection of variables. A typical pair plot contains:

- distributions or other univariate summaries along the diagonal;
- scatterplots in the off-diagonal panels; and
- optional colors or symbols representing groups or classes.

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[290] **A/B testing** is a controlled experimental design in which a population of users, customers, or other experimental units is randomly divided into at least two groups: **Group A (the control)** receives the existing version of a product, intervention, or feature, while **Group B (the treatment)** receives a modified version. The outcome of interest is then compared between groups to estimate the causal effect of the modification, typically using a statistical hypothesis test or confidence interval. Although A/B testing does not have a single inventor, its statistical foundations are closely related to **randomized experiments** developed by **Ronald A. Fisher (1935)**, particularly his work on experimental design and randomization. For example, an online retailer might randomly show half of its visitors the existing checkout page (A) and half a redesigned checkout page (B), then compare the purchase-conversion rates. Similarly, a streaming service could randomly expose users to two different recommendation algorithms and compare measures such as viewing time or the proportion of recommended content watched. In each case, random assignment helps ensure that observed differences in the outcome can be attributed to the experimental change rather than systematic differences between the groups. R. A. Fisher, *The Design of Experiments*, Edinburgh: Oliver and Boyd, 1935.

[291] A **directed acyclic graph (DAG)** is a graphical model used to represent hypothesized **causal relationships** among variables, where variables are represented as nodes and directed arrows indicate the assumed direction of causal influence; *acyclic* means that following the direction of the arrows can never lead back to the starting variable. DAGs are particularly important in causal inference because they help researchers identify **confounders**, **mediators**, **colliders**, and **appropriate variables** for **statistical adjustment**. The modern formal use of DAGs for causal reasoning is strongly associated with **Judea Pearl (1995)**. For example, if age (A) affects both exercise (X) and blood pressure (Y), a DAG can be represented as $A \to X$, $A \to Y$, and $X \to Y$; this indicates that age is a potential confounder when estimating the causal effect of exercise on blood pressure. Another example is $\text{smoking} \to \text{lung cancer}$, with $\text{age} \to \text{smoking}$ and $\text{age} \to \text{lung cancer}$, where the DAG makes explicit why age may need to be considered when estimating the causal effect of smoking on lung cancer. J. Pearl, *Causal diagrams for empirical research*, Biometrika, Volume 82(4), pp 669-688, 1995.

[292] An **instrumental variable (IV)** is a variable used in **causal inference** to estimate the causal effect of a potentially **endogenous treatment or exposure** when direct adjustment for confounding is inadequate or impossible. An instrument Z must satisfy three key conditions: **relevance**, meaning Z is associated with the treatment X; **exclusion**, meaning Zaffects the outcome Y only through X; and **independence (or exogeneity)**, meaning Z is not associated with unmeasured factors that directly affect Y. In a simple model, the IV estimator can be expressed as $\hat{\beta}_{\text{IV}} = \frac{\text{Cov}(Z,Y)}{\text{Cov}(Z,X)}$. For example, distance to a college can sometimes be used as an instrument for college attendance when estimating the causal effect of education on earnings, because proximity may affect the likelihood of attending college while, under the required assumptions, not directly affecting earnings except through education. Another classic example is using physician prescribing preference as an instrument for receiving a particular medical treatment: a physician's tendency to prescribe a treatment may influence whether a patient receives it, while the physician's preference is assumed not to directly determine the patient's outcome after accounting for relevant factors. IV methods are especially useful in observational studies where treatment assignment is not randomized, but their validity depends critically on the instrument assumptions, particularly the difficult-to-verify exclusion restriction.

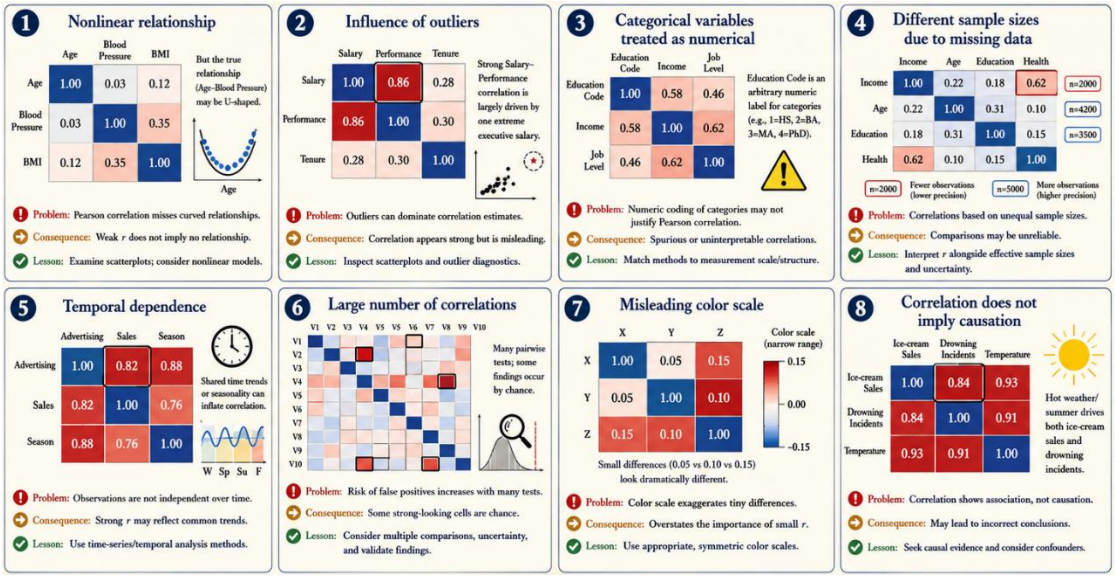

Figure 39: Instructive limitations of correlation heatmaps.

Pair plots can be useful for:

- low- to moderate-dimensional datasets;
- detecting class separation;
- discovering nonlinear associations;
- identifying clusters;
- comparing marginal distributions; and
- examining joint behavior between variables.

However, the number of panels grows rapidly with the number of variables. With p variables, a full pairwise display contains approximately p^2 panels, depending on whether both symmetric halves are shown.

As dimensionality increases, the result can quickly become visually overwhelming. Analysts should therefore use a **hypothesis-driven or domain-informed subset of variables** rather than automatically generating pair plots containing hundreds of features.

Example # 84: Infographic of pair plots

A pair plot combines univariate distributions along the diagonal with pairwise scatterplots in the off-diagonal panels, optionally using color or symbols to represent groups or classes (**Figure 40**). It can reveal linear and nonlinear associations, clusters, class separation, marginal-distribution differences, and joint behavior across variables (**Figure 40**). Because the number of panels grows approximately with p^2, pair plots are most useful for low- to moderate-

dimensional data and should be constructed from a hypothesis-driven or domain-informed subset of variables as dimensionality increases.

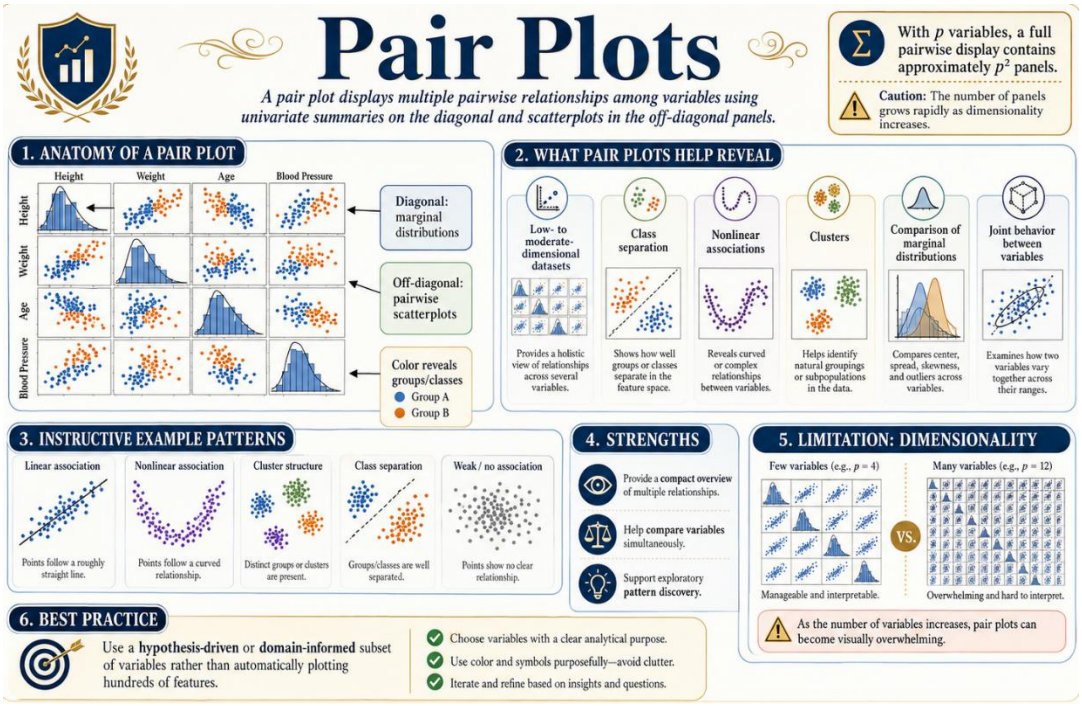

Figure 40: Pair plots for multivariable exploratory analysis.

3.3.7 Time-series plots

A **time-series plot** preserves the temporal ordering of observations (t_i, x_i). Unlike a conventional scatterplot of independent observations, a time-series visualization **explicitly represents the order** in which observations occurred. This can reveal:

- long-term trend;
- seasonality;
- cycles;
- abrupt changes;
- regime shifts;
- missing periods;
- anomalies;
- changes in variability;
- sensor drift.

Temporal structure is especially important because **observations collected close together** in time **may not be** statistically independent.

Raw and smoothed series

A raw time series can be displayed together with a moving average or other smoothing procedure. For example, a simple moving average with window (w) can be written as:

$$\mathrm{MA}_t = \frac{1}{w}\sum_{j=0}^{w-1} x_{t-j}$$

Smoothing **can make** broad trends **easier to identify**, but it **can also conceal** short-lived events, abrupt changes, or local anomalies. Both the original observations and the smoothing method should therefore be considered when interpreting a time-series visualization.

Avoid connecting unrelated observations

Lines imply continuity and order. A line plot should therefore not connect observations merely because they happen to appear consecutively in a dataset. Connecting unrelated categories or independent observations can create a visual impression of continuity that does not exist in the underlying data.

Temporal leakage

When constructing features for prediction, temporal summaries must use only information that would have been available at the prediction time. For example:

$$\mathrm{Feature}_t = g(x_{t-1}, x_{t-2}, \cdots)$$

A centered rolling average that incorporates future observations may be appropriate for retrospective descriptive analysis, but it can introduce **temporal leakage** if used as a predictor for a prospective task.

Example # 85: Infographic of time-series plots

Time-series plots preserve the ordering of observations and can reveal trends, seasonality, cycles, abrupt changes, regime shifts, missing periods, anomalies, changing variability, and measurement drift (**Figure 41**). Raw observations may be complemented by smoothing to highlight broader patterns, although excessive smoothing can obscure short-lived events(**Figure 41**). The figure also emphasizes that lines should connect only meaningfully

ordered observations and that predictive features must use information available at or before the prediction time to avoid temporal leakage.

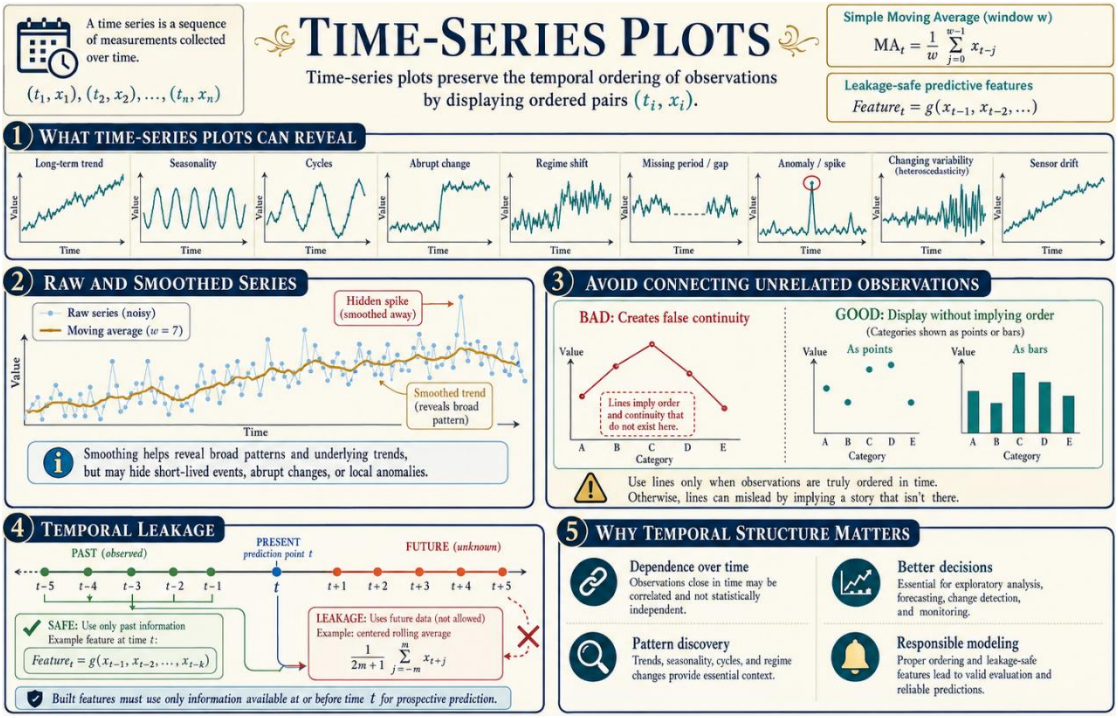

Figure 41: Time-series plots for revealing temporal structure in numerical data.

3.3.8 Categorical summaries

For a categorical variable C, the **number of observations** in category k, and the corresponding **proportion**, can be written as:

$$n_k = \#\{i: C_i = k\} \text{ and } p_k = \frac{n_k}{n}$$

Useful displays include:

- count plots;
- proportion plots;
- grouped bar charts;
- stacked bar charts;
- mosaic plots; and
- contingency tables.

Counts versus proportions

Counts answer: How many observations are in each category?
Proportions answer: What fraction of observations belongs to each category?

Both can be useful, but they answer different questions. When comparing groups with **substantially different sample sizes**, proportions are often more informative because raw counts may simply reflect differences in group size.

Ordering categories

Categories may be ordered according to:

- a natural ordering;
- alphabetical order;
- frequency;
- outcome rate; or
- domain-defined importance.

Ordinal variables should generally retain their meaningful order. For nominal categories, the chosen ordering should support the analytical purpose and make comparisons easy.

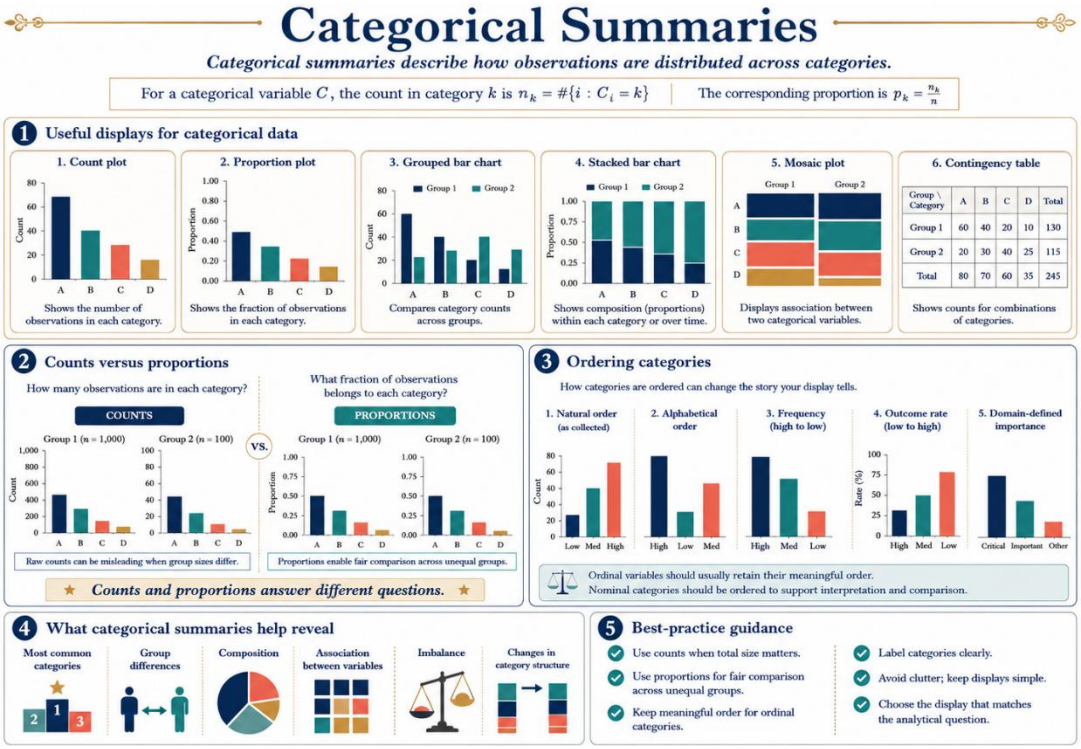

Figure 42: Categorical summaries for describing and comparing category structure.

Example # 86: Infographic of categorical summaries

Counts and proportions summarize how observations are distributed across categories and can be visualized using count plots, proportion plots, grouped or stacked bar charts, mosaic plots, and contingency tables (Figure 42).

Counts emphasize absolute frequency, whereas proportions are often more appropriate when comparing groups of unequal size. The ordering of categories also affects interpretation: ordinal variables should retain their meaningful order, while nominal categories may be arranged alphabetically, by frequency, outcome rate, or domain-defined importance to support clearer comparison.

3.3.9 Principles of responsible visualization

A visualization is not simply a decorative representation of data. It is an **analytical argument**: decisions about scales, axes, colors, aggregation, filtering, and annotation influence what the viewer notices and how the evidence is interpreted. A responsible visualization should therefore:

- use a chart type appropriate to the variables and analytical question;
- clearly label axes, units, and relevant categories;
- state the sample size when it is important for interpretation;
- avoid misleading or unnecessarily distorted scales;
- display uncertainty when uncertainty is relevant to the claim;
- disclose important filtering, exclusions, transformations, or aggregation;
- distinguish raw observations from fitted trends or model-based quantities;
- use consistent scales when visual comparisons between groups are intended;
- avoid unnecessary three-dimensional effects and other decorative distortions; and
- remain interpretable for users with color-vision limitations and, where appropriate, when reproduced in grayscale.

Good visualization does not mean making a chart look more impressive. It means making the **relevant evidence easier to see without introducing misleading visual cues**. A well-designed visualization can expose structure that would otherwise remain hidden, while a poorly designed visualization can obscure or distort the same evidence. For this reason, visualization should be treated as an **s** rather than merely as a final presentation step.

Example # 87: Infographic on responsible visualization in air-quality monitoring

The infographic (**Figure 43**) contrasts misleading and responsible ways of presenting daily $PM_{2.5}$ measurements[293], highlighting appropriate chart selection, clear labeling, honest scales, uncertainty display, disclosure of missing data and other analytical choices, consistent comparisons, and accessible design. Supporting views demonstrate how complementary summaries can strengthen interpretation. The example and numerical values are **illustrative and synthetic**, designed solely to demonstrate good visualization practice rather than represent a specific real-world dataset.

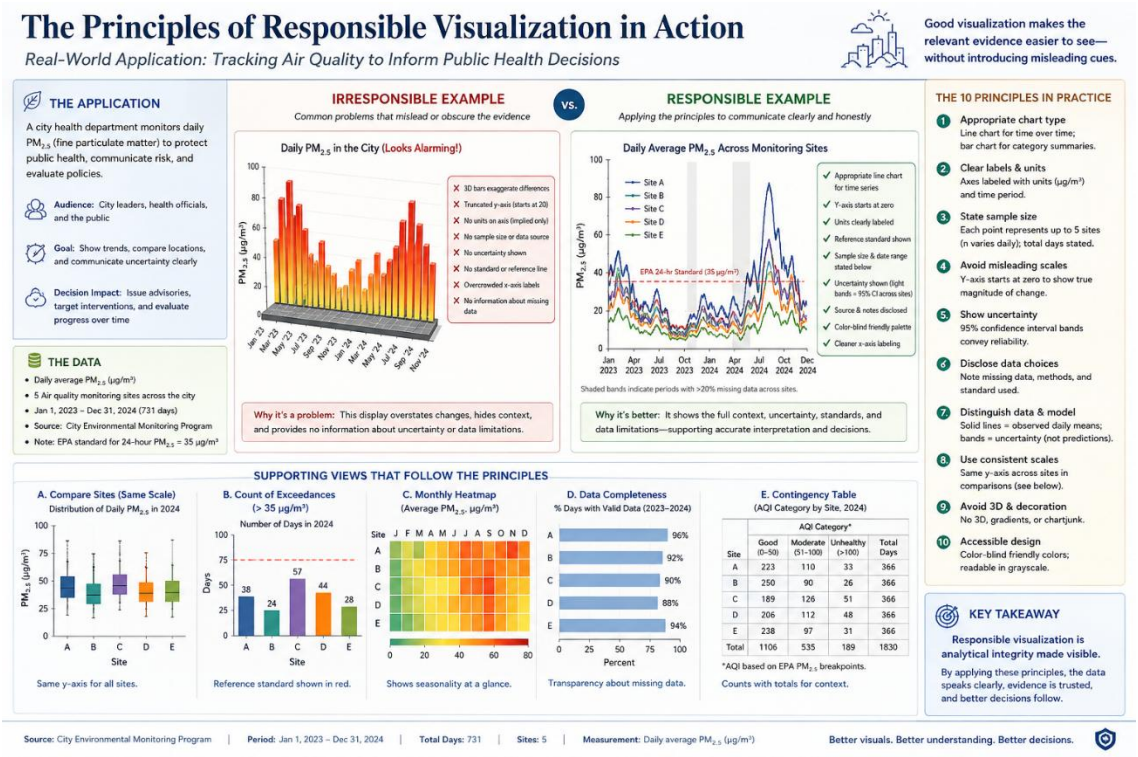

Figure 43: Principles of responsible visualization illustrated through a synthetic air-quality monitoring example.

[293] **$PM_{2.5}$** stands for **particulate matter with an aerodynamic diameter of 2.5 micrometers (µm) or smaller**. These particles are extremely small – roughly **30 times smaller than the width of a human hair** – and can come from sources such as vehicle exhaust, industrial emissions, combustion, wildfires, and atmospheric chemical reactions. Because of their small size, $PM_{2.5}$ particles can penetrate deeply into the lungs and, in some cases, enter the bloodstream. For this reason, $PM_{2.5}$ concentration is widely used as an important indicator of **air quality and public-health risk**, typically reported in micrograms per cubic meter (µg/m^3).

Descriptive Statistics

Descriptive statistics provide a **compact numerical summary** of the **characteristics of a dataset**. Whereas visualizations allow analysts to see distributions and relationships, descriptive statistics **quantify important aspects** of those patterns, including **location**, **spread**, **relative position**, **asymmetry**, and **association**. They are therefore an essential complement to graphical EDA. However, descriptive statistics **should not be interpreted in isolation**: a single number can conceal substantial structure, and the appropriate summary depends on the distribution of the data, the measurement scale, the presence of unusual observations, and the analytical objective.

3.4.1 Mean

For observations $x_1, \cdots, x_n$, the sample mean is given by $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$. The mean **incorporates every observation** and therefore provides a measure of the **overall arithmetic center** of the data. However, because every value contributes to the calculation, the mean **can be strongly affected by extreme observations**.

For example, consider the observations $\{2, 3, 3, 4, 100\}$. The $\bar{x} = 22.4$. Although mathematically correct, this value **does not represent where most observations are concentrated**. Four of the five observations are between 2 and 4, while a single extreme value of 100 pulls the mean substantially upward.

Hence, mean should be **interpreted in the context of the distribution** rather than treated automatically as the most representative measure of center.

3.4.2 Weighted mean

When observations contribute unequally to an estimate, a **weighted mean** may be appropriate. If observation i has weight w_i, the weighted mean is given by $\bar{x}_w = \frac{\sum_i x_i w_i}{\sum_i w_i}$. Weighted means are commonly used for:

- survey weights;
- class or observation weights;
- exposure durations;

- combining estimates from groups with different levels of representation; and
- other settings in which observations do not contribute equally.

The meaning of a weighted mean **depends critically** on what the **weights represent**. Analysts should therefore document the source and interpretation of the weights rather than treating them as arbitrary numerical adjustments.

3.4.3 Median

The **median** is the **middle value** of an **ordered dataset**. It divides the observations so that approximately half lie below it and approximately half lie above it. For an **odd number** of observations n, the median is the observation at position $\tilde{x} = x_{\left(\frac{n+1}{2}\right)}$.

For an **even number** of observations, the median is commonly defined as the **average of the two central ordered observations**, although software and statistical procedures may use different conventions in specialized settings.

For the ordered set of observations $\{2, 3, 3, 4, 100\}$. The median $\tilde{x} = 3$. Unlike the mean, the median is **relatively resistant to extreme observations**. The value 100 changes the mean substantially but does not change the median in this example.

3.4.4 Mean vs Median

A substantial difference between the mean and median may provide evidence of:

- skewness;
- extreme observations;
- mixtures of subpopulations; or
- possible data-quality problems.

Neither the mean **nor** the median is **universally superior**. The appropriate choice **depends on the distribution** and the **analytical purpose**. For **approximately symmetric distributions** without substantial outliers, the mean can provide an efficient summary of location. For **strongly skewed distributions** or distributions containing **influential extreme observations**,

the median may provide a more representative description of the typical observation. In many EDA settings, reporting **both the mean and median** is useful because their relationship itself provides information about the distribution.

Example # 88: Infographic on household income analysis

Household income provides a clear example of how the mean and median can convey different aspects of a distribution. In a right-skewed income distribution, a small number of very high incomes can pull the mean upward, while the median remains closer to the experience of the typical household (**Figure 44**). A substantial mean-median difference may therefore signal skewness, extreme observations, heterogeneous subpopulations, or possible data-quality issues. In practice, reporting both measures often provides a more informative summary than relying on either alone.

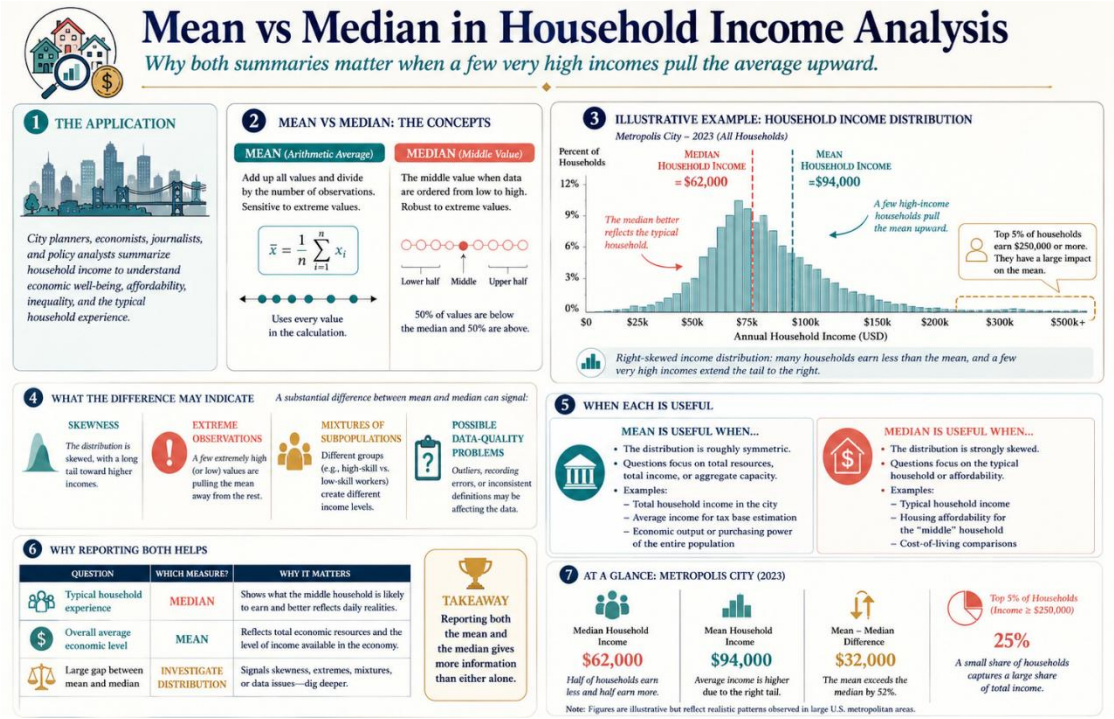

Figure 44: Mean versus median illustrated through household income.

3.4.5 Variance and standard deviation

Measures of central tendency describe where observations are located, but they do not describe how widely the observations are dispersed. **Variance and standard deviation** quantify this spread. The sample variance is given by:

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$$

The sample standard deviation is the square-root of the variance i.e., s. The use of $(n-1)$ rather than n in the sample variance is the **conventional unbiased estimator**[294] of the population variance under the usual **independent-sampling framework**.

Variance is expressed in **squared units**. For example, if X is measured in kilograms, its variance is expressed in kilograms squared. Standard deviation is expressed in the **same units as the original variable**, making it generally easier to interpret.

Sensitivity to outliers

Variance and standard deviation are **sensitive to extreme observations** because deviations from the mean are squared. A single unusually large or small observation can therefore have a **disproportionate effect** on the estimated spread.

For distributions containing **substantial skewness** or **extreme observations**, robust measures of spread may provide useful complementary information.

A robust measure of spread is the interquartile range $\text{IQR} = Q_{0.75} - Q_{0.25}$. It measures the **range containing the middle 50%** of the observations and is relatively resistant to extreme values.

Another robust measure is the **median absolute deviation (MAD)**:

$$\text{MAD} = \text{median}(|x_i - \text{median}(x)|)$$

Because the IQR and MAD are **less sensitive to extreme observations** than variance and standard deviation, they can be particularly informative for skewed or heavy-tailed distributions.

Example # 89: Infographic on salaries in a technology company

[294] An **unbiased estimator** is a statistic whose **expected value equals the true population parameter** it is intended to estimate; formally, an estimator $\hat{\theta}$ is unbiased for θ if $\mathbb{E}[\hat{\theta}] = \theta$. In the context of the **population variance**, the conventional unbiased estimator under the usual **independent-sampling framework** is the sample variance S^2, defined as $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$, where X_i are **independent observations**, $\bar{X}$ is the sample mean, and $(n-1)$ represents the degrees of freedom. Under these assumptions, $\mathbb{E}[S^2] = \sigma^2$, so repeated use of across random samples will, on average, equal the true population variance. Proof is as follows:

$$\mathbb{E}[S^2] = \frac{1}{n-1}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] \overset{\text{iid}}{=} \frac{1}{n-1}\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \mu)^2 - n(\bar{X} - \mu)^2\right] = \frac{1}{n-1}\left(n\sigma^2 - n\text{Var}(\bar{X})\right) \overset{\text{via independence}}{=} \frac{1}{n-1}\left(n\sigma^2 - n\frac{\sigma^2}{n}\right) = \sigma^2$$

The infographic (**Figure 45**) shows how a strongly right-skewed salary distribution with a small number of very high earners can substantially increase the mean, variance, and standard deviation. In contrast, the median, interquartile range (IQR), and median absolute deviation (MAD) provide more robust summaries of the typical center and spread. Department-level boxplots further illustrate how variability and extreme observations can differ across groups. The numerical values are illustrative and synthetic, emphasizing why traditional and robust measures are often most informative when reported together.

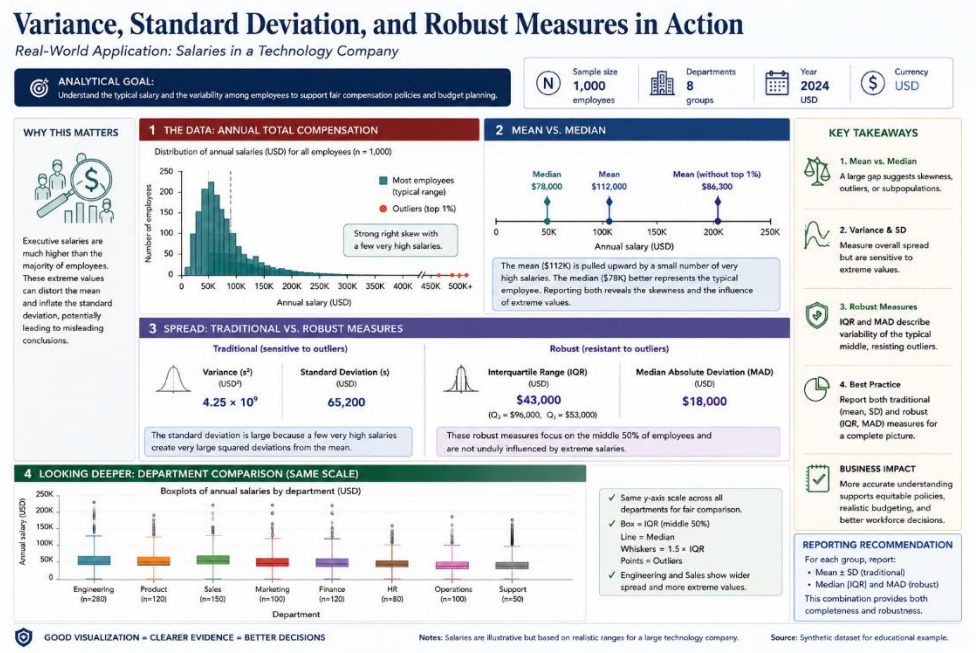

Figure 45: Variance, standard deviation, and robust measures of spread illustrated through employee compensation.

3.4.6 Quantiles

A **quantile**[295] describes the **relative position** of an observation **within a distribution**. The q-quantile (Q_q) is a value below which approximately a proportion q of observations fall. Common quantiles include:

- $Q_{0.25}$: first quartile;
- $Q_{0.5}$: median;

[295] A **quantile** is a value that divides a probability distribution or ordered dataset into specified proportions. More formally, the p-quantile $Q(p)$ is a value such that approximately p proportion of observations are less than or equal to it, where $0 < p < 1$. For example, the **0.50 quantile** is the median, meaning approximately 50% of observations lie below it; the **0.90 quantile** is the 90th percentile, meaning approximately 90% of observations lie below it.

- $Q_{0.75}$: third quartile[296];
- $Q_{0.90}$: 90th percentile[297];
- $Q_{0.95}$: 95th percentile.

Quantiles are particularly useful because they **describe the distribution without requiring the data** to follow a particular parametric form. They can be used for:

- summarizing skewed distributions;
- defining operational or clinical thresholds;
- service-level analysis;
- examining the lower and upper tails;
- screening unusually large or small observations; and
- comparing distributions across groups.

Quantiles also form the **basis of several common descriptive tools**. For example, the **interquartile range** is calculated from the first and third quartiles, while **boxplots use quartiles** to represent the central portion of a distribution.

Quantile estimation

For a **finite sample**, the exact numerical value assigned to a quantile depends on how the ordered observations are mapped to the desired percentile. Different statistical **software packages** can therefore produce **slightly different sample quantiles** because they use **different interpolation** or **estimation conventions**.

This usually has little practical importance for large datasets, but it can matter for small samples or when results must be reproduced exactly. Analysts should therefore be aware of the quantile definition used by their software when precise reproducibility is important.

Example # 90: Infographic on emergency department waiting-time analysis

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[296] A **quartile** is a **specific type of quantile** that divides an ordered dataset into **four parts of approximately equal size**. The three quartiles are Q_1, Q_2, and Q_3, corresponding approximately to the **0.25**, **0.50**, and **0.75 quantiles**, respectively. For example, if household incomes have $Q_1 = \$30{,}000$, $Q_2 = 50{,}000$, and $Q_3 = \$80{,}000$, then approximately 25% of households have incomes at or below \$30,000, 50% at or below \$50,000, and 75% at or below \$80,000.

Thus, **all quartiles are quantiles, but not all quantiles are quartiles**: the 25th, 50th, and 75th percentiles are quartiles, whereas the 10th, 90th, or 95th percentiles are quantiles but not quartiles.

[297] A **percentile** is simply a quantile expressed on a $\mathbf{0-100}$ **scale**: the p^{th} percentile corresponds to the $p/100$ quantile; thus, the 90th percentile is the 0.90 quantile, and the 25th percentile is the 0.25 quantile.

Quantiles describe the relative position of observations within a distribution and are especially informative for skewed service-time data. In this example (**Figure 46**), the median summarizes the typical patient experience, while the 90th and 95th percentiles **characterize upper-tail delays** and support operational targets, service-level monitoring, and anomaly screening. **Group-specific quantiles** also enable fair comparison across triage[298] categories, while the interquartile range and boxplot show how quantiles underpin robust summaries of central spread. The numerical values shown are illustrative.

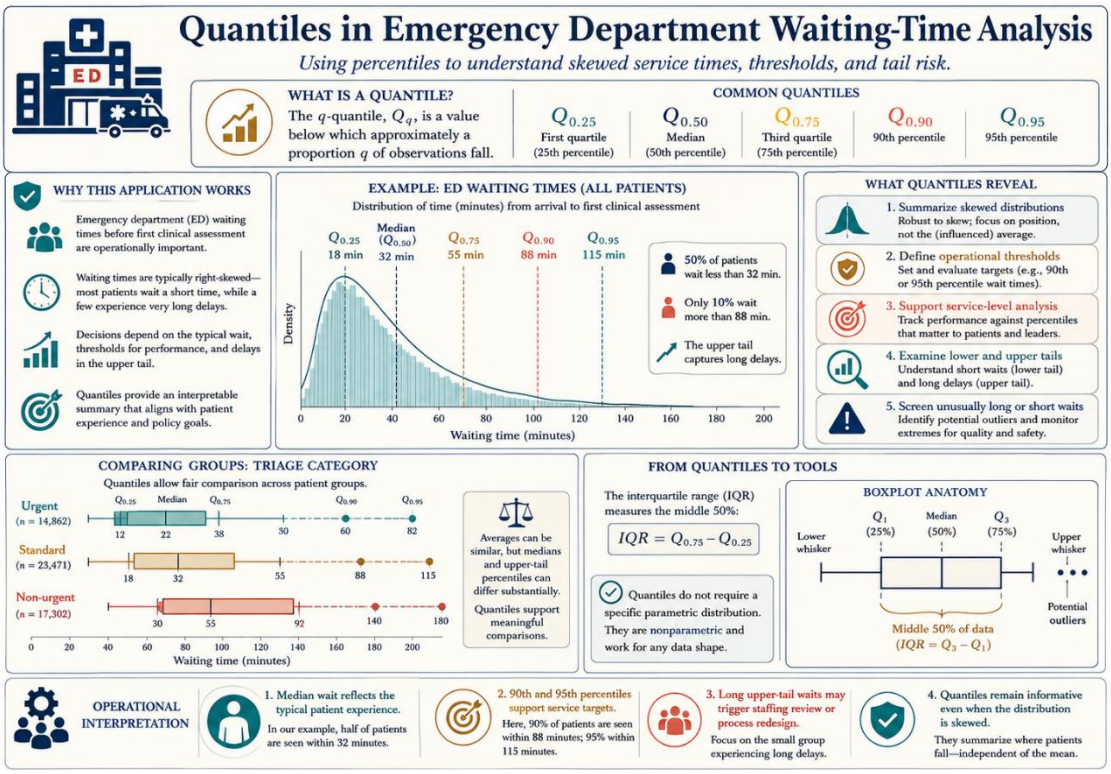

Figure 46: Quantiles illustrated through emergency-department waiting times.

Example # 91: The "80th percentile" discrepancy in SLA reporting

A cloud infrastructure company is testing a new server architecture. They measure the network latency (in milliseconds) for a small pilot sample of **5 servers** to verify compliance with a strict Service Level Agreement (SLA). The recorded latencies, sorted in ascending order, are: $[120, 135, 142, 150, 165]$.

[298] **Triage** is the process of **assessing patients** when they arrive at a **healthcare facility** and **prioritizing them** according to the **urgency** and **severity of their condition**.

The SLA contract states: *The 80th percentile of latency* ***must not exceed*** *155 ms*. Because the sample size is very small ($n = 5$), the 80th percentile falls **between** the 4th and 5th data points (150 and 165). The software **must interpolate**, and this is where the **discrepancy occurs**.

Different statistical packages use different default interpolation formulas, leading to two entirely different business outcomes from the exact same data (**Table 31**).

Table 31: Comparison of 80th percentile calculations for SLA assessment. The table contrasts two commonly used percentile definitions – Python/NumPy (Hyndman and Fan Type 7) and Excel PERCENTILE.EXC / R Type 6 – and shows how the choice of method can change the estimated 80th percentile and therefore the final service-level agreement outcome.

Software / Method	Formula Used	Calculation Steps	Calculated 80th Percentile	SLA Outcome
Python (numpy) (Hyndman & Fan Type 7)	$1 + (n-1) \times p$	Index = $1 + (5-1) \times 0.80 = 4.2$ Value = $150 + 0.2 \times (165 - 150) = 153.0$	153.0 ms	✓ PASSES (153 < 155)
Excel (PERCENTILE.EXC) R (Type 6)	$(n+1) \times p$	Index = $(5+1) \times 0.80 = 4.8$ Value = $150 + 0.8 \times (165 - 150) = 162.0$	162.0 ms	✗ FAILS (162 > 155)

The practical takeaways are as follows:

- **Sample size sensitivity**
  - In large datasets ($n > 10{,}000$), the index lands **almost exactly** on a specific data point, making the difference between methods mathematically negligible. In small samples, the **interpolation convention** dictates the result.
  - Always flag quantile calculations on small samples ($n < 30$) for manual review or explicit method declaration.
- **Reproducibility risk**
  - If a data scientist uses Python (Type 7) but the compliance auditor uses Excel's default (Type 6), they will argue over a 9 ms difference that is purely an **artifact of software defaults**, not reality.
  - Document the exact quantile method (e.g., **80th percentile calculated using inclusive linear interpolation, Type 7**) in all reports and code.

- **Regulatory/SLA thresholds**
  - When a contract or regulation hinges on a specific percentile, ambiguity in the calculation method can lead to financial penalties or failed audits.
  - Define the exact interpolation algorithm in the SLA contract itself, not just the target value.

3.4.7 Skewness

Skewness describes the **asymmetry** of a **probability distribution**. A commonly used moment-based population measure is:

$$\gamma_1 = \frac{\mathbb{E}[(X-\mu)^3]}{\sigma^3}$$

The interpretation is as follows:

- $\gamma_1 > 0$: longer or heavier right tail;
- $\gamma_1 < 0$: longer or heavier left tail;
- $\gamma_1 \approx 0$: approximate symmetry, though not necessarily normality.

Skewness is **particularly useful** when a distribution has a **pronounced tail** on one side. Common examples of variables that may exhibit right skewness include:

- income;
- waiting times;
- transaction amounts;
- healthcare costs; and
- numbers of social-media followers.

Skewness statistics can themselves be **highly sensitive** to **extreme observations**. They should therefore be interpreted alongside graphical displays such as histograms, density plots, or boxplots rather than being treated as definitive descriptions of distributional shape.

Example # 92: Infographic on Health-care cost distributions

Healthcare expenditures often exhibit **strong right skew** because **most patients incur relatively modest costs** while a **small number of complex cases generate very high expenditures**. The infographic shows how such asymmetry produces a long right tail, can separate the mean from the median,

and may strongly influence moment-based skewness estimates (Figure 47). It also emphasizes that skewness statistics are sensitive to extreme observations and should therefore be interpreted alongside graphical displays such as histograms, density plots, and boxplots. The numerical values shown are illustrative and synthetic.

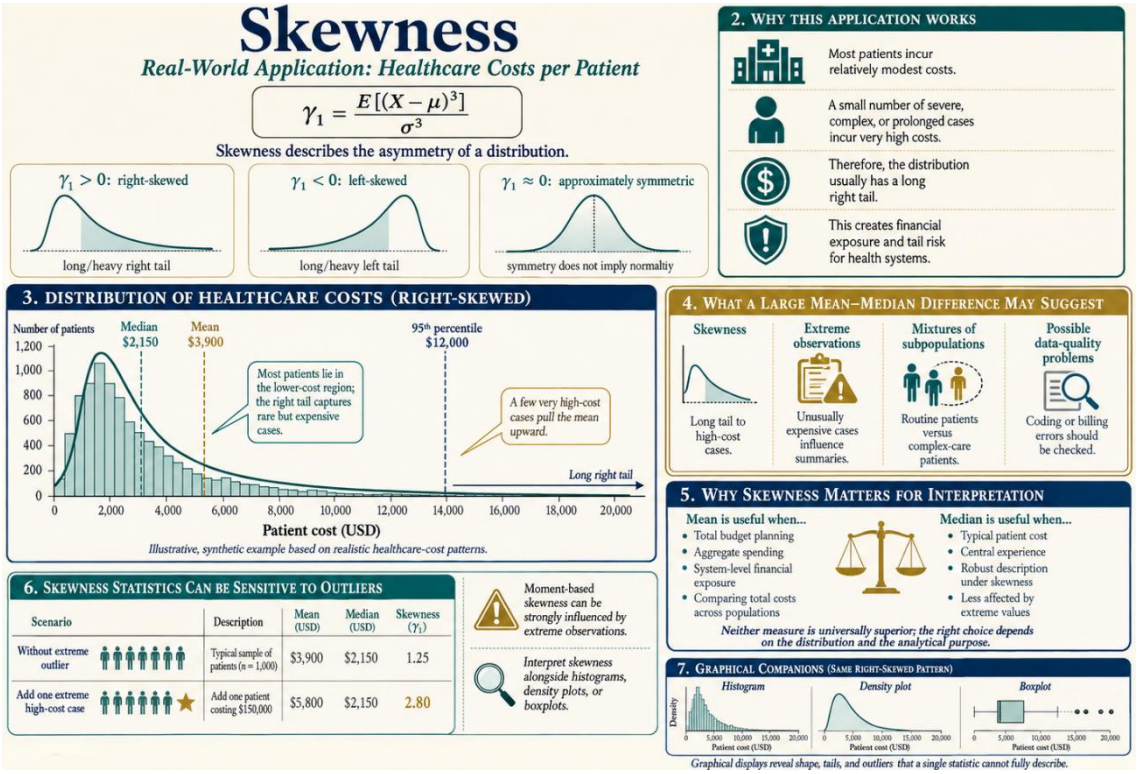

Figure 47: Skewness illustrated through healthcare-cost distributions.

Transformations for skewed variables

In some applications, a transformation can make a highly skewed variable easier to analyze. For a **nonnegative variable**, a logarithmic transformation can reduce strong right skewness and compress large values.

$$x' = \log(1 + x)$$

However, transformations **should not be applied** merely to make a distribution look more symmetric. The **transformed scale** should have a **meaningful interpretation**, and the analyst should consider how the transformation affects the intended statistical or predictive task.

Example # 93: House prices are skewed

House prices are nonnegative and often **exhibit substantial right skew** because a **small number of very expensive properties** extend the upper tail.

In this example (Figure 48), the transformation compresses large values, reduces skewness, and can make broad structure easier to summarize and model. The figure also emphasizes that such transformations should be used only when the transformed scale remains meaningful for the analytical or predictive task, rather than merely to make the distribution appear more symmetric.

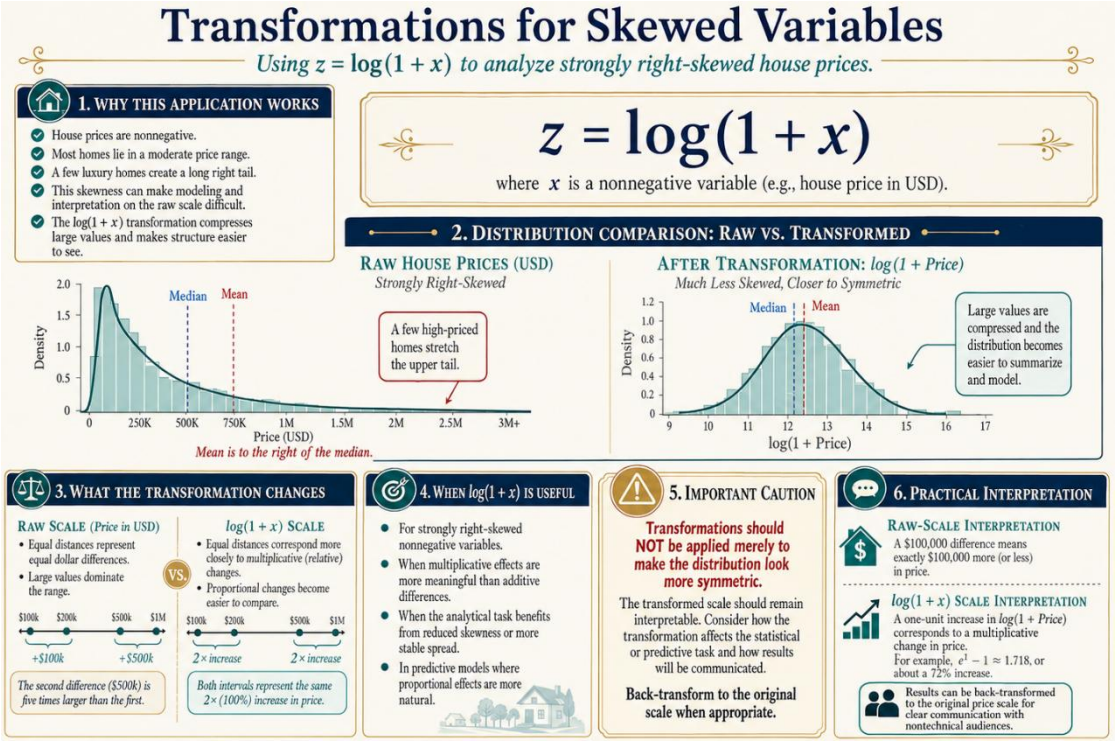

Figure 48: Logarithmic transformation of a strongly right-skewed variable illustrated through house prices.

3.4.8 Covariance

The **sample covariance** between and is:

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

Covariance describes the direction in which two variables tend to vary together. In general,

- **positive covariance** indicates that larger values of one variable tend to occur with larger values of the other;

- **negative covariance** indicates that larger values of one variable tend to occur with smaller values of the other; and
- **covariance near zero** indicates little linear co-movement.

Covariance is measured in the **product of the two variables' units**. For example, if X is measured in kilograms and Y in centimeters, covariance has units of kilogram-centimeters. This **dependence on measurement units** makes the magnitude of covariance **difficult to compare across different pairs of variables**. A **change of scale** in either variable **changes the covariance**, even when the underlying pattern of association remains the same.

Example # 94: Infographic on diversified investment portfolio

In a multi-asset portfolio, the covariance matrix summarizes all pairwise relationships while its diagonal elements contain individual asset variances, providing the basis for portfolio-variance calculations and diversification analysis (Figure 49). Because covariance depends on measurement units and scale, its magnitude is not directly comparable across different variable pairs; correlation is generally preferred when comparing association strength.

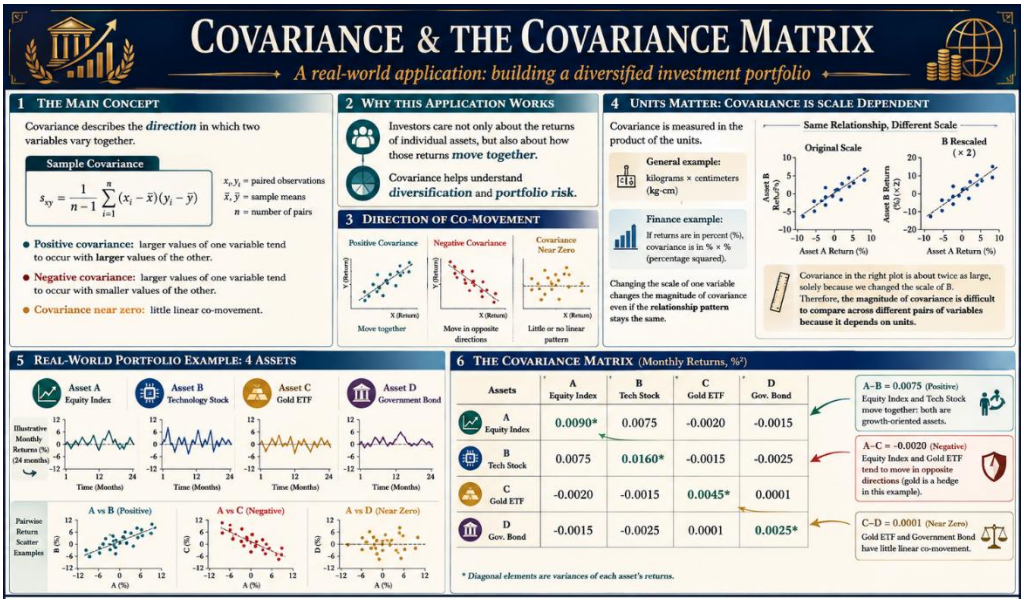

Figure 49: Covariance and the covariance matrix illustrated through portfolio diversification.

3.4.9 Correlation

Pearson's correlation coefficient standardizes covariance by the standard deviations of the two variables:

$$r_{XY} = \frac{s_{XY}}{s_X s_Y} = \frac{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2}} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

The coefficient satisfies $-1 \leq r_{XY} \leq 1$. Unlike covariance, correlation is dimensionless, making it easier to compare the strength of linear association across variable pairs.

Interpretation

In the idealized case of a perfect linear relationship:

- $r = 1$: perfect positive linear relation;
- $r = -1$: perfect negative linear relation;
- $r = 0$: no linear relationship.

The final point is particularly important. A correlation of zero **does not imply** that two variables are **independent** or **unrelated** in every sense. For example, if $Y = X^2$ and X has a distribution that is symmetric around zero. There can be a **perfectly deterministic relationship** between X and Y, yet the Pearson correlation can be zero because the relationship is nonlinear and symmetric.

To reiterate, **a Pearson correlation of zero means that there is no linear association, not necessarily that there is no relationship**.

Limitations

Pearson correlation can be affected by:

- outliers;
- restricted ranges;
- mixtures of distinct groups;
- nonlinear relationships;
- selection effects or selection bias; and
- temporal trends or dependence.

A **high correlation** can also arise from a **common underlying trend** rather than from a **direct relationship** between the variables. Consequently, correlation should be examined alongside scatterplots and relevant contextual information. Most importantly: **Correlation measures association; it does not establish causation**.

Rank correlation

Spearman correlation applies Pearson correlation to ranks and captures monotonic association:

$$\rho_S = \mathrm{Corr}\big(\mathrm{rank}(X), \mathrm{rank}(Y)\big)$$

It is particularly useful when the relationship is expected to be **monotonic but not necessarily linear**, or when the **variables are ordinal**. Because it operates on ranks, it can also be less sensitive to the influence of extreme values than Pearson correlation, although it is **not immune to outliers** or other data problems.

Spearman correlation is useful when:

- the relationship is monotonic but nonlinear;
- measurements are ordinal;
- the original scales are not naturally comparable; or
- Pearson correlation is inappropriate because the relationship is not adequately represented by a linear association.

As with Pearson correlation, Spearman correlation should be **interpreted as a measure of association** rather than evidence of a causal relationship.

Example # 95: Infographic on lifestyle factors and cardiovascular health

The infographic (**Figure 50**) shows how correlation can summarize relationships among variables such as physical activity, body mass index, sodium intake, stress, and systolic blood pressure, while also emphasizing important limitations, including sensitivity to outliers, restricted ranges, nonlinear relationships, subgroup mixtures, temporal trends, and selection effects. It further contrasts Pearson correlation with Spearman rank correlation for monotonic but nonlinear or ordinal relationships and reinforces that correlation measures association rather than causation.

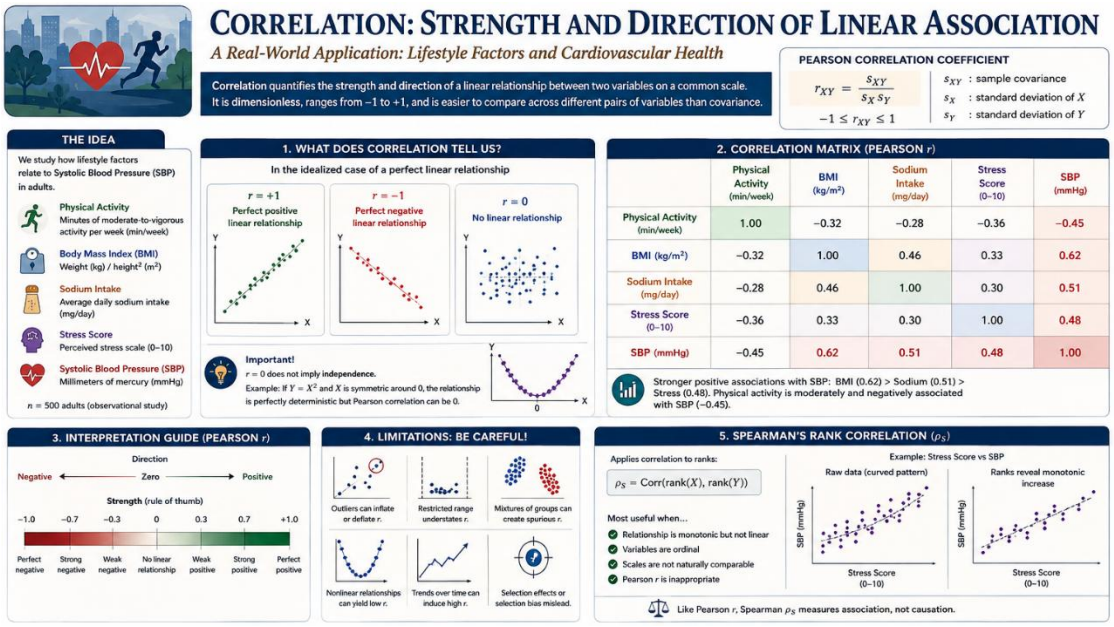

Figure 50: Correlation illustrated through lifestyle factors and cardiovascular health.

Feature Engineering

Feature engineering is the process of **converting raw data into representations** that are more useful for statistical analysis and predictive modeling. Raw variables are often recorded in forms that do not directly express the patterns a model needs to learn. An analyst may therefore need to **transform**, **combine**, **encode**, **aggregate**, or otherwise represent the original information in a more meaningful form. Good feature engineering combines **statistical reasoning**, **domain knowledge**, and careful control of **information flow**. However, feature engineering is not simply a search for variables that improve predictive performance: every transformation must be defined in a way that is **valid at the time of prediction**, **reproducible during deployment**, and **resistant to information leakage**.

3.5.1 What is feature engineering?

Feature engineering is the process of **transforming raw observations** into **representations** that make relevant structure more accessible to a statistical or machine-learning model. Let x_{raw} denote the original inputs. Feature engineering constructs a transformed representation $\phi(x_{\text{raw}}) = x_{\text{model}}$. The transformation $\phi(\cdot)$ may consist of **one or several operations**, including **scaling**, **encoding**, **aggregation**, mathematical **transformation**, **feature selection**, or **construction** of new variables. A feature transformation may be used to:

- improve numerical stability;
- incorporate domain knowledge;
- express nonlinear relationships;
- represent temporal history;
- reduce dimensionality;
- encode categorical information;
- make patterns easier for a model to learn; and
- improve interpretability.

For example, a **raw timestamp** may be transformed into **hour of day**, **day of week**, or **time since a previous event**. A person's date of birth may be transformed into age at the time of an observation. Several transaction records

may be aggregated into a measure of recent transaction frequency. These transformations do not necessarily add new information; rather, they **represent existing information** in a form that may be **more useful for the modeling task**.

Kuhn and Johnson[299] describe feature engineering as the development of **improved representations of predictors** within a **disciplined predictive-modeling workflow**. The key point is that feature engineering should be treated as **part of the modeling process** rather than as an isolated preprocessing exercise.

3.5.2 Feature engineering must occur inside the modeling pipeline

A fundamental rule of feature engineering is that **any transformation** whose **parameters are learned** from data **must be fitted** using the **training data only**. Suppose a transformation has parameters ψ. These parameters should be estimated as $\hat{\psi} = \mathrm{Fit}(X_{\mathrm{train}})$. The **fitted transformation** is then **applied consistently** to the **training**, **validation**, and **test data**:

$$X'_{\mathrm{train}} = T\left(X_{\mathrm{train}}; \hat{\psi}\right), X'_{\mathrm{validation}} = T\left(X_{\mathrm{validation}}; \hat{\psi}\right), X'_{\mathrm{test}} = T\left(X_{\mathrm{test}}; \hat{\psi}\right)$$

The validation and test sets **must not be used** to estimate $\hat{\psi}$. This principle applies to many operations, including:

- scaling and standardization;
- missing-value imputation;
- categorical encoding;
- feature selection;
- dimensionality reduction;
- vocabulary construction;
- target encoding; and
- data-driven feature transformations.

For example, if the mean and standard deviation used for standardization are calculated from the entire dataset before the train-test split, **information from the test set** has **influenced the representation** used during model

[299] M. Kuhn and K. Johnson, *Feature engineering and selection: A practical approach for predictive models*, CRC Press, 2026.

development. The resulting evaluation may therefore be **optimistically biased**.

The safest approach is to place data-dependent preprocessing **inside the modeling pipeline**, so that each training procedure estimates its preprocessing parameters using **only the data available to that procedure**.

3.5.3 Scaling and standardization

Scaling changes the **numerical representation of a feature** without necessarily changing the information it contains. It is particularly important for algorithms that depend on distances, coefficient magnitudes, or gradient-based optimization.

Standardization

A commonly used transformation is **standardization**: $z_i = \frac{x_i - \mu_{\text{train}}}{\sigma_{\text{train}}}$. Here, μ_{train} and σ_{train} are estimated from the training data. After standardization, the training feature will typically have approximately: $\text{mean}(z) = 0, \sigma(z) = 1$. Another name for this is **mean-variance normalization**.

Standardization is often useful for:

- logistic regression;
- support-vector machines;
- neural networks;
- principal component analysis;
- k-nearest neighbors;
- regularized regression; and
- gradient-based optimization.

Why scaling matters

Consider two features:

- annual income ranging from 0 to 10^7;
- debt ratio ranging from 0 to 1.

For a **distance-based algorithm**, the **income** feature **can dominate** the numerical distance simply because its **units** and **range** are **much larger**. This does not mean that income is inherently more informative; it is partly an **artifact of scale**.

Scaling can also affect **regularized regression**. **Penalties** applied to coefficients are **meaningful only** in relation to the **scales of the corresponding predictors**. Without appropriate scaling, the same numerical penalty can affect variables differently.

Robust scaling

When extreme observations are present, a robust alternative is to center using the median and scale using the interquartile range:

$$x_i' = \frac{x_i - \text{median}(x)}{\text{IQR}(x)}$$

This approach is **less sensitive to extreme values** than mean-and-standard-deviation scaling.

Not all algorithms require scaling to the same degree. For example, many tree-based models rely primarily on threshold comparisons and therefore are generally much less sensitive to the absolute numerical scale of individual predictors.

3.5.4 Normalization

The term **normalization** is used inconsistently across fields and can refer to several different transformations. It is therefore important to specify exactly what operation is being performed.

Min–max normalization

A common transformation maps a variable to an interval such as $[0,1]$:

$$x_i' = \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}$$

The **minimum** and **maximum** should be **estimated** from the **training data** when the transformation is used in a predictive workflow. Min–max scaling is **sensitive to extreme observations** because a single unusually large or small value can determine the range and compress the remaining observations.

Unit-vector normalization

For a vector x, unit-vector normalization can be written as:

$$x_i' = \frac{x}{\|x\|_2}$$

This makes $\|x'\|_2 = 1$ provided $x \neq 0$. This type of normalization is often useful when the **direction of a vector is more important than its magnitude**, such as:

- document vectors;
- embeddings;
- cosine-similarity systems.

Distributional transformations

Mathematical transformations can also be used to **change the shape or scale of a distribution**. Examples include:

$$x' = \log(1 + x); x' = \sqrt{x}$$

Such transformations **may reduce** skewness, stabilize variance, or make a relationship easier for a model to represent. Transformation decisions should consider:

- the distributional shape;
- the measurement scale;
- model assumptions;
- interpretability; and
- consistency between training and deployment.

A transformation that improves a numerical metric but makes the **resulting feature difficult to interpret** or reproduce **may not be an appropriate** engineering choice.

Example # 96: Choosing appropriate data transformations

- **Household income**
  - *Situation:* A dataset contains household income ranging from $20,000 to several million dollars. Most households have moderate incomes, while a small number have extremely high incomes.
  - *Problem:* Income is strongly **right-skewed**, and extreme values can dominate some analyses.
  - *Transformation:* The analyst applies a **log transformation**, such as $\log(\text{income})$.
  - *Potential benefit:* The distribution becomes less skewed, and differences between very large incomes are compressed.

- *Interpretability:* The transformed variable no longer represents income in dollars directly, so the analyst must explain what the transformed values mean.
- *Lesson:* A log transformation can be useful when the original variable is highly skewed, but its effect on **interpretation** must be considered.

- **Hospital length of stay**
  - *Situation:* Most patients stay in hospital for $1-5$ days, but a small number remain for several months.
  - *Problem:* Length of stay is highly right-skewed.
  - *Transformation:* The analyst uses $\log(1+\text{length of stay})$ to handle the skewness while accommodating zero values if necessary.
  - *Potential benefit:* The extreme long stays have less influence, and a statistical model may represent the distribution more effectively.
  - *Model consideration:* The transformation may be particularly useful if a model assumes approximately constant variance.
  - *Lesson:* Transformation can improve model behavior, but the analyst should still report results in terms that are meaningful for healthcare interpretation.

- **Website visits and sales**
  - *Situation:* A retailer examines the relationship between **website visits and sales**. A few very large advertising campaigns generate hundreds of thousands of visits.
  - *Problem:* Both variables are strongly skewed, and the relationship appears curved rather than linear.
  - *Transformation:* The analyst applies logarithmic transformations to both variables.
  - *Potential benefit:* The transformed scatterplot shows a relationship that is closer to linear and easier for a linear regression model to represent.
  - *Interpretation:* The model can now be interpreted in terms of **relative or percentage changes**, rather than simply changes in the original units.
  - *Lesson:* A transformation can change not only the distribution but also the **meaning of the model coefficients**.

- **Age and mortality risk**

- *Situation:* A predictive model uses age as a feature. The relationship between age and mortality risk is relatively weak at younger ages but increases rapidly at older ages.
- *Problem:* A simple linear age term may not adequately represent the relationship.
- *Transformation:* Instead of using only age, the analyst may introduce a nonlinear representation, such as age^2, splines, or age groups.
- *Potential benefit:* The model can capture the changing relationship between age and risk.
- *Interpretability:* A complex transformation may improve predictive performance but make individual coefficients harder to explain.
- *Lesson:* Transformation is not limited to reducing skewness; it can also help a model represent **nonlinear relationships**.

- **Training vs deployment**
  - *Situation:* A fraud-detection model is trained using transaction amounts transformed with $\log(amount)$.
  - *Problem:* The model performs well during development, but the production system supplies raw transaction amounts.
  - *Consequence:* The model receives features on a different scale from the one used during training, causing predictions to become unreliable.
  - *Solution:* The exact same transformation must be applied during **training**, **validation**, **testing**, and **deployment**.
  - *Lesson:* A transformation is part of the model pipeline and must be **consistently reproduced in production**.

- **A transformation that improves accuracy but reduces interpretability**
  - *Situation:* An analyst tests several complex transformations and finds one that increases predictive accuracy from 82% to 84%.
  - *Problem:* The transformation is difficult to explain, depends on arbitrary choices, and cannot be reliably reproduced in the operational environment.
  - *Consequence:* Although the numerical metric improves, the resulting model may be difficult to maintain, audit, or interpret.
  - *Lesson:* **Higher predictive performance alone does not guarantee that a transformation is appropriate.** Interpretability, reproducibility,

operational consistency, and the measurement scale should also be considered.

3.5.5 Binning

Binning converts a continuous variable into a set of intervals or categories. Given thresholds $c_1, \cdots, c_{K-1}$:

$$B(x) = \begin{cases} 1, & x < c_1 \\ 2, & c_1 \le x < c_2 \\ 3, & c_2 \le x < c_3 \\ \quad \vdots & \\ K, & x \ge c_{K-1} \end{cases}$$

Types of bins

Common approaches include:

- equal-width bins;
- equal-frequency or quantile bins;
- domain-defined bins; and
- supervised bins based on outcome information.

Potential benefits

Binning can:

- capture threshold-like relationships;
- reduce sensitivity to small measurement fluctuations;
- improve interpretability;
- simplify communication; and
- allow simple models to represent certain nonlinear patterns.

For example, an age variable might be divided into domain-relevant age groups when those categories correspond to meaningful clinical, policy, or operational thresholds.

Example # 97: AQI and respiratory hospitalization risk

The infographic (**Figure 51**) contrasts equal-width, equal-frequency, domain-defined, and supervised binning, then illustrates how continuous Air Quality Index (AQI) values can be mapped into meaningful risk categories for respiratory-health monitoring and public communication. The example highlights how binning can reveal threshold-like behavior, simplify

interpretation, and support operational decisions, while emphasizing that bin boundaries should be selected deliberately and validated for the analytical objective.

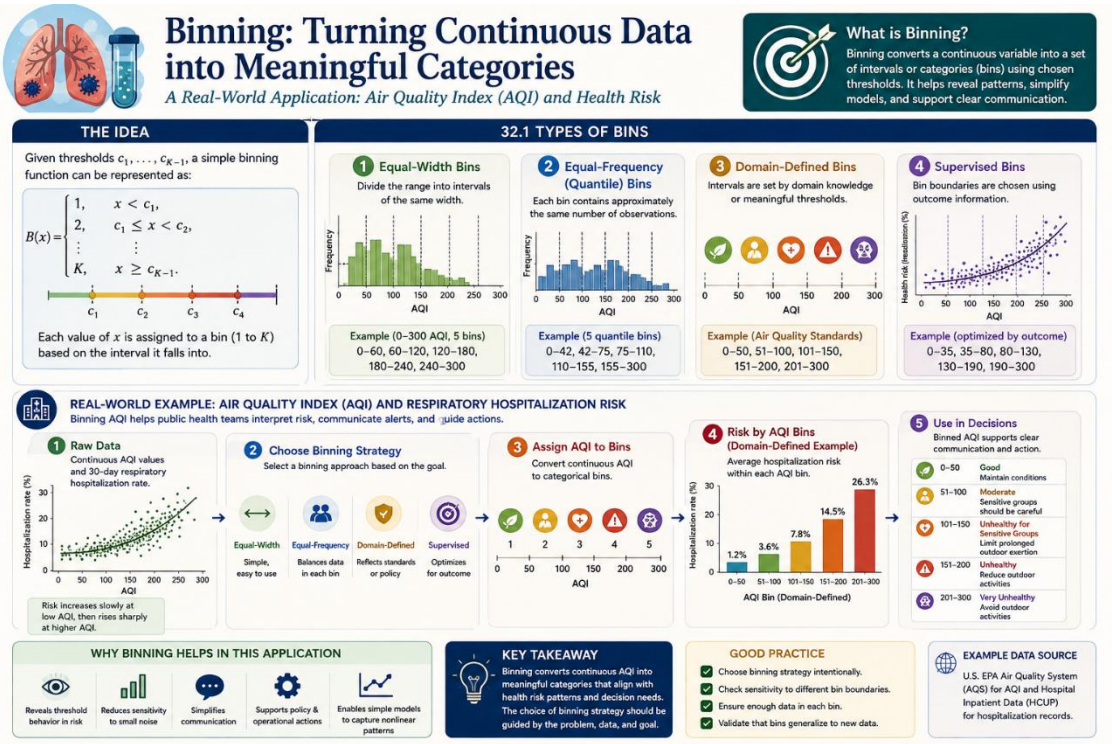

Figure 51: Binning illustrated through air-quality risk categorization.

Risks

Binning also has important disadvantages. It can:

- discard information;
- introduce artificial discontinuities;
- create instability near category boundaries;
- conceal important variation within bins; and
- introduce target leakage when outcome-informed bins are constructed using information from evaluation data.

When scientifically meaningful thresholds exist, **domain-defined bins are generally preferable to arbitrary cut points**. If data-driven or supervised binning is used, the procedure must be fitted within the training process.

Example # 98: Infographic on clinical modeling

Converting continuous systolic blood pressure into categories can simplify interpretation, but it may also discard information, create artificial discontinuities, destabilize predictions near cut points, and conceal clinically important variation within bins. **Figure 52** further shows how outcome-informed binning can introduce target leakage if cut points are selected using evaluation data. Accordingly, scientifically meaningful domain-defined thresholds are preferable when available, while data-driven or supervised binning should be learned strictly within the training process and validated on independent data.

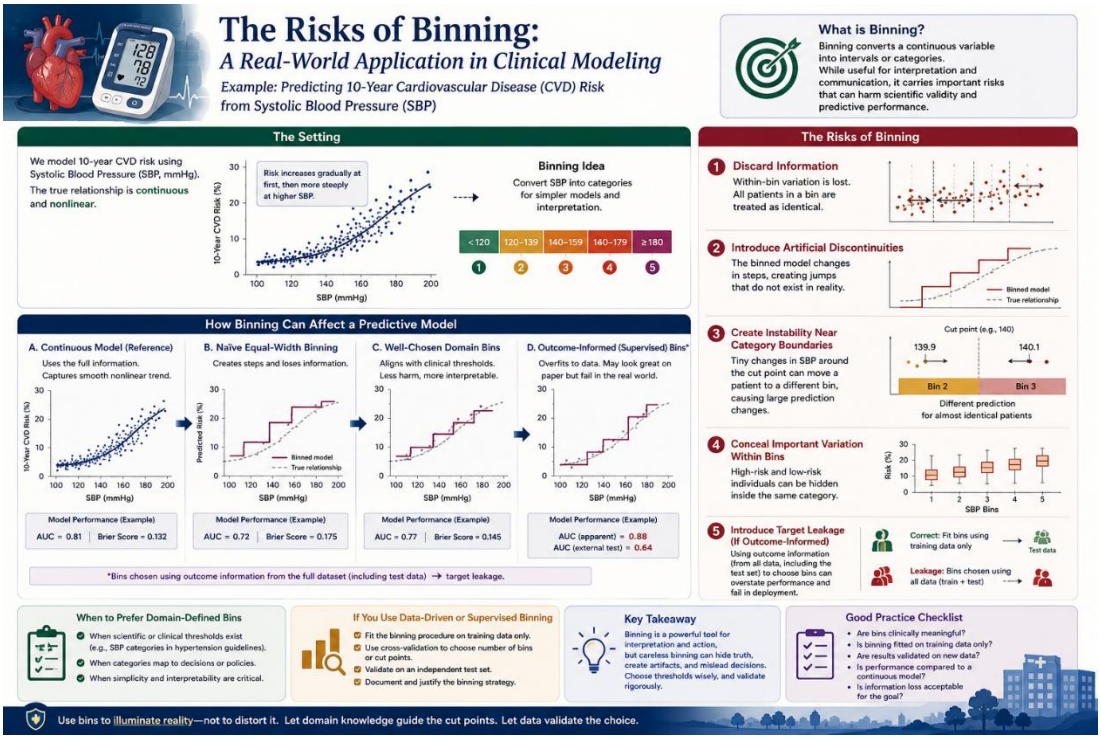

Figure 52: Risks of binning illustrated through cardiovascular risk modeling.

3.5.6 Encoding categorical variables

Many statistical and machine-learning algorithms require categorical variables to be represented numerically. **Categorical encoding** converts categories into numerical representations while attempting to preserve the information relevant to the modeling task.

Ordinal encoding

For genuinely ordered categories, an ordinal representation may be appropriate:

$$\text{low} = 1, \text{medium} = 2, \text{high} = 3$$

This encoding should be used only when the categories have a meaningful order. Applying it to nominal categories such as colors, countries, or product types can introduce an artificial numerical relationship that does not exist.

One-hot encoding

For a categorical variable with K categories, one-hot encoding creates binary indicators:

$$x_{ik} = \mathbf{1}\{C_i = k\}$$

Each observation receives a value of 1 for the category to which it belongs and 0 for the others. For linear models that include an intercept, one category is commonly omitted to avoid exact linear dependence among the predictors. The omitted category then serves as the reference category.

Example # 99: When one-hot encoding works and when it does not

Suppose $\text{device_type} \in \{\text{Android}, \text{iOS}, \text{Web}\}$. One-hot encoding of device_type is:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Pros and Cons: It is **simple**, **safe**, widely used. However, it is **not scalable** in the sense that if a category has high cardinality, its one-hot encoding will yield huge sparse vectors.

For instance, the category zipcode has a large cardinality (Pakistan has $1{,}972$ unique zip codes and USA have $40{,}000$ unique zip codes). One-hot encoding of zipcode yields a binary sparse matrix of dimensionality $40{,}000 \times 40{,}000$ which can **blow up memory** and cause **overfitting**.

Example # 100: One-hot encoding in linear models

Imagine a clinical data with a feature column **Gender** which has three categories $\{\text{Male}, \text{Female}, \text{Other}\}$. One-hot encoding gives:

$$\text{Male} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}; \text{Femal} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}; \text{ Other} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Notice that for every person, $\text{Male} + \text{Female} + \text{Other} = 1$. A linear model with a y-intercept takes the form: $y = \beta_0 + \beta_1\text{Male} + \beta_2\text{Female} + \beta_3\text{Other}$. But we already know that $\text{Male} + \text{Female} + \text{Other} = 1$. So, the model has redundant information. The computer can't uniquely figure out what $\beta_0, \beta_1, \beta_2, \beta_3$ should be.

Hence, we leave one category out, so the model **doesn't have redundant columns**, and the category we leave out becomes the baseline against which the other categories are compared. Now, the linear model becomes $y = \beta_0 + \beta_1\text{Male} + \beta_2\text{Female}$.

For Other, $\text{Male} = 0$ & $\text{Female} = 0$: $y = \beta_0 + \beta_1(0) + \beta_2(0) = \beta_0$.
For Male, $\text{Male} = 1$ & $\text{Female} = 0$: $y = \beta_0 + \beta_1(1) + \beta_2(0) = \beta_0 + \beta_1$.
For Female, $\text{Male} = 0$ & $\text{Female} = 0$: $y = \beta_0 + \beta_1(0) + \beta_2(1) = \beta_0 + \beta_2$.

So β_1 tells you **how Male differs from Other** and β_2 tells you **how Female differs from Other**.

Frequency encoding

A category can instead be represented by its frequency:

$$f_k = \frac{\#\{i: C_i = k\}}{n}$$

Frequency encoding is **compact** and can be **useful for high-cardinality variables**. However, categories with similar frequencies **may receive identical or similar representations** even when they **have very different meanings or relationships** with the outcome.

Example # 101: Encoding cities

Suppose we have 10 customers and their cities are: $\{\text{LHR}, \text{LHR}, \text{LHR}, \text{LHR}, \text{KHI}, \text{KHI}, \text{KHI}, \text{ISL}, \text{ISL}, \text{PEW}\}$. Frequency encoding asks: **How often does each category appear?** So, we replace the city names of the customers with their frequencies (**Table 32**).

How does this compare to one-hot encoding?

Imagine you have **100,000 different cities** or **categories**. One-hot encoding would potentially create **100,000 columns**. Frequency encoding can represent each category using just **one number**. Hence, frequency encoding is compact and can be useful for high-cardinality variables.

Table 32: Example of frequency encoding for a categorical variable. Customer city is replaced by the proportion of observations belonging to each city in the training data. Frequency encoding converts category labels into numerical values while preserving information about their relative prevalence.

Customer ID	Customer City	Frequency Encoding
1	Lahore	0.4
2	Lahore	0.4
3	Lahore	0.4
4	Lahore	0.4
5	Karachi	0.3
6	Karachi	0.3
7	Karachi	0.3
8	Islamabad	0.2
9	Islamabad	0.2
10	Peshawar	0.1

What can potentially go wrong with frequency encoding?

Consider a dummy data table in **Table 33**. Frequency encoding gives both the same number i.e., 0.1. So, the model sees **exactly the same encoded value**, even though the two cities might have completely different relationships with the outcome (house price).

Table 33: Illustration of a limitation of frequency encoding.

City	Frequency	Average house price
City A	0.10	$100,000
City B	0.10	$1,000,000

Target encoding

In **target encoding**, a category is represented using an **estimate of the outcome** associated with that category:

$$\hat{\mu}_k = \mathbb{E}[Y|C = k]$$

This can be powerful for high-cardinality categorical variables, but it is particularly vulnerable to **target leakage and overfitting**. If the outcome of an observation **contributes directly to the encoded value** used for that same observation, the model can **effectively receive information about the target** during feature construction.

Example # 102: Customer loan default prediction

Imagine a bank wants to predict whether a customer will **default on a loan**. The target variable is: $1 =$ customer defaulted and $0 =$ customer did not default. The bank has a categorical feature called **occupation**. A sample training data is provided in **Table 34**. The bank uses **target encoding**, replacing each occupation with its average default rate.

Table 34: Illustrative customer dataset for target encoding.

Customer	Occupation	Default (Target)
A	Teacher	0
B	Teacher	0
C	Teacher	1
D	Driver	1
E	Driver	1
F	Driver	0

For **Teacher** we have $\frac{0+0+1}{3} = 0.33$. So, every Teacher receives an encoded value of 0.33. For **Driver** we have $\frac{1+1+0}{3} = 0.67$. So, every Driver receives an encoded value of 0.67.

The problem is that Customer C's **own outcome** (Default = 1) **was used** to calculate the **feature value** that Customer C receives. This becomes particularly serious when a category has only one observation.

Suppose Customer G is the **only Engineer** in the training data and Customer G defaulted. The target encoding for Engineer would be $\frac{1}{1} = 1.00$. But because Customer G is the **only Engineer**, the encoded feature essentially tells the model: *This customer defaulted*.

The model has therefore received information about the target through the feature itself. It may appear to perform extremely well during training, but this performance is misleading because the feature construction has **leaked the answer into the input data**.

What is the solution?

A safer approach is to calculate Customer G's encoding using **other observations** rather than Customer G's own outcome. If there are no other Engineers, the encoder could instead use a smoothed or global target value.

Using the global target value: Because there are no other Engineers, we cannot reliably calculate an Engineer-specific default rate. So, we use the overall training default rate of $\frac{3}{6} = 0.5$. Hence, the target encoding of an Engineering is 0.5.

The value 0.50 **does not reveal** that Customer G defaulted. It simply reflects the **general default rate** in the training population.

Using smoothing: We use the following formula:

$$\mathrm{TE}(c) = \frac{n_c\,\bar{y}_c + \lambda\bar{y}}{n_c + \lambda}$$

For Customer G we have: $n_c = 1$ because there is only one Engineer; $\bar{y}_c = 1.00$ because that engineer has defaulted; $\bar{y} = 0.5$ is the overall default rate; $\lambda = 10$ is the assumed smoothing value. This yields:

$$\mathrm{TE}(\text{Engineer}) = \frac{1 \cdot\ 1.00 + 10 \cdot 0.5}{1 + 10} = 0.455$$

So, the smoothed encoding would be approximately **0.455**.

Note: The important point is that **smoothing reduces the influence of Customer G's individual outcome**, but it **does not completely remove** it.

Therefore, smoothing should generally be combined with **out-of-fold/cross-fitted encoding** when constructing training features.

Safe target encoding therefore generally requires techniques such as:

- cross-fitting[300];
- smoothing[301];
- out-of-fold estimates[302]; and
- explicit handling of categories not observed during training[303].

Example # 103: Defaulting a lone

Suppose a bank wants to predict whether a customer will **default on a loan**. The target variable is: $1 =$ customer defaulted and $0 =$ customer did not default. The bank has a categorical feature called **occupation**. A sample training data is provided in Table 35.

Table 35: Illustrative customer dataset for occupation-based target encoding.

Customer	Occupation	Default
A	Teacher	0
B	Teacher	1
C	Teacher	0
D	Driver	1
E	Driver	0
F	Driver	1
G	Engineer	1
H	Engineer	0

[300] **Cross-fitting** is a technique where the data are divided into several folds, and the **target encoding** for each observation is **calculated** using data from the **other folds**, not the observation's own fold. This prevents the target value of an observation from directly influencing its encoded value. For example, if the data are divided into five folds, observations in Fold 1 are encoded using target information from Folds 2 to 5.

[301] **Smoothing** reduces the reliability problem associated with categories that have very few observations. Instead of using only the category's target mean, the encoding is combined with the overall target mean. For example, if a city has only two customers and both defaulted, its unsmoothed encoding would be 1.00. Smoothing might reduce this to 0.40 by combining the category's estimate with the overall default rate.

[302] An **out-of-fold (OOF) estimate** is a target encoding calculated for an observation using **information from the folds** in which that observation was **not included**. For example, if an observation belongs to Fold 2, its encoding is calculated using Folds 1, 3, 4, and 5. OOF estimates are therefore a practical way of implementing cross-fitting and preventing target leakage during model training.

[303] **Handling categories not observed during training:** This means explicitly deciding what to do when a category appears in validation, testing, or production data but was never present in the training data. For example, if Lahore, Karachi, and Islamabad occurred during training but Multan appears only during testing, the encoder cannot calculate a target mean for Multan. A common solution is to assign it the **overall training target mean** or a dedicated unknown-category value.

We divide the data into two folds – *Fold 1* and *Fold 2* (Table 36).

Table 36: Two-fold partition for out-of-fold target encoding.

Fold 1

Customer	Occupation	Default
A	Teacher	0
B	Teacher	1
D	Driver	1
E	Driver	0

Fold 2

Customer	Occupation	Default
C	Teacher	0
F	Driver	1
G	Engineer	1
H	Engineer	0

Step 1: Encode Fold 1 using Fold 2

When encoding customers in **Fold 1**, we **do not use their own targets**. We calculate the occupation averages using only Fold 2 (Table 37).

- Teacher target encoding based on Fold 2 is: $\frac{0}{1} = 0$.
- Driver target encoding based on Fold 2 is: $\frac{1}{1} = 1.00$.
- Engineer target encoding based on Fold 2 is: $\frac{1+0}{2} = 0.5$.

Table 37: Out-of-fold target encoding for Fold 1.

Customer	Occupation	Actual Default	OOF Encoding
A	Teacher	0	**0.00**
B	Teacher	1	**0.00**
D	Driver	1	**1.00**
E	Driver	0	**1.00**

Customer B's **default value** of 1 **was not used** to calculate Customer B's encoding**.** Its encoding came entirely from Fold 2.

Step 2: Encode Fold 2 using Fold 1

To encode customers in **Fold 2**, we use only Fold 1 (Table 38).

- Teacher target encoding based on Fold 1 is: $\frac{0+1}{1} = 0.5$.
- Driver target encoding based on Fold 1 is: $\frac{1+0}{2} = 0.5$.
- Since we have no Engineer observations in Fold 1, we use an appropriate fallback, such as the global training mean (Table 35). The global default rate is $\frac{4}{8} = 0.5$.

Table 38: Out-of-fold target encoding for Fold 2.

Customer	Occupation	Actual Default	OOF Encoding
C	Teacher	0	**0.50**
F	Driver	1	**0.50**
G	Engineer	1	**0.50**
H	Engineer	0	**0.50**

The final training data

After cross-fitting, every training observation has an encoding calculated **without using its own target** (Table 39).

Table 39: Combined out-of-fold target encoding values for the training dataset.

Customer	Occupation	Default	OOF Target Encoding
A	Teacher	0	**0.00**
B	Teacher	1	**0.00**
C	Teacher	0	**0.50**
D	Driver	1	**1.00**
E	Driver	0	**1.00**
F	Driver	1	**0.50**
G	Engineer	1	**0.50**
H	Engineer	0	**0.50**

Rare and unseen categories

Practical systems must also **account for categories** that **occur infrequently** or that were **not present during training**. Possible strategies include:

- combining rare levels into an Other category;
- hashing categories;
- hierarchical grouping;
- learned embeddings; and
- explicit handling of unseen categories.

The appropriate method depends on the number of categories, their meaning, the model, and the expected behavior of the system after deployment

Example # 104: Customer loan default prediction with rare/unseen categories

Imagine an online bank building a model to predict whether a customer will **default on a loan**. One of the features is **employer**. During training, most customers work for common employers such as large banks, universities, or government departments. However, some employers appear only once or twice (Table 40).

Table 40: Illustrative customer-employer dataset for categorical encoding.

Customer	Employer	Default (1 = Yes, 0 = No)
A	National Bank	0
B	National Bank	0
C	National Bank	1
D	City University	0
E	City University	1
F	Small Company X	1
G	Small Company X	0
H	Small Company Y	1

Here, **Small Company X** occurs only twice and **Small Company Y** occurs only once. These are **rare categories**. If the model creates a separate representation for every employer, it may learn **unreliable patterns from categories with very few observations**. Small Company Y, for example, appears to have a 100% default rate, but that conclusion is based on only one customer.

Combining rare levels into Other: One simple solution is to define a minimum frequency, such as **5 observations**. Any employer appearing fewer than five times is replaced with Other.

Thus Small Company X → Other and Small Company Y → Other. The model no longer treats these tiny groups as completely separate categories. This **can reduce overfitting**, although it also **removes potentially useful differences** between the **rare employers**.

Hashing categories: Suppose the bank has **millions of different employers**. Creating a separate feature for every employer could be expensive. A hashing technique converts each employer name into **one of a fixed number of numerical *buckets***. For example, suppose there are 1,000 buckets:

- National Bank → Bucket 327
- City University → Bucket 814
- Small Company X → Bucket 192

The model therefore does not need to maintain a separate feature for every possible employer. Hashing is particularly useful when the number of categories is extremely large or when new categories are expected to appear frequently.

Hierarchical grouping: Sometimes categories have a **meaningful hierarchy**. Instead of treating every employer as completely independent, we can **group employers** according to **broader characteristics**. For example,

Small Company X → Technology company → Private sector → Employer category

The model could therefore use **information at several levels**:

- specific employer;
- industry;
- public/private sector;
- geographical region.

If a particular employer appears only twice, its **industry-level information** may still be reliable. This allows the model to **retain useful information** even when the specific category is rare.

Learned embeddings: For models such as neural networks, categories can be represented using **learned embeddings**. Instead of assigning each employer a simple one-hot value, the model learns a **numerical vector representing** each employer. For example,

- National Bank → $[0.21, -0.14, 0.73, \dots]$
- City University → $[0.18, -0.10, 0.69, \dots]$

Employers with **similar patterns** can potentially **develop similar representations**. Embeddings are especially useful when there are many categories and the model can learn meaningful relationships between them.

However, an employer that **appears only once** may **not provide enough information** to learn a reliable embedding, so **rare-category strategies may still be necessary**.

Explicit handling of unseen categories: Now suppose the model is deployed. A completely new employer appears: **Customer I → NewTech Solutions**. But NewTech Solutions did **not exist in the training data**.

The model cannot simply calculate a training-based statistic for this employer because there are **no training observations for it**. A robust system therefore needs a **predefined rule**. For example:

- NewTech Solutions → Unknown category or
- NewTech Solutions → global average / default representation

The important point is that the system should **know what to do with an unseen category before deployment** rather than failing or assigning an arbitrary value.

Putting the strategies together: The bank might therefore implement a policy summarized in **Table 41**.

The central issue is that **rare and unseen categories are not necessarily the same problem**. A rare category exists in the training data but has very little information, whereas an unseen category has **no training observations at all**.

3.5.7 Interaction features

An **interaction** occurs when the effect or predictive contribution of one variable depends on the value of another variable.

Table 41: Recommended strategies for handling employer categories under different data conditions.

For example, a linear model containing an interaction between x_1 and x_2 can be written as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$$

The partial effect of x_1 is: $\frac{\partial \hat{y}}{\partial x_1} = \beta_1 + \beta_3 x_2$. Thus, when $\beta_3 \neq 0$, the contribution of x_1 depends on x_2. Examples include:

- a medication effect depending on age;
- the effect of rainfall depending on soil type;
- the relationship between income and household expenditure depending on household size; and
- a sensor's response depending on operating temperature.

Interaction features can **incorporate important domain knowledge** and allow relatively **simple models** to represent **conditional relationships**. However, **adding many interactions** increases model **complexity** and the **risk of overfitting**. Interactions should therefore be **justified and validated** rather than added indiscriminately.

Example # 105: Predicting annual household expenditure

The infographic (Figure 53) shows how an interaction term between **annual income** and **household size** allows a linear model to represent situations in which the effect of one predictor depends on the level of another. In the example, adding the term $\mathbf{Income} \times \mathbf{HouseholdSize}$ changes the effect of income from a constant quantity to one that increases with household size, so households with the same income can have substantially different predicted expenditures. This highlights the essential idea of interaction: predictors may work **jointly rather than independently**, and when the interaction coefficient is nonzero, the model no longer assumes purely additive effects.

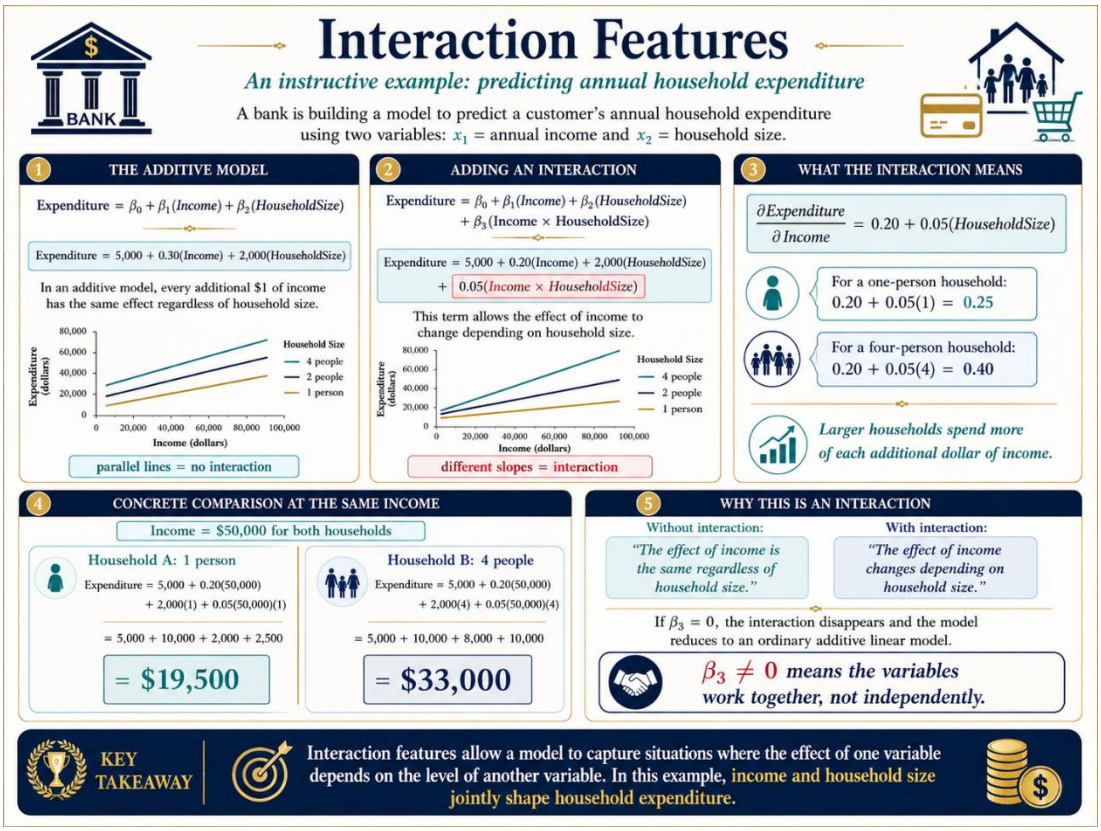

Figure 53: Interaction features illustrated through household expenditure prediction.

3.5.8 Polynomial features

Polynomial features allow models that are **linear in their parameters** to represent **certain nonlinear relationships**. For a single predictor:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_d x^d$$

For two predictors, second-degree features might include: $x_1, x_2, x_1^2, x_2^2, x_1 x_2$.

Advantages

Polynomial features provide:

- a relatively simple way to **approximate nonlinear relationships**;
- **interpretable nonlinear terms** at **low degrees**; and
- **compatibility** with ordinary linear estimation and regularization methods.

Example # 106: Spending and house-price relationships

The infographic (**Figure 54**) shows how a squared income term can capture diminishing growth in predicted spending, how multivariable polynomial expansions can include both curvature terms and interactions, and why model degree must be chosen carefully. Low-degree models may underfit, whereas unnecessarily high degrees can overfit noise; validation, cross-validation, and regularization can help control this trade-off.

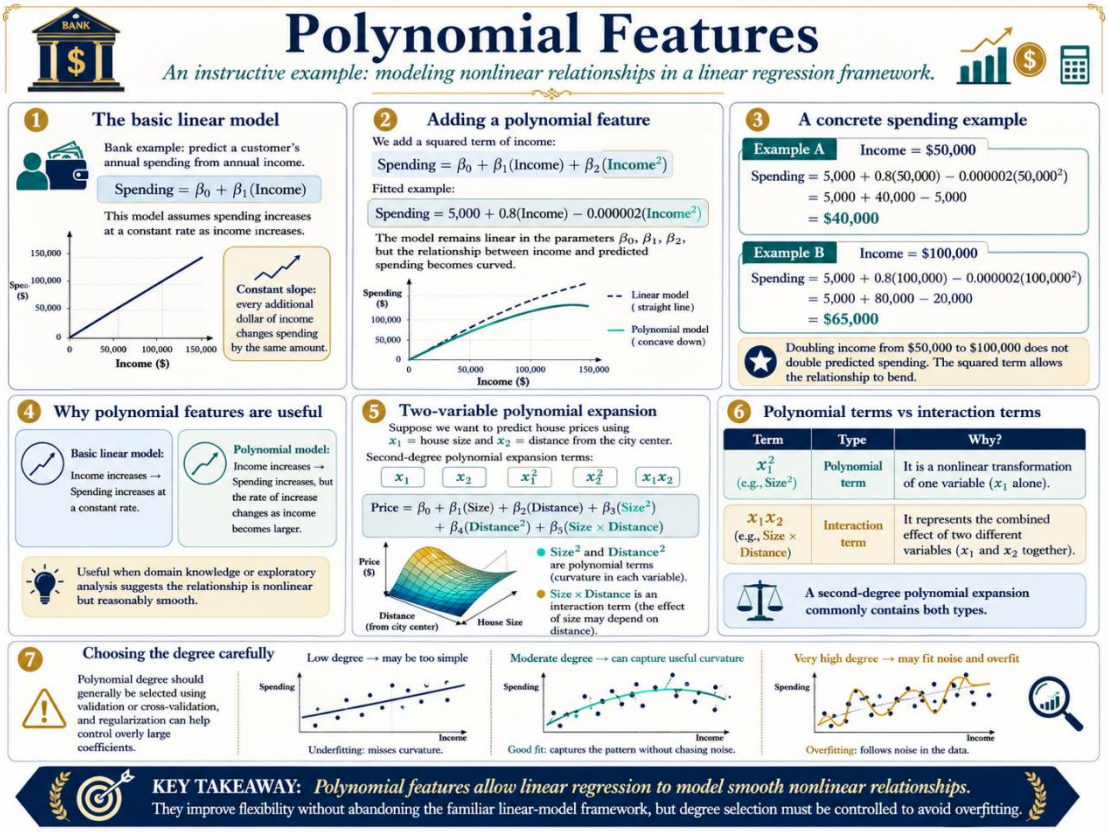

Figure 54: Polynomial features illustrated through nonlinear spending and house-price relationships.

Risks

They can also introduce:

- rapid growth in the number of features;
- multicollinearity;
- sensitivity to scale;
- unstable behavior outside the range of the training data; and
- overfitting at high polynomial degrees.

As the number of **original variables increases**, the **number of polynomial terms can grow rapidly**. Regularization and cross-validation are therefore particularly important when polynomial expansions are used.

Example # 107: Spending and house-price relationships

The infographic (**Figure 55**) illustrates these risks in a house-price prediction setting and emphasizes the role of appropriate scaling, validation, cross-validation, and regularization in selecting a polynomial representation that captures useful nonlinear structure without unnecessary complexity.

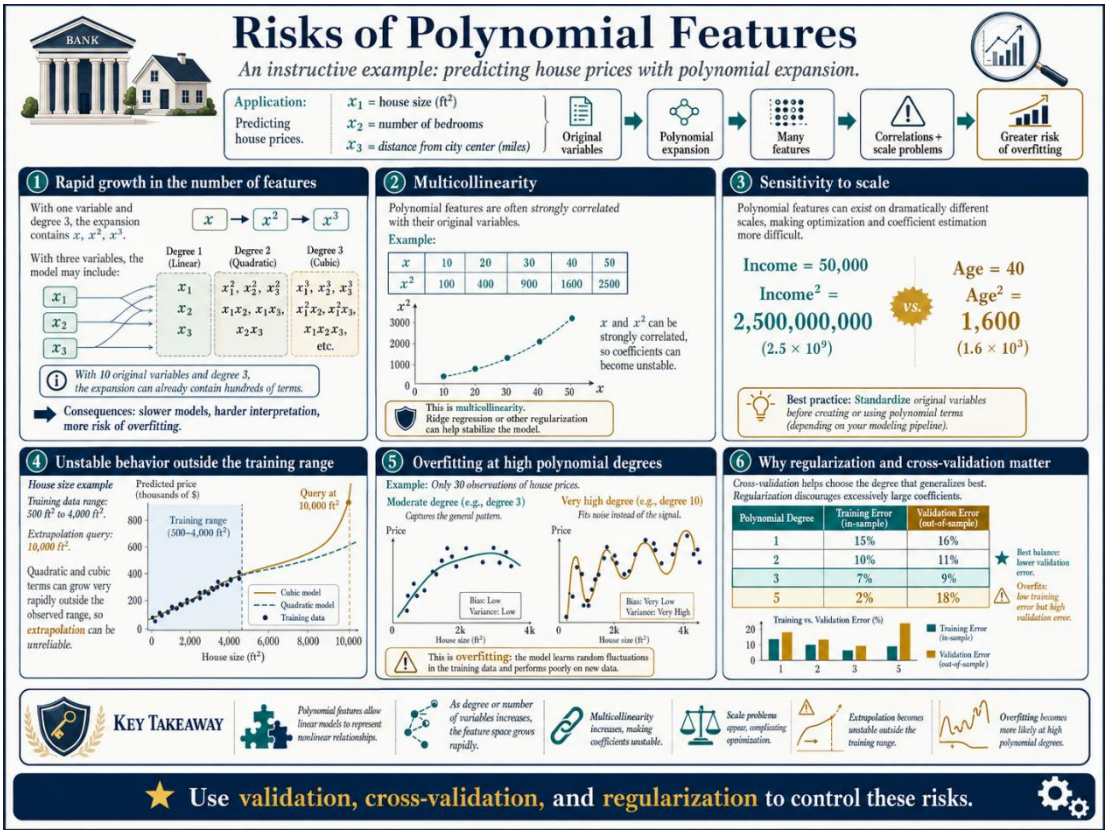

Figure 55: Risks associated with polynomial feature expansion.

3.5.9 Time-Window features

For temporal data, **individual observations** may **not contain enough information** to represent recent history. **Time-window features** summarize information from **previous observations** and can capture **persistence**, **trends**, **recent activity**, and **event history**. For a variable x_t, common features include:

- **Lag:** $L_k(t) = x_{t-k}$
- **Rolling mean:** $M_w(t) = \frac{1}{w}\sum_{j=1}^{w} x_{t-j}$
- **Rolling standard deviation:** $S_w(t) = \sqrt{\frac{1}{w-1}\sum_{j=1}^{w}\left(x_{t-j} - M_w(t)\right)^2}$
- **Difference:** $\Delta x_t = x_t - x_{t-1}$
- **Percentage change:** $r_t = \frac{x_t - x_{t-1}}{x_{t-1}}$
- **Event count:** $C_w(t) = \sum_{j=1}^{w} \mathbf{1}\{\text{event at } t-j\}$
- **Time since event:** $T_{\text{since}}(t) = t - t_{\text{last event}}$

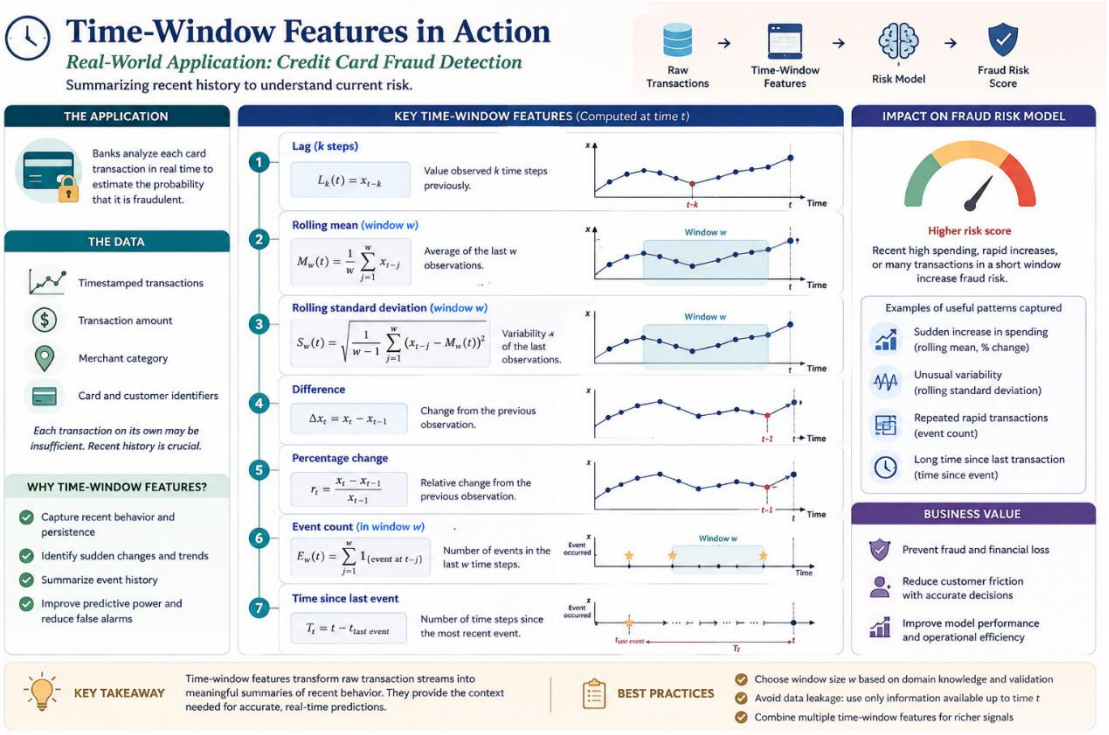

Figure 56: Time-window features illustrated through credit-card fraud detection.

Example # 108: Credit card fraud detection

The infographic (**Figure 56**) illustrates lag values, rolling means, rolling standard deviations, differences, percentage changes, event counts, and time

since the last event, showing how these features can capture persistence, trends, variability, rapid activity, and unusual behavioral changes. In predictive applications, such features should be computed using only information available up to the prediction time to avoid temporal leakage, while window sizes should be selected using domain knowledge and validation.

Calendar features

Depending on the application, temporal variables may also include:

- **Hour:** Extracts the time of day. For example, $7\ \text{PM} \rightarrow \text{hour} = 19$. A food-delivery model may learn that orders increase around dinner time.
- **Day of week:** Identifies the weekday. For example, $\text{Saturday} \rightarrow \text{day} = 5$[304]. A retail model may learn that sales are higher on weekends.
- **Month:** Identifies the month. For example, $\text{December} \rightarrow \text{month} = 12$. A retailer may learn that sales increase during the holiday season.
- **Holiday indicator:** Shows whether a date is a public holiday. For example, $\text{Holiday} \rightarrow 1$, otherwise 0. A transport model may learn that travel patterns differ on holidays.
- **Season:** Groups dates into seasons such as summer or winter. An electricity model may learn that energy consumption increases during very hot summers because of air conditioning.
- **Working-day indicator:** Shows whether the day is normally a working day. For example, $\text{Monday} \rightarrow 1$, $\text{Sunday} \rightarrow 0$. A commuting model may learn that demand is much higher on working days.

Example # 109: Timestamp to features

A timestamp such as **Saturday, 15 August, 7 PM** could be transformed into:

$$\textbf{Hour} = 19; \textbf{Day} = 5; \textbf{Month} = 8; \textbf{Holiday} = 0; \textbf{Season} = 2^{305}; \textbf{Working day} = 0$$

The critical requirement is **temporal availability**. A feature used to predict an event at time *t* must be calculated **only from information that would actually**

[304] Saturday → 5 because a **zero-based numbering system** is being used, where Monday = 0, Tuesday = 1, Wednesday = 2, Thursday = 3, Friday = 4, Saturday = 5, and Sunday = 6. This is a common convention in programming and machine-learning datasets, but it is not the only option; you could also use Monday = 1 through Sunday = 7. The important point is to use the same numbering consistently.

[305] Spring = 1, Summer = 2, Autumn = 3, Winter = 4.

have been available before the prediction was made. A rolling statistic that accidentally incorporates future observations creates **temporal leakage**. The resulting model may perform well during development while being impossible to reproduce in real-world prospective use.

Example # 110: 30-day hospital readmission prediction

Using 30-day hospital readmission prediction as an example, the infographic (**Figure 57**) contrasts a valid backward-looking feature computed only from information available at discharge with an invalid centered rolling statistic that incorporates future observations. Temporal leakage can produce artificially optimistic development performance while yielding features that cannot be reproduced prospectively at deployment. Reliable time-dependent modeling therefore requires strict alignment of every feature to the prediction time, backward-looking windows, chronological data splits, time-aware validation, and clear documentation of feature timestamps and availability.

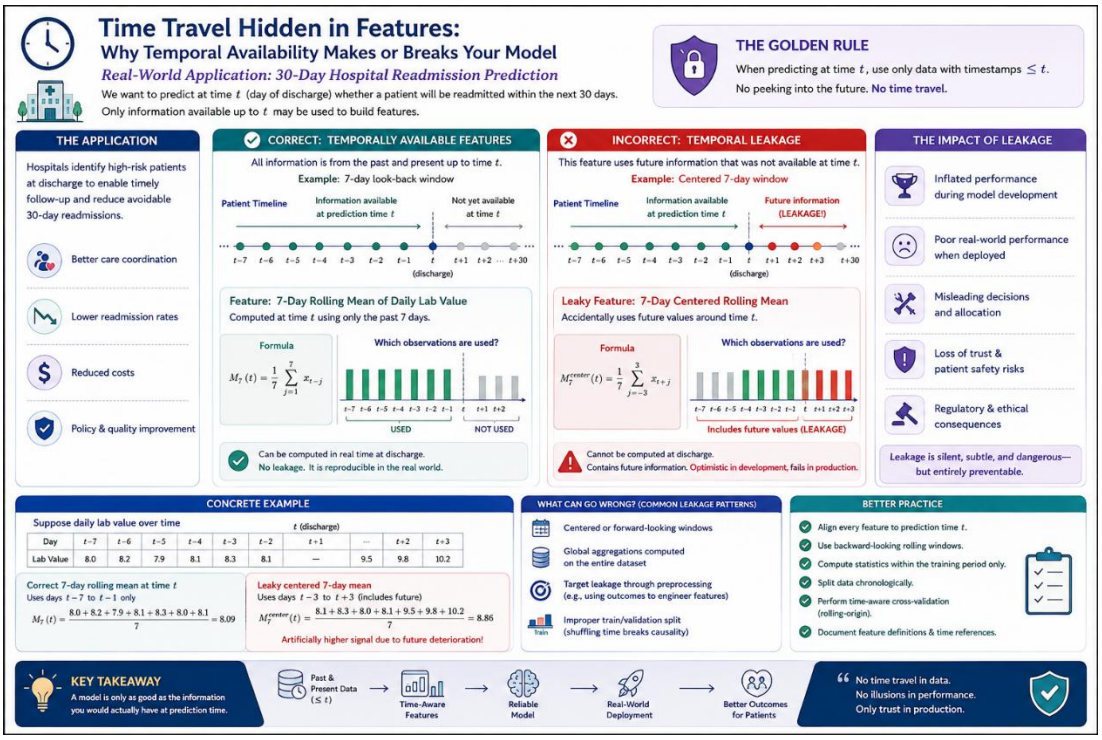

Figure 57: Temporal availability and the prevention of temporal leakage in predictive modeling.

3.5.10 Cyclical feature encoding

Many calendar variables have a **cyclical structure** that ordinary numerical encoding fails to represent correctly. For example, **hour 23** and **hour 0** are **adjacent in time**, but their numerical difference is $|23 - 0| = 23$. Treating hour simply as an integer therefore suggests that 23 and 0 are far apart, even though they are consecutive points on a 24-hour cycle. A common solution is to represent a cyclical variable using sine and cosine transformations:

$$x_{\sin} = \sin\left(\frac{2\pi x}{P}\right);\ x_{\cos} = \cos\left(\frac{2\pi x}{P}\right)$$

where P is the period. Examples include:

- For hour of day, $P = 24$
- For day of week, $P = 7$
- For month of year, $P = 12$

The resulting **two-dimensional representation** preserves **circular proximity**. Thus, the end and beginning of the cycle are represented as neighbors rather than as distant numerical values.

Example # 111: Real-world examples of cyclical feature encoding

Time of Day: Predicting Electricity Demand

Suppose a model predicts household electricity consumption based on the **hour of the day**.

- $23{:}00$ and $00{:}00$ are only **one hour apart** in reality.
- With ordinary numerical encoding, they appear far apart i.e. 23.
- Using a 24-hour period:

$$x_{\sin} = \sin\left(\frac{2\pi \times \text{hour}}{24}\right);\ x_{\cos} = \cos\left(\frac{2\pi \times \text{hour}}{24}\right)$$

The encoded values place 23:00 and 00:00 close together, allowing the model to recognize the continuity of the daily cycle (**Figure 58**).

Day of the Week: Predicting Restaurant Demand

Consider predicting the number of restaurant customers based on the **day of the week**.

- Sunday and Monday are consecutive days.
- Numerical encoding might represent them as 7 and 1, making them appear very distant.

- Using a period of $P = 7$:

$$x_{\sin} = \sin\left(\frac{2\pi \times \text{day}}{7}\right);\ x_{\cos} = \cos\left(\frac{2\pi \times \text{day}}{7}\right)$$

This represents **Sunday and Monday as neighboring points on a circle**, which better reflects the weekly pattern.

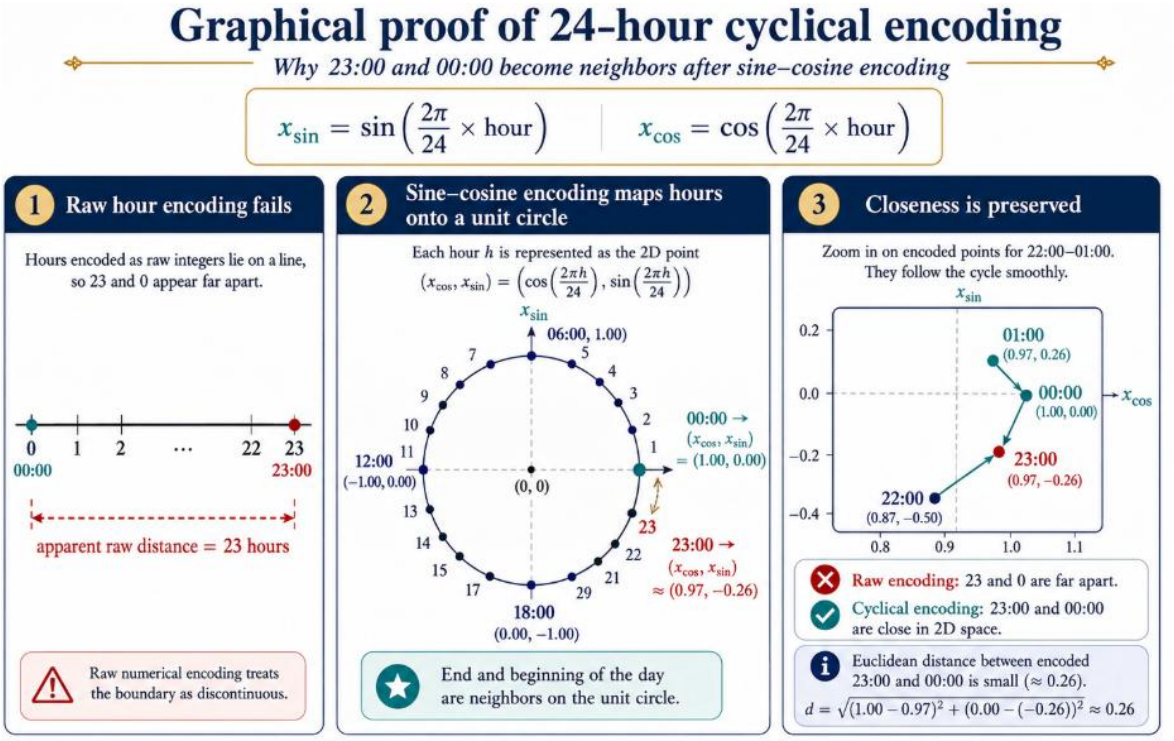

Figure 58: Graphical proof of 24-hour cyclical encoding using sine and cosine transformations.

Positional encoding in Transformers

Transformers process all **tokens in a sequence** simultaneously, so they need additional information to understand the **position or order** of each token. A common solution is to **add positional information** to each token embedding using sine and cosine functions:

$$\text{PE}(\text{pos}, 2i) = \sin\left(\frac{\text{pos}}{10000^{2i/d}}\right);\ \text{PE}(\text{pos}, 2i+1) = \cos\left(\frac{\text{pos}}{10000^{2i/d}}\right)$$

where pos is the **position of the token**, i is the **dimension index**, and d is the **embedding dimension**.

For example, consider the sentence: The cat chased the mouse. The words cat and mouse have **different meanings** depending **partly on where they occur**

in the sentence. Positional encoding assigns each token a **unique position-dependent pattern**:

- **The** → position 1
- **cat** → position 2
- **chased** → position 3
- **the** → position 4
- **mouse** → position 5

These positional encodings are **added to the corresponding word embeddings**. Consequently, the Transformer can **distinguish between sequences** such as The cat chased the mouse and The mouse chased the cat, even though both contain exactly the same words.

Thus, positional encoding provides Transformers with information about **where each token occurs in the sequence**, allowing the model to learn relationships that depend on word order.

3.5.11 Feature-engineering risks

Feature engineering can **substantially improve a model**, but it **can also introduce errors** that are difficult to detect if the feature-construction process is not carefully controlled.

The infographic (**Figure 59**) summarizes eight important sources of failure in engineered features: **target leakage**, **temporal leakage**, **identity leakage**, **proxy discrimination**, **unstable high-cardinality features**, **train-serving inconsistency**, **excessive dimensionality**, and **spurious interactions**. Together, these risks illustrate how seemingly useful features can undermine model validity, fairness, generalization, and deployment reliability.

For this reason, every engineered feature should have a clear specification documenting at least:

- its definition;
- its unit;
- its source variables;
- its computation window;
- its availability time;
- its missing-value rule;
- its transformation or processing version; and

- its substantive or modeling justification.

Figure 59: Major risks in feature engineering.

Feature engineering should ultimately be judged not only by whether it improves a validation score, but also by whether the feature is **valid**, **reproducible**, **interpretable where required**, **available at prediction time**, and **robust under realistic deployment conditions**. A feature that cannot be constructed reliably when the model is actually used is not a useful predictive feature, regardless of how well it performs during development.

Example # 112: Example on proxy discrimination

A lending model uses **ZIP code** as a feature when deciding whether to approve a loan. Although ZIP code does not explicitly contain a person's race, it **can strongly correlate** with **racial or socioeconomic composition**. The model may therefore make different decisions for applicants based on a sensitive characteristic indirectly represented by the feature.

Example # 113: Unstable high-cardinality features

An *e*-commerce model uses **customer ID** as a feature to predict whether a customer will purchase a product. Because most customer IDs appear only a few times in the training data (i.e., **each individual customer ID is present in only a small number of training examples**), the model may learn patterns specific to individual customers rather than general purchasing behavior. When new customers appear, these patterns provide little useful information and can lead to poor generalization.

Statistical Foundations

Statistical inference provides the framework for **moving from observed data to conclusions** about a **broader population** or **process**. The central challenge is that analysts rarely observe every unit of interest. Instead, they **work with a sample** and must **distinguish** what is **directly observed** from **what is being inferred**. This distinction introduces uncertainty, and sound statistical reasoning requires that the **target population**, **sampling process**, **estimator**, **sources of bias**, and **sources of variability** be made explicit. These foundations are especially important in EDA because descriptive patterns observed in a dataset do not automatically generalize beyond the population and conditions that produced the data.

3.6.1 Population versus Sample

A **population** is the complete collection of units, observations, or potential outcomes about which an analysis intends to make a statement.

A **sample** is the subset of that population that is actually observed and analyzed.

For example, if the goal is to understand the spending behavior of all customers using a service, all such customers constitute the target population, while the customers whose transactions are available in the dataset constitute the sample.

Population quantities are generally called **parameters**, whereas quantities calculated from the **observed sample** are called **statistics** (Figure 60).

The purpose of statistical inference is to **use sample information** to **learn about population parameters** while explicitly **accounting for uncertainty**.

This distinction is fundamental. A **statistic** describes the **data that were observed**; a **parameter** describes a **population quantity** that **may not be directly observable**. The quality of an inference therefore depends not only on how accurately the statistic was calculated, but also on **how the sample relates** to the **population** about which the conclusion is intended to apply.

Figure 60: Population parameters versus sample statistics across real-world settings.

3.6.2 The target population must be explicit

Before interpreting a statistical result, the analyst should explicitly define the **target population**. Possible target populations include:

- all customers currently using a particular product;
- future patients treated at a specified hospital;
- all adults in Pakistan;
- all devices produced by a particular manufacturing process; or
- future transactions occurring under the current fraud environment.

A dataset **can be highly representative of one population** while being **poorly representative of another**. For example, suppose a study analyzes records from patients who visited a particular hospital during a specified period. The data may support conclusions about:

- patients who visited that hospital during the study period,

but they do not automatically support conclusions about:

- all patients in the country;
- patients treated at rural hospitals;
- patients who never sought medical care; or
- future patients after clinical protocols or disease prevalence have changed.

Generalization is therefore not determined simply by sample size. It depends on the relationship between the **observed sample**, **the sampling mechanism**, and the **intended target population**.

A useful analysis should state explicitly: **Who or what does this dataset represent, and to whom or what is the conclusion intended to apply?**

Example # 114: Generalization in the real-world

Medical Research: A study tests a new treatment on **10,000 patients from one large urban hospital**. The sample size is large, but the results may not generalize to the entire population if the hospital mainly serves older or urban patients. The problem is not the sample size; it is that the **observed sample differs** from the **intended target population** because of how patients were sampled.

Election polling: A poll collects responses from **100,000 people who voluntarily complete an online survey**. Despite the huge sample, the results **may poorly predict** the views of all voters because people who choose to participate online may differ systematically from those who do not. A **smaller, carefully designed probability sample** could potentially generalize better.

Machine Learning: A bank trains a credit-risk model using **one million applications from existing customers**. The dataset is very large, but the model may perform poorly when applied to new customers from different regions or economic conditions. The training sample was generated by a particular **sampling mechanism**, people who already applied to or interacted with the bank, which may not represent the **target population** the model is intended to serve.

3.6.3 Sampling methods

The method used to **select observations** determines the **relationship between** the **sample** and the **population**. Different **sampling designs** provide different forms of **statistical information** and require different **analytical considerations**.

Simple random sampling

In **simple random sampling**, units are selected so that **each eligible population unit** has an **equal probability of selection**, under the standard design. This design provides a straightforward foundation for many classical statistical procedures because **selection is not deliberately concentrated** in **particular parts** of the population.

Example # 115: Randomly selecting university students to estimate study time

Suppose a university wants to estimate the **average number of hours students spend studying each week**. The university has a list of all **10,000 enrolled students**. It uses a computer to randomly select **500 students**, giving every student the same probability of being selected. For example,

- Student A → equal chance of being selected
- Student B → equal chance of being selected
- Student C → equal chance of being selected
- *and so on* for all 10,000 students.

The university **does not deliberately select** more students **from particular** years, departments, or faculties. The sample is simply drawn randomly from the entire eligible population.

Stratified sampling

In **stratified sampling**, the population is divided into **predefined (meaningful) groups**, or **strata**, and observations are **sampled within each stratum**. Strata might be defined by:

- geographic region;
- age group;
- industry;
- institution type; or
- another variable relevant to the sampling objective.

Stratification can ensure that **important population groups are represented** and can **improve statistical efficiency**[306] when the strata are meaningfully related to the outcome of interest.

Example # 116: Surveying customer satisfaction across different regions

A company wants to measure **customer satisfaction across the entire country**. Its customers are divided into predefined geographic strata:

- **North**
- **South**
- **East**
- **West**

Instead of randomly selecting customers from the country as a whole, the company **samples customers separately within each region**. For example, it might select 250 customers from each region, giving all four regions representation in the final sample.

Cluster sampling

[306] **Statistical efficiency** refers to **how much useful statistical information** an estimator **obtains from a given amount** of data. An estimator is **more efficient** if it achieves **smaller variance (or standard error)** than another estimator while **targeting the same quantity**.

In **cluster sampling**, **naturally occurring groups** or clusters are sampled rather than selecting individuals independently from the entire population. Examples of clusters include:

- schools;
- hospitals;
- villages;
- households; or
- manufacturing facilities.

Units may then be observed within the selected clusters. Because **observations within the same cluster may be correlated**, treating them as **completely independent** can **produce standard errors** that are too small. The analysis therefore needs to account for the clustering structure.

Example # 117: Patient outcomes across hospitals

A health researcher wants to estimate the **average recovery time after a particular surgery**. Instead of randomly selecting individual patients from hospitals across the country, the researcher randomly selects **20 hospitals** and collects data from patients within those hospitals. For example,

- Hospital A → 100 patients
- Hospital B → 100 patients
- Hospital C → 100 patients
- *and so on* for 20 hospitals.

This is **cluster sampling** because hospitals are treated as naturally occurring clusters, and patients are observed within the selected hospitals.

However, patients from the **same hospital may have more similar recovery outcomes** than patients from different hospitals. They may share the **same surgeons**, **treatment protocols**, **equipment**, **staffing levels**, and hospital **environment**.

For example, if Hospital A has an unusually effective recovery program, many of its 100 patients may recover faster than average. Those 100 observations therefore contain **less independent information than 100 completely unrelated patients**.

If a researcher incorrectly treats all 2,000 patients as fully independent, the calculated **standard error may be too small**, making the estimate appear more precise than it really is. The analysis should instead account for the **hospital-level clustering**, such as by using cluster-robust standard errors or a multilevel model.

Convenience sampling

In **convenience sampling**, observations are selected primarily because they are easily accessible. Examples include:

- surveying people who happen to be present at a particular location;
- analyzing readily available administrative records; or
- using volunteers recruited through an easily accessible online platform.

Convenience samples can be **useful for exploratory work**, but their **representativeness cannot be assumed**.

Example # 118: Surveying shoppers at a nearby mall

A retail company wants to understand **customer preferences for a new product**. Instead of randomly selecting customers from its entire customer base, the company surveys **100 people who happen to be shopping at a nearby mall** because they are easy to reach.

For example, the researcher might approach people who are available between 2 p.m. and 5 p.m. and ask them to complete a short questionnaire. This is **convenience sampling** because participants are selected primarily based on their **accessibility**, rather than through a random sampling procedure.

The results may be useful for **exploratory research** i.e., identifying possible product preferences or generating hypotheses. However, the findings may not represent the company's entire customer population. People shopping at that particular mall and during those particular hours could differ systematically from other customers.

Voluntary-response sampling

In **voluntary-response sampling**, individuals decide whether to participate. Participation may be related to **attitudes**, **experiences**, **outcomes**, or **levels of interest** in the subject being studied. Consequently, respondents can differ

systematically from nonrespondents. For example, a customer-feedback survey may disproportionately attract people with extremely positive or extremely negative experiences.

Example # 119: Online survey about public transportation

A city government wants to understand **residents' satisfaction with public transportation**. It posts an online survey on its website and social-media accounts and invites residents to participate.

People decide for themselves whether to complete the survey. Individuals who have had **very positive or very negative experiences** with public transportation may be especially motivated to respond, while people who are indifferent may simply ignore the survey. For example,

- A resident who recently experienced a major delay may be highly motivated to complain.
- A resident who is extremely satisfied with a new bus service may be motivated to praise it.
- A resident with little interest in public transportation may not respond at all.

This is **voluntary-response sampling** because participation is determined by the **individual's decision to participate**, rather than by a sampling procedure that gives people a controlled probability of selection.

Case-control sampling

In **case-control designs**, observations are **selected partly** according to **outcome status**. Cases have the outcome of interest, while controls do not.

This design can be highly useful when studying **rare outcomes** because researchers can **deliberately obtain sufficient numbers** of cases. However, because the sampling process depends on outcome status, the **proportion of cases** in the sample **does not generally represent** the **prevalence of the outcome** in the source population. The statistical analysis must therefore account for how the observations were selected.

Example # 120: Studying a rare disease

Researchers want to identify factors associated with a **rare type of cancer**. In the general population, only a very small proportion of people develop this

cancer, so randomly selecting people from the population might produce very few cancer cases. Instead, the researchers deliberately select:

- **200 patients** who have the cancer as **cases**
- **200 similar people** who do not have the cancer as **controls**

They then **compare characteristics** such as previous exposures, occupations, or lifestyle factors between the two groups.

This is a **case-control design** because individuals are selected partly according to their **outcome status**. The design is useful because researchers can **deliberately obtain enough cancer cases** to study, even though the cancer is rare in the source population.

However, the sample contains **50% cases** and **50% controls**, even though the cancer may occur in far less than 1% of the population. Therefore, the **50% case proportion** in the study **does not represent** the cancer's **prevalence** in the population. If researchers ignored this sampling process and treated the sample as though it were a random sample of the population, they **could draw incorrect conclusions** about population-level quantities.

3.6.4 Sampling bias

Sampling bias occurs when the **mechanism determining** whether a unit enters the observed dataset is **systematically related** to the **quantity being studied**. Let

$$R_i = \begin{cases} 1, & \text{unit } i \text{ is included} \\ 0, & \text{otherwise} \end{cases}$$

If inclusion is **systematically related** to the outcome or another important characteristic, then the observed sample may differ from the target population. For example, if inclusion depends on Y, conceptually: $\mathrm{Corr}(R, Y) \neq 0$, then the sample mean **need not provide an unbiased estimate** of the population mean. Examples include:

- online polls that exclude people without internet access;
- medical datasets containing only patients who sought care;
- customer surveys dominated by highly satisfied or highly dissatisfied users;
- sensor datasets that exclude devices that failed completely; and

- social-media studies whose participants represent platform users rather than the general population.

Sampling bias is a form of **systematic error**. **Increasing** the number of **observations** can **reduce random sampling variability**, but it does not necessarily remove systematic selection bias.

This leads to an important distinction: **A large sample can provide a very precise estimate of the wrong population quantity**. Therefore, sample size **should never be treated as a substitute** for appropriate sampling or careful definition of the target population.

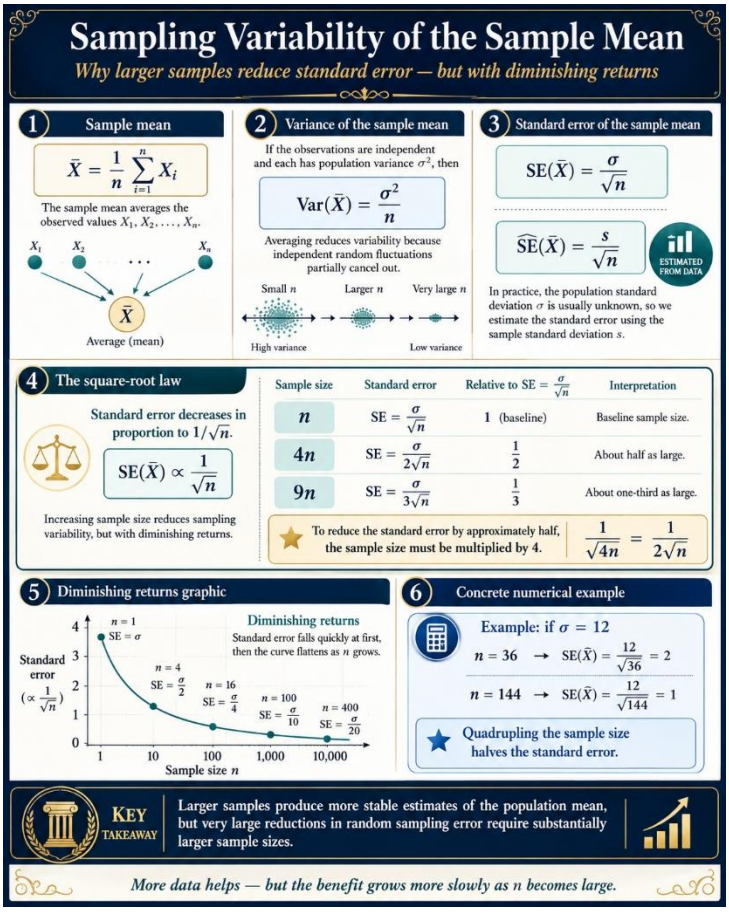

Figure 61: Sampling variability and standard error of the sample mean.

3.6.5 Sampling variability

Even when sampling is appropriately designed, **different samples** drawn from the **same population** will generally produce **different estimates**. This phenomenon is called **sampling variability**. Figure 61 shows that the

standard error of the sample mean decreases according to the **square root law**, so progressively larger increases in sample size are required to achieve further reductions in sampling variability.

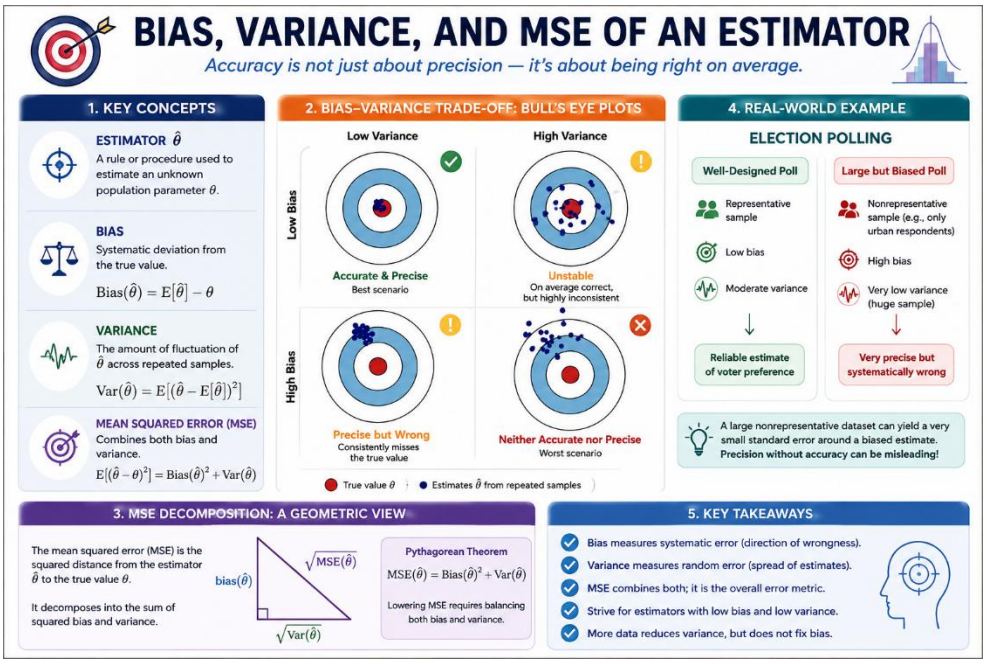

Figure 62: Bias, variance, and mean-squared error illustrated through bull's-eye plots.

3.6.6 Bias and variance of an estimator

An **estimator** is a **rule** or **procedure** used to **estimate an unknown** population parameter. Let $\hat{\theta}$ be an estimator of the population parameter θ.

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta; \text{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$$

Bias describes **systematic deviation from the target**, whereas **variance** describes **how much** the estimator would **fluctuate across repeated samples** (Figure 62). The **mean squared error (MSE)** can be decomposed[307] as:

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \text{Bias}(\hat{\theta})^2 + \text{Var}(\hat{\theta})$$

[307] **Proof** is as follows:

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \mathbb{E}[\hat{\theta}^2 - 2\theta\hat{\theta} + \theta^2] \overset{\text{Expectation is linear}}{=} \mathbb{E}[\hat{\theta}^2] - 2\mathbb{E}[\theta\hat{\theta}] + \mathbb{E}[\theta^2] \overset{\theta \text{ is fixed}}{=} \mathbb{E}[\hat{\theta}^2] - 2\theta\mathbb{E}[\hat{\theta}] + \theta^2$$

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \mathbb{E}[\hat{\theta}^2] - 2\theta\mathbb{E}[\hat{\theta}] + \theta^2 + \mathbb{E}[\hat{\theta}]^2 - \mathbb{E}[\hat{\theta}]^2 = \left(\mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2\right) + \left(\mathbb{E}[\hat{\theta}]^2 - 2\theta\mathbb{E}[\hat{\theta}] + \theta^2\right)$$

$$\mathbb{E}\left[(\hat{\theta} - \theta)^2\right] = \left(\mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2\right) + \left(\mathbb{E}[\hat{\theta}] - \theta\right)^2 \overset{\text{by definition}}{=} \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2$$

This decomposition is important because an **estimator with low variance** is **not necessarily accurate**. An estimator can produce highly consistent results while being consistently wrong if it has substantial bias.

The **same principle applies to datasets**. A **very large nonrepresentative** dataset can produce a **very precise estimate** (i.e., a small estimated standard error) while **remaining systematically biased** for the intended target population.

Example # 121: Estimating average income from a large online dataset

Suppose a researcher wants to estimate the **average annual income of adults in a country**. They obtain income data from **10 million users of a professional networking platform**.

Because the dataset is enormous, the estimated average income may have a **very small standard error**. The estimate is statistically precise because there are millions of observations.

However, the dataset may be **systematically unrepresentative** of the target population. People who use a professional networking platform may be more likely to be employed in certain occupations, have higher levels of education, or work in formal sectors.

For example, suppose the dataset produces: **Estimated average income = $60,000, with a very small standard error**. The small standard error means the researcher has estimated the **average income of this particular sample very precisely**. It does **not** mean that $60,000 is necessarily a good estimate of the average income of all adults in the country.

If the true population average were $40,000, the huge dataset could still produce a highly precise estimate around $60,000 because of **systematic sampling bias**. The key idea is that **Precision and representativeness are different concepts**.

3.6.7 Sources of uncertainty

Uncertainty in statistical analysis has **many sources**, and **ordinary sampling variability** is only one of them. Important sources include:

- **Finite sampling:** only a subset of the target population is observed;

- **Measurement error:** recorded values differ from the quantities of interest;
- **Missing data:** some observations or variables are unavailable;
- **Label uncertainty**: outcome labels may be noisy, ambiguous, or based on imperfect proxies;
- **Parameter estimation:** model parameters are estimated rather than known;
- **Model specification:** the chosen model may not adequately represent the underlying process;
- **Feature selection:** the selected predictors may depend on data-driven decisions;
- **Hyperparameter selection:** tuning decisions introduce additional uncertainty;
- **Distribution shift:** the population or data-generating process may change over time;
- **Analyst decisions:** preprocessing, exclusions, transformations, and modeling choices can influence results; and
- **Unknown causal structure:** observed associations may have several competing explanations.

These sources of uncertainty do not necessarily appear in a conventional standard error. For example, a **confidence interval** based on a fixed statistical model **may account** for **sampling variability conditional on that model**, while **ignoring uncertainty** introduced by **trying many models**, **selecting variables** after examining the data, or **changing the analysis** specification.

Example # 122: Sources of uncertainty in real-world examples

Finite sampling in estimating average blood pressure
A researcher wants to estimate the **average blood pressure of all adults in a country** but can measure only 5,000 people. Even if the measurements are perfectly accurate and the sample is randomly selected, the 5,000 people will **not exactly match the entire population**. A different random sample could produce a somewhat different estimate.
Remark: Uncertainty arises because we observe only a **subset of the target population**.

Measurement error in measuring house prices

A researcher studies the relationship between **house size** and **property value** using recorded house prices. However, some recorded prices may contain errors: for example, a transaction price might be entered incorrectly, or the recorded property size may differ from the actual size. The analysis is therefore uncertain because the **observed values** are **imperfect measurements** of the **quantities of interest**.
Remark: The data themselves may not measure the underlying quantities perfectly.

Missing data in patient follow-up
A clinical study follows 1,000 patients after treatment, but 150 patients do not return for their final assessment. The researcher therefore does not know the final outcomes for those 150 patients. If patients who fail to return are systematically different from those who do, the missing data can affect the estimated treatment effect.
Remark: Uncertainty arises because some relevant observations are **unavailable**.

Label uncertainty in medical diagnosis
An AI system is trained to identify pneumonia from medical images. The training labels come from doctors' diagnoses. However, doctors may sometimes disagree about whether an image shows pneumonia. Some cases may be ambiguous, and the diagnosis may depend on additional information that is not available in the image. The model is therefore being trained using labels that are **not perfectly reliable**.
Remark: The outcome labels themselves can contain **noise or ambiguity**.

Parameter estimation in predicting house prices
A regression model predicts house prices using size, location, and number of bedrooms: $\text{Price} = \beta_0 + \beta_1(\text{Size}) + \beta_2(\text{Bedrooms}) + \beta_3(\text{Location}) + \epsilon$. The coefficients $\beta_0, \beta_1, \beta_2, \beta_3$ are unknown and must be estimated from the available data. A different sample could produce somewhat different coefficient estimates.
Remark: Model parameters are **estimated from data**, so the estimated parameters themselves are uncertain.

Model specifications in predicting hospital readmission

A hospital builds a model to predict whether a patient will be readmitted within 30 days. The researcher chooses logistic regression, assuming a particular relationship between the predictors and readmission risk. But the real process might involve nonlinear relationships or interactions: for example, age might matter differently for patients with different chronic conditions. If the chosen model does not adequately represent the underlying process, predictions and conclusions can be uncertain.
Remark: Uncertainty can arise because the **model itself may be an imperfect representation of reality**.

Feature selection in predicting employee turnover
A company has **500 potential predictors** of employee turnover. An analyst examines the data and selects the 20 variables that appear most predictive. However, the selected variables depend on the particular dataset and the selection approach. With another dataset (and/or another approach), a different set of variables might have been selected.
Remark: When predictors are chosen using the data, the **selection process itself introduces uncertainty**.

Hyperparameter selection in training an image classifier
A machine-learning researcher trains an image classifier and must choose the **learning rate**, **tree depth**, **regularization strength**, and **number of training iterations**. They try many combinations and select the one that performs best on a validation set. The final model therefore depends partly on the tuning decisions made during development. A different validation sample could lead to different choices.
Remark: Model performance can depend on **data-driven tuning decisions**, creating additional uncertainty.

Distribution shift in fraud detection
A bank trains a fraud-detection model using transactions from 2025. In 2026, fraudsters change their behavior and begin using new techniques. The distribution of transaction patterns is therefore different from the distribution used to train the model. A model that performed very well originally may perform substantially worse on the new data.
Remark: Uncertainty arises because the **data-generating process can change over time**.

Analyst decisions in estimating the effect of a treatment

Suppose an analyst studies whether a treatment improves recovery time. Before fitting the model, they must decide:

- which patients to exclude;
- how to handle missing values;
- whether to transform recovery time;
- which variables to adjust for; and
- which statistical model to use.

Different reasonable choices may produce different results.

Remark: Results can depend on **analytical choices made during the research process**, not only on sampling variability.

Unknown causal structure in education and income

A researcher observes that people with **more years of education tend to have higher incomes**. However, several explanations are possible. Education may increase income, but other factors, such as family background, ability, location, or social networks, may influence both education and income. Therefore, the observed association does not by itself establish the causal effect of education on income.

Remark: When the underlying causal structure is uncertain, the same observed association may be consistent with **multiple competing explanations**.

Therefore, a reported confidence interval should not automatically be interpreted as a complete measure of all uncertainty surrounding a scientific or predictive conclusion. A rigorous analysis should distinguish among at least three questions:

- How much would the estimate vary if we repeated the sampling process?
- How much uncertainty is introduced by measurement, modeling, and analytical decisions?
- How likely is the estimated relationship to remain valid when the data-generating environment changes?

Recognizing these distinctions is essential for **interpreting statistical evidence responsibly**. Statistical uncertainty is not merely a property of a

formula; it is a consequence of the entire process through which data are **generated**, **sampled**, **measured**, **processed**, and **analyzed**.

Example # 123: Predicting hospital readmissions
Suppose researchers develop a model to estimate the **risk that a patient will be readmitted to a hospital within 30 days**. The model reports that patients with a particular risk factor have a **20% higher estimated readmission risk**, with a 95% confidence interval of $15\% - 25\%$.

It would be tempting to conclude that the confidence interval captures all of the uncertainty surrounding this finding. However, it **addresses only a particular source of uncertainty**. The researchers should consider at least three questions:

- **Sampling variability:** If the researchers repeated the study with a different sample of similar patients, how much would the estimated 20% effect change?
  Remark: The confidence interval provides **information related to this question** under the **assumptions of the statistical procedure**.
- **Measurement, modeling, and analytical uncertainty:** Are patients' diagnoses and readmission outcomes measured accurately? Would the estimated relationship change if a different model, set of predictors, or missing-data procedure were used?
  Remark: The confidence interval **does not automatically capture** all of these possibilities.
- **Changing environment:** Will the relationship remain valid next year if treatment protocols, patient populations, or hospital practices change?
  Remark: A confidence interval **calculated** from **today's data cannot guarantee** that the **relationship will remain stable** under future conditions.

For example, the researchers might obtain a narrow confidence interval of $15\% - 25\%$, suggesting high statistical precision. Yet a change in treatment practices could cause the true relationship to weaken or disappear in the future.

Confidence Intervals

Confidence intervals provide a principled way to express **uncertainty about an unknown population parameter**. Rather than reporting **only a point estimate**, such as a sample mean or regression coefficient, an **interval estimate provides a range of values** generated by a specified statistical procedure. The **interpretation** of that range **depends critically** on the **inferential framework** and its assumptions. In the classical frequentist framework, confidence intervals are understood through their **long-run coverage properties**: the procedure is designed so that, over repeated samples generated under the same conditions, a specified proportion of the intervals will contain the parameter of interest. In the **Bayesian framework,** uncertainty is expressed through the **posterior distribution**[308] **of the parameter.** A 95% credible interval indicates that, given the observed data and prior assumptions, the parameter has a 95% posterior probability of lying within the interval.

3.7.1 What is a confidence interval?

Suppose θ is an unknown population parameter and a statistical procedure produces a lower limit $L(X)$ and an upper limit $U(X)$ from observed data X. A $100(1-\alpha)\%$ **confidence interval** is constructed so that, under the assumptions of the procedure,

$$P_\theta\big(L(X) \le \theta \le U(X)\big) = 1 - \alpha$$

where α is the **significance level** or **error rate** and $1-\alpha$ is the confidence level. The probability statement refers to the **procedure across repeated samples**, not to the fixed parameter changing randomly from one sample to another. For example, a **95% confidence procedure** is designed to have **approximately 95% coverage** in **repeated applications** under the conditions for which the procedure is valid.

The framework of confidence intervals is closely associated with the work of **Jerzy Neyman**, who formalized confidence intervals in terms of **repeated-**

[308] **Posterior distribution:** The probability distribution of an unknown parameter **conditional on the observed data**, obtained by updating its prior distribution using the likelihood: $\underbrace{p(\theta|\mathcal{D})}_{\text{posterior}} \propto \underbrace{p(\mathcal{D}|\theta)}_{\text{likelihood}}\ \underbrace{p(\theta)}_{\text{prior}}$. For example, after observing 7 heads and 3 tails, the posterior distribution of a coin's head probability θ with a $\text{Beta}(1,1)$ prior is $\theta|D \sim \text{Beta}(8,4)$.

sampling coverage and confidence procedures in his 1937 paper[309] on the theory of confidence intervals. The essential idea is that the interval-producing procedure has a known long-run performance under repeated sampling.

3.7.2 Point estimates versus interval estimates

A **point estimate** provides a single value: $\hat{\theta}$. A **confidence interval** provides an interval: $[L(X), U(X)]$. For example, an estimated population mean might be reported as $\hat{x} = 72.4$ whereas an **interval estimate** might be between $[69.1, 76.3]$. The **point estimate** communicates the **central estimate**, while the **interval** communicates how much **sampling uncertainty** is associated with the **estimation procedure**.

A confidence interval therefore should not be viewed simply as a statement that *the parameter is somewhere between these two numbers*. Its meaning comes from the **procedure that generated the interval** and the **assumptions** under which its coverage property holds.

3.7.3 Confidence interval for a mean

Suppose observations are **approximately independent** and arise from a population for which a **normal-based mean procedure** is appropriate, but the **population variance is unknown**. A commonly used confidence interval for the population mean μ is:

$$\bar{x} \pm t_{1-\alpha/2,n-1} \frac{s}{\sqrt{n}}$$

where $\bar{x}$ is the sample mean, s is sample standard deviation, n is the sample size, $t_{1-\alpha/2,n-1}$ is the appropriate critical value from the t-distribution[310], $n-1$ is the corresponding degrees of freedom, and $1-\alpha$ is the confidence level. The interval can be **verbally written** as:

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[309] **Jerzy Neyman** (1894-1981) was a Polish-American mathematician and statistician who founded the modern theory of statistical inference.

J. Neyman, *Outline of a theory of statistical estimation based on the classical theory of probability*, Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences 236, No. 767, 1937.

[310] ***t*-distribution:** A symmetric, bell-shaped probability distribution used to estimate a population mean when the population standard deviation is unknown, with heavier tails than the normal distribution; e.g., for $n = 10$, use a t-distribution with $n - 1 = 9$ degrees of freedom.

$$\text{estimate} \pm \text{critical value} \times \text{standard error}$$

Thus:

$$\text{confidence interval} = \text{point estimate} \pm \text{margin of error}$$

where margin of error is given by $t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}$.

Interpretation

Consider a 95% confidence interval for a population mean.

The **frequentist interpretation** is as follows:

If the entire process of sampling observations and constructing the confidence interval were repeated many times under the same conditions, approximately 95% *of the resulting intervals would contain the true population mean*.

For example, suppose repeated samples produce the following intervals: $[48, 52], [51, 55], [49, 53], [54, 58], [50, 54], \cdots$. The confidence level describes the **long-run proportion of intervals that successfully cover the fixed parameter**. It is therefore not strictly correct, under the classical frequentist interpretation, to say: *There is a* 95% *probability that the true mean is inside this particular interval*. Once the data have been observed and the interval calculated, the parameter is treated as fixed. The interval either contains that parameter or it does not. The $\mathbf{95\%}$ **probability** refers to the performance of the **interval-generating procedure over repeated samples**.

From a **Bayesian perspective**, uncertainty is expressed in terms of a **posterior distribution** for the population mean rather than the long-run coverage of a confidence-interval procedure. After observing the data and combining them with a prior distribution, a Bayesian analysis can produce a 95% **credible interval** for μ. This interval can be interpreted as:

Given the observed data, the chosen prior distribution, and the statistical model, there is a 95% *posterior probability that the true population mean lies between* 48 *and* 52.

Thus, while a frequentist 95% confidence interval describes the long-run performance of an interval-generating procedure, a Bayesian 95% credible

interval directly describes the posterior uncertainty about the parameter, conditional on the observed data and prior assumptions.

3.7.4 Factors affecting interval width

The **width of a confidence interval** reflects the amount of uncertainty represented by the procedure. For the mean interval,

$$\bar{x} \pm t_{1-\alpha/2,n-1}\frac{s}{\sqrt{n}}$$

the **width depends** primarily on the **confidence level**, **variability**, **sample size**, and **assumptions** about the **sampling process**.

Higher confidence levels

Increasing the confidence level $(1-\alpha)$ **decreases the significance level** α, thereby **increasing the critical *t*-value**. Because this value multiplies the standard error, a **larger critical *t*-value** directly **expands the margin of error**.

From a frequentist perspective, a **95% confidence interval procedure** is designed so that, under repeated sampling and its underlying assumptions, **approximately 95% of the intervals constructed** will capture the true parameter μ. **Increasing the confidence level** requires a procedure with **greater long-run coverage**, which results in a **wider interval**.

Similarly, from a Bayesian perspective, a **higher credibility level** requires a **wider credible interval** to encompass a **larger probability mass** of the **posterior distribution**. For example, a 95% credible interval contains 95% posterior probability for the parameter, conditional on the observed data and the specified model.

Greater variability

Greater sample variability increases the standard error: $\mathrm{SE}(\bar{x}) = \frac{s}{\sqrt{n}}$. Therefore, greater variability generally produces a **wider interval**.

Smaller sample sizes

Because $\mathrm{SE}(\bar{x}) \propto \frac{1}{\sqrt{n}}$ **smaller samples** generally produce **larger standard errors** and **wider confidence intervals**. Conversely, increasing n generally narrows the interval, although the reduction occurs at a square-root rate.

Dependence and clustering

The simple formula $\frac{s}{\sqrt{n}}$ implicitly relies on an **appropriate sampling** and **dependence structure**. If observations are strongly correlated (e.g., several measurements come from the same patient) the effective amount of independent information may be considerably smaller than the raw number of observations suggests.

Consequently, treating $10{,}000$ **correlated observations** as though they were $10{,}000$ **independent observations** can produce confidence intervals that are **artificially narrow**. Appropriate methods may need to account for:

- **Clustering:** Groups observations within naturally occurring clusters and adjusts uncertainty for within-cluster similarity; e.g., students sampled from the same schools tend to be correlated.
- **Repeated measurements:** Accounts for multiple observations taken from the same individual or unit; e.g., blood pressure measured several times for each patient.
- **Stratification:** Divides the population into predefined subgroups (strata) and accounts for the sampling within each subgroup; e.g., sampling separately by age group or gender.
- **Survey weights:** Assigns observations different weights to reflect unequal selection probabilities or to make the sample representative of the target population; e.g., giving under-sampled rural respondents greater weight.
- **Temporal dependence:** Accounts for correlation between observations collected close together in time; e.g., daily temperature measurements, where today's value is related to yesterday's value.
- **Other sampling-design features:** Uses methods that reflect additional complexities of how data were collected; e.g., multistage sampling or finite-population sampling.

Rather than assuming that all n observations provide **independent information**, they **adjust the estimated standard error** to reflect clustering,

repeated measurements, survey design, temporal dependence, or unequal sampling probabilities. Consequently, the **margin of error** $t_{1-\alpha/2,n-1} \times \mathrm{SE}$ **may become larger** (or, in some designs, smaller)[311], producing a confidence interval that more accurately represents the true uncertainty in $\bar{x}$. The sample mean $\bar{x}$ itself may also be **reweighted or otherwise estimated differently** when survey weights or complex sampling designs are involved.

Narrowness does not guarantee validity

A narrow confidence interval indicates relatively high **precision under the assumptions of the procedure**. It does not, by itself, establish that the estimate is accurate. For example, if a sample is **systematically biased** because of **nonrepresentative selection**, increasing the sample size can produce a very narrow interval around a biased estimate.

Thus, **precision does not imply validity**. For example, if a survey consistently samples only high-income people, increasing the sample from 100 to $10{,}000$ may produce a very narrow confidence interval, but the estimate can still be far from the true population value because the sample is not representative.

A confidence interval **cannot automatically correct** for a **flawed sampling mechanism**, **severe measurement bias**, or an **inappropriate model**.

3.7.5 Bootstrap confidence intervals

The **bootstrap**[312] is a computational approach for **approximating the sampling distribution** of a **statistic**[313] using **repeated resampling** from the **observed dataset**. Suppose the observed sample is:

$$X = (x_1, x_2, x_3, \cdots, x_n)$$

For $b = 1, \cdots, B$:

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[311] **May become larger** when observations provide less independent information (e.g., repeated measurements from the same patient), **or smaller** when the sampling design increases efficiency (e.g., stratifying a population into internally similar groups), thereby respectively increasing or reducing the standard error.

[312] The **bootstrap** is a **computational resampling** method introduced by **Bradley Efron (1979)** that approximates the sampling distribution of a statistic by repeatedly drawing samples with replacement from the observed dataset; examples include estimating the standard error of a mean, constructing confidence intervals for a median, and assessing uncertainty in regression coefficients or correlation coefficients.

B. Efron, *Bootstrap methods: Another look at the Jackknife*, The Annals of Statistics, Vol 7, No. 1, pp 1-26, 1979.

[313] A **statistic** is a **numerical quantity** computed from **sample data** to **summarize a characteristic** of the sample; examples include the sample mean, sample median, sample variance, sample proportion, and correlation coefficient.

- Draw a bootstrap sample of size n **with replacement** from the observed data.
- Calculate the statistic of interest, denoted by $\hat{\theta}^{*(b)}$.
- **Repeat** the process **many times** to obtain an **empirical bootstrap distribution**: $\hat{\theta}^{*(1)}, \hat{\theta}^{*(2)}, \cdots, \hat{\theta}^{*(B)}$.

The resulting distribution **approximates how the statistic** might vary under repeated sampling.

Percentile bootstrap interval

A **percentile bootstrap confidence interval** is constructed by taking the appropriate **lower** and **upper quantiles** directly from the bootstrap distribution of the estimated statistic. It is given by

$$[Q_{\alpha/2}(\hat{\theta}^*), Q_{1-\alpha/2}(\hat{\theta}^*)]$$

where $Q_q(\hat{\theta}^*)$ denotes the q-quantile of the bootstrap distribution and $\hat{\theta}^*$ represents the bootstrap estimates obtained from repeated resampling. For a **95% percentile** bootstrap confidence interval, we set $\alpha = 0.05$, so the interval is approximately the 2.5^{th} percentile to the 97.5^{th} percentile of the bootstrap estimates. Thus, the **middle** 95% of the bootstrap distribution is **retained as the estimated confidence interval**, while the lowest and highest 2.5% are excluded.

Why the bootstrap is useful

The bootstrap can be particularly useful when the sampling distribution of a statistic is difficult to derive analytically. It can be applied to statistics such as:

- means;
- medians;
- quantiles;
- correlations;
- regression coefficients;
- prediction measures; and
- other complicated statistics.

However, the **validity of a bootstrap** procedure **does not follow** simply from **repeatedly resampling** the observed data. The resampling scheme **must be**

consistent with the way the original data were generated, particularly with respect to **any dependence**, **clustering**, **pairing**, **ordering**, or **other sampling structure** present in the data. For example,

- **Independent data:** When observations are independent, the standard bootstrap i.e., resampling individual observations with replacement is generally appropriate.
- **Time-series data:** Since **observations collected close** together in time **may be correlated**; randomly resampling individual observations would break this temporal dependence, so a **block bootstrap**, which **resamples consecutive groups of observations**, may be more appropriate.
- **Clustered data:** If data are collected from (say) students within schools, students from the same school may be more similar to one another than to students from other schools. Resampling individual students independently could therefore underestimate uncertainty, whereas resampling entire schools or appropriately accounting for the clustered structure preserves the dependence.
- **Paired data:** For paired data, such as before-and-after measurements on the same individuals, the two measurements must remain paired during resampling rather than being sampled separately.

Thus, a bootstrap method should be designed to **mimic the relevant features of the original sampling process**; otherwise, the resulting bootstrap distribution, standard errors, or confidence intervals may give a **misleading representation of statistical uncertainty**.

Respect the unit of sampling

A fundamental principle of bootstrap resampling is that the **resampling unit should correspond to the unit at which the data were sampled or at which the relevant dependence structure exists**. Simply resampling every recorded observation independently can be inappropriate when several observations belong to the same underlying sampling unit.

For example, suppose a medical dataset contains **2,000 patients** and **50,000 radiological images**. If each patient contributes multiple images, the 50,000 images **cannot generally be treated** as 50,000 **independent sampling units**

because **images from the same patient are likely to be correlated**. Independently resampling individual images could therefore break the patient-level dependence and make the effective sample size appear much larger than it actually is. A more **appropriate bootstrap** would typically **resample patients**, with all relevant images belonging to a selected patient retained together within each bootstrap sample. The same principle applies to other forms of structured data:

- **Household data:** If households are the sampling units, resample households rather than individual household members. All members of a selected household should remain together.
- **Clustered school data:** If schools constitute the relevant sampling units, resample schools rather than treating individual students as independent observations.
- **Repeated-measures data:** When the same subject is measured multiple times, resample at the subject level or otherwise preserve the dependence among measurements from the same subject.
- **Time-series data:** Because observations close together in time may be dependent, independently resampling individual observations can destroy the temporal structure. A block bootstrap or another appropriate dependent-data bootstrap should be used instead.
- **Paired data:** When observations naturally occur in pairs, such as before-and-after measurements on the same individual, the pair should generally be resampled as a single unit rather than resampling the two measurements independently.

Hence, the **bootstrap resampling unit** should reflect the **relevant data-generating process**, **sampling design**, and **dependence structure** of the original data.

Bootstrap is not a cure for biased data

Bootstrap resampling is a method for quantifying uncertainty, but it **does not automatically correct problems or biases** in the underlying data. If the observed dataset is **systematically unrepresentative** or **contains errors**, repeatedly resampling that same dataset will generally reproduce those problems rather than remove them.

- **Sampling bias**, where the observations included in the dataset are systematically different from the target population;
- **Measurement bias**, where variables are measured inaccurately or systematically in one direction;
- **Nonrepresentative samples**, where the observed sample does not adequately reflect the population of interest;
- **Systematic missingness**, where missing observations are related to characteristics of the observations or subjects;
- **Data leakage**, where information that would not be available at the time of prediction is inadvertently incorporated into the analysis; or
- **Distribution shift**, where the distribution of future or target data differs from the distribution represented in the observed dataset.

For example, suppose a survey is conducted using a sample that substantially **underrepresents older adults**. A bootstrap procedure can repeatedly resample from that dataset and provide an estimate of the uncertainty of the sample mean, but **it cannot make the sample representative of the full population**. Similarly, if a measurement instrument **systematically overestimates a quantity**, bootstrap resampling will **reproduce the biased measurements** rather than correct the measurement error.

Thus, the bootstrap should primarily be understood as a tool for **quantifying sampling uncertainty** conditional on the **observed data**, the **statistic being estimated**, and the **chosen resampling scheme**. It does not replace careful study design, appropriate sampling, reliable measurement, suitable handling of missing data, or methods for addressing systematic sources of bias. In short, the bootstrap **can quantify uncertainty around the information contained in the data**, but it **cannot manufacture information** or **remove systematic problems** that are already present in the data-generating process.

Hypothesis Testing

Hypothesis testing provides a **formal framework** for **using sample data** to **evaluate claims about a population** or **data-generating process**. The central idea is to specify a **null hypothesis** representing a **reference condition** and an **alternative hypothesis** representing a **departure from that condition** and then assess how compatible the observed data are with the null hypothesis. This assessment is typically based on a **test statistic**, its **null distribution**, and the resulting p**-value**, while also considering **effect sizes**, **confidence intervals**, **assumptions**, **study design**, and **statistical power**. Importantly, hypothesis testing does not by itself establish that a hypothesis is true or false; rather, it **provides a structured way** to **quantify evidence** against **a specified null model** while making the uncertainty and limitations of the analysis explicit.

3.8.1 Null and alternative hypotheses

A hypothesis test begins by **formulating two competing statements** about a population parameter or other quantity of interest. These statements are typically called the **null hypothesis** and the **alternative hypothesis**.

The **null hypothesis** specifies a reference value or condition that is tested:

$$H_0: \theta = \theta_0$$

The **alternative hypothesis** represents the **departure from the null hypothesis** that the analysis is designed to detect. For a **two-sided test**, the alternative is:

$$H_1: \theta \neq \theta_0$$

This asks whether the **parameter differs from the null value in either direction**. For a **one-sided test**, the alternative may instead specify a particular direction, for example:

$$H_1: \theta > \theta_0$$

when the scientific question concerns whether the **parameter is greater than the reference value**.

Common examples include:

- $H_0: \mu_A = \mu_B$ for comparing two population means;
- $H_0: p_A = p_B$ for comparing two population proportions; and
- $H_0: \rho = 0$ for testing whether the population correlation is zero.

The hypotheses should ideally be **specified before examining the results**, together with the **outcome**, **analysis population**, **statistical test**, **significance level**, and **relevant analysis assumptions**. Pre-specification helps reduce the risk of choosing hypotheses or analytical methods after seeing the data in ways that could produce misleadingly favorable conclusions.

3.8.2 Test statistic

A **test statistic** is a numerical quantity calculated from the observed data that **measures how far the data are from what would be expected** under the null hypothesis. A common standardized form is:

$$T = \frac{\text{estimate} - \text{null value}}{\text{standard error}}$$

The **numerator** measures the **observed departure from the null value**, while the **denominator** expresses that **departure relative to the estimated sampling variability**.

For example, when **testing a population mean** against a **specified value** μ_0, the **one-sample *t*-statistic** is:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

where $\bar{x}$ is the sample mean, s is the sample standard deviation, and n is the sample size.

The **null distribution** of the test statistic describes the values of T that would be expected if the null hypothesis were true and the assumptions of the statistical model were appropriate. The **observed test statistic** is then compared with this distribution to assess how unusual the observed result would be under H_0.

3.8.3 The p-value

A **p-value** is the **probability**, calculated **under the null hypothesis** and the **specified statistical model**, of obtaining a **test statistic at least as incompatible with the null hypothesis** as the value actually observed. For a two-sided test, this can be written as:

$$p = P_{H_0}(|T| \geq |T_{\text{observed}}|)$$

Thus, a small p-value indicates that the observed data would be relatively unusual under the null hypothesis and the assumptions of the model.

Importantly, a p-value is **not the probability that the null hypothesis is true**. It is a probability about the data, or more precisely, about the test statistic **conditional on the null hypothesis and the statistical model**.

The American Statistical Association has emphasized that p-values **should be interpreted** as **measures** of the **compatibility** or **incompatibility of the observed data** with a **specified statistical model**, rather than as definitive measures of whether a hypothesis is True or False. Statistical conclusions should therefore **not depend solely** on whether a p-value crosses an arbitrary threshold such as 0.05.

A p-value should generally be considered alongside the **estimated effect size**, **confidence interval**, **sample size**, **study design**, **assumptions**, **data quality**, and **scientific context**.

3.8.4 What a p-value does not mean

A p-value of 0.03 does not mean that:

- there is a 3% probability that H_0 is true;
- there is a 97% probability that H_1 is true;
- there is a 3% probability that the observed result occurred *by chance*;
- the result is scientifically or practically important;
- the study has a 97% probability of replicating;
- the statistical model is correctly specified; or
- the observed association is necessarily causal.

For example, suppose a study reports $p = 0.03$ for a difference between two treatment groups. This **does not mean** that there is a 3% probability that there is no treatment difference. Rather, it means that **if the null hypothesis and the specified model were correct**, results **at least as extreme** as the observed result would occur with probability 0.03 under the relevant testing procedure.

A small p-value can arise for several different reasons, including:

- a genuinely large and scientifically meaningful effect;
- a very small effect detected with a very large sample;
- incorrect or violated model assumptions;
- biased sampling or measurement;
- unmodeled dependence between observations;
- selective reporting or analysis choices; or
- multiple hypothesis tests without appropriate adjustment or interpretation.

Consequently, the magnitude of a p-value alone does not establish the **size**, **importance**, **validity**, or **causal interpretation** of an observed effect.

3.8.5 Type I and Type II errors

Hypothesis testing involves two **fundamental types of decision errors** summarized in Table 42.

Table 42: Decision outcomes in classical hypothesis testing. The table summarizes the four possible outcomes obtained by comparing the statistical decision with the true state of nature: a **Type I error** occurs when H_0 is rejected although it is true (α), whereas a **Type II error** occurs when H_0 is not rejected although H_1 is true (β). The corresponding correct decisions occur with probabilities $(1-\alpha)$ and $(1-\beta)$, with $(1-\beta)$ representing the **statistical power** of the test.

Reality	Reject H_0	Do not reject H_0
H_0 is true (Null hypothesis is correct)	**Type I error** (False positive) Conclude H_1 when H_0 is true Probability: α	**Correct decision** (True negative) Correctly retain H_0 Probability: $1-\alpha$
H_1 is true (Alternative hypothesis is correct)	**Correct detection** (True positive) Correctly reject H_0 Probability: $1-\beta$ (Power)	**Type II error** (False negative) Fail to reject H_0 when H_1 is true Probability: β

A **Type I error** occurs when the **null hypothesis is rejected even though it is true**. Its probability is represented by:

$$\alpha = P(\text{reject } H_0 \mid H_0 \text{ true})$$

The value of α is the **Type I error rate** or **significance level** of the testing procedure. A commonly chosen value is 0.05, although other levels may be appropriate depending on the context.

A **Type II error** occurs when the **null hypothesis is not rejected even though the relevant alternative is true**. Its probability is:

$$\beta = P(\text{do not reject } H_0 \mid H_1 \text{ true})$$

The **complement of the Type II error** probability is the **statistical power**:

$$\text{Power} = 1 - \beta$$

Power is the probability that a **specified effect will be detected** by the statistical test **under the assumed conditions**. Power depends on several factors, including:

- **Effect size:** larger effects are generally easier to detect;
- **Sample size:** larger samples generally provide greater precision;
- **Variability:** greater random variability generally makes effects harder to detect;
- **Significance level α:** changing the rejection threshold changes the probability of detection;
- **Study design and efficiency:** efficient designs can provide more information for a given sample size; and
- **Measurement quality:** noisy or unreliable measurements can reduce the ability to detect real effects.

Failure to reject H_0 does not demonstrate that H_0 is true. A **non-significant result may occur** because the **null hypothesis is a reasonable description** of the underlying population, but it **may also occur** because the study has **insufficient sample size**, **substantial variability**, **imprecise measurements**, or otherwise **inadequate power to detect** the effect of interest. Therefore, a non-significant result should be **interpreted as insufficient evidence** to reject

the null hypothesis, **rather than as proof** that the null hypothesis has been established.

Example # 124: Medical diagnostic test

Suppose a new test is being evaluated for detecting a disease (**Table 43**).

- H_0: The patient **does not have the disease**.
- H_1: The patient **has the disease**.

Table 43: Type I and Type II errors illustrated through medical diagnosis.

Reality (True patient condition)	**Test rejects H_0** (Test says disease is present)	**Test does not reject H_0** (Test says disease is absent)
Patient is healthy (No disease)	**Type I error** Test says disease is present (false positive)	**Correct decision** Test correctly says disease is absent (true negative)
Patient has disease (Disease is present)	**Correct detection** Test correctly says disease is present (true positive)	**Type II error** Test fails to detect disease (false negative)

Type I Error

A healthy patient is incorrectly classified as having the disease. For example, if $\alpha = 0.05$ then, under the testing procedure, there is a 5% probability of rejecting H_0 when H_0 is actually true.

Type II Error

A patient who actually has the disease receives a negative result. If $\beta = 0.20$ then the probability of failing to detect the disease under the specified alternative is 20%. Therefore, $\text{Power} = 1 - \beta = 0.80$. So the test has **80% power** to detect the specified disease effect **under the assumed conditions**.

Example # 125: New teaching method

A university wants to determine whether a new teaching method improves students' average exam scores.

- H_0: The new method **does not improve** the average score.
- H_1: The new method **improves** the average score.

Suppose the new method really does improve scores, but the study finds a non-significant result. This is a **Type II error** since the study failed to reject H_0 even though the relevant alternative was true. For example, $\beta = 0.25$ means there is a 25% probability of failing to detect the specified improvement under the assumed conditions. Thus $\text{Power} = 1 - \beta = 0.75$. The study has **75% power** to detect that specified improvement.

Importantly, the **non-significant result does not prove** that the teaching method has no effect. The study might simply have had **too few students** or **too much variation** in exam scores.

Example # 126: Quality control in a factory
Suppose a factory produces bottles that should contain exactly $500\ \text{mL}$ of liquid.

- H_0: The production process is **correctly centered** at $500\ \text{mL}$.
- H_1: The production process is **not correctly centered** at $500\ \text{mL}$.

Type I Error
The production process is actually correctly centered, but the statistical test concludes that it is defective. This could cause the factory to **unnecessarily stop production** or investigate a process that is actually functioning properly.

Type II Error
The production process is actually producing bottles incorrectly, but the test fails to detect the problem. This could allow defective production to continue.

This example illustrates that **both errors can have practical consequences**, and the appropriate significance level depends partly on the consequences of the decisions.

Example # 127: Effect of sample size on power
Suppose researchers want to determine whether a new drug reduces blood pressure.

With a **small sample**, random variation may be large relative to the treatment effect. The study may fail to detect a real reduction. For example, $\beta = 0.40$ gives $\text{Power} = 1 - \beta = 0.60$.

With a **large sample**, the estimate becomes more precise. Suppose this reduces the Type II error probability to $\beta = 0.10$. Then $\text{Power} = 1 - \beta = 0.90$. Thus, **increasing sample size** generally **increases the ability** to detect a specified effect.

Example # 128: Why a non-significant result does not prove H_0
Imagine a researcher tests:

- $H_0: \mu = 100$.
- $H_1: \mu \neq 100$.

The test produces $p = 0.18$. At the 5% significance level, $0.18 > 0.05$, so the researcher **does not reject** H_0.

It would be **incorrect to conclude** that the test proved that $\mu = 100$.

The **appropriate conclusion** is that there is **insufficient evidence to reject** H_0 at the 5% significance level.

The result **could reflect a genuine lack of effect**, but it could also **reflect a small sample**, **high variability**, **noisy measurements**, or **insufficient statistical power**.

Effect Size and Practical Significance

Statistical analysis should not stop at determining whether an observed result is statistically detectable. An important complementary question is **how large the observed effect is** and **whether that magnitude is meaningful in the context of the application**. **Effect sizes** provide a quantitative description of the **magnitude of differences or associations**, while **confidence intervals** communicate the **uncertainty** surrounding those estimates. **Practical significance** then places the **estimated effect in its substantive context** by considering factors such as clinical or scientific relevance, costs, risks, benefits, and implementation constraints. Together, these concepts help **distinguish statistical evidence from substantive importance**, providing a more informative interpretation of empirical results than a p-value alone.

3.9.1 What is an effect size?

An effect size quantifies the **magnitude of a difference**, **relationship**, or **association observed in the data**. Whereas a hypothesis test primarily addresses whether the data provide evidence against a specified null hypothesis, an **effect size describes how large the observed effect** is on a **scientifically or practically interpretable scale**.

Effect sizes can be expressed in their original measurement units or on standardized scales. Common examples include:

- **Mean difference**, which measures the difference between two group means;
- **Standardized mean difference**, which expresses a mean difference relative to a measure of variability;
- **Risk difference**, which measures the absolute difference in probabilities of an outcome;
- **Risk ratio**, which compares the probabilities of an outcome between two groups;
- **Odds ratio**, which compares the odds of an outcome between two groups;
- **Correlation**, which describes the strength and direction of a statistical association between two variables;

- **Regression coefficients**, which quantify the expected change in an outcome associated with a change in a predictor, conditional on the model; and
- **Changes in prediction error**, which quantify how much predictive performance changes between competing models or interventions.

For example, a study might find that a treatment lowers average blood pressure by 4 mmHg, that the risk of an adverse event decreases by 2% points, or that two variables have a correlation of $r = 0.35$. These quantities describe the magnitude of the observed relationship more directly than a p-value does.

Effect sizes should ordinarily be **reported together** with **confidence intervals** or **other measures of uncertainty**. The **effect estimate describes** the **observed magnitude**, while the **interval communicates** the **precision** with which that magnitude has been estimated[314].

3.9.2 Common effect size measures

Effect size measures quantify the **magnitude** and **direction** of an **observed difference or association**. Unlike a p-value, which addresses the strength of evidence against a specified null model, an effect size describes **how large the observed effect is on a particular measurement scale**. The appropriate measure depends on the type of outcome, study design, and scientific question. Common measures include mean differences, standardized mean differences, risk differences, risk ratios, odds ratios, and correlations.

Mean difference

For two groups, the **difference in sample means** is: $\Delta = \bar{x}_1 - \bar{x}_2$. For example, if the mean recovery time is 8 days in one group and 10 days in another, the estimated mean difference is: $\Delta = \bar{x}_1 - \bar{x}_2 = 8 - 10 = -2 \text{ days}$. The negative sign indicates that the first group has a lower mean recovery time by 2 days, given the direction in which the difference was defined.

Standardized mean difference

[314] G. M. Sullivan and R. Feinn, *Using Effect Size – or Why the P Value Is Not Enough*, Journal of Graduate Medical Education, 4(3), pp 279-282, 2012.

When measurements are on a scale where a **unit difference is difficult to interpret** or when **comparisons across studies are desired**, the mean difference can be standardized:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_{\text{pooled}}}$$

Here, s_{pooled} is a pooled estimate of the within-group standard deviation[315]. The resulting quantity expresses the difference in standard deviation units.

Risk difference

For a binary outcome, the risk difference is:

$$\text{RD} = P(Y = 1|T = 1) - P(Y = 1|T = 0)$$

For example, if an outcome occurs in 8% of a treatment group and 12% of a comparison group: $\text{RD} = 0.08 - 0.12 = -0.04$. This corresponds to an **absolute risk reduction** of 4% for the direction defined above.

Risk ratio

The risk ratio is:

$$\text{RR} = \frac{P(Y = 1|T = 1)}{P(Y = 1|T = 0)}$$

Using the same example, $\text{RR} = \frac{0.08}{0.12} \approx 0.67$. Thus, the estimated risk in the treatment group is approximately 0.67 times the risk in the comparison group.

Odds ratio

The odds ratio is:

$$\text{OR} = \frac{p_1/(1 - p_1)}{p_0/(1 - p_0)}$$

[315] s_{pooled} is the **pooled within-group standard deviation**, combining the variability of the two groups into a single estimate:

$$s_{\text{pooled}} = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

For example, If Group 1 has $n_1 = 20$, $s_1 = 4$ and Group 2 has $n_2 = 30$, $s_2 = 5$, then $s_{\text{pooled}} \approx 4.63$, meaning the standardized mean difference is measured relative to a typical within-group spread of about 4.63 units.

Unlike the risk ratio, the odds ratio **compares odds rather than probabilities**[316]. It is particularly common in logistic regression and case-control studies.

Correlation

The Pearson correlation coefficient is:

$$r = \frac{\mathrm{Cov}(X, Y)}{s_X s_Y}$$

It describes the direction and strength of a **linear association** between two variables, with values ranging from -1 to 1. A value near 1 indicates a strong positive linear association, a value near -1 indicates a strong negative linear association, and a value near 0 indicates little linear association.

The **interpretation of an effect size** depends on its **measurement scale** and **scientific context**. There is no universal numerical threshold at which an effect becomes *important* across all fields. For this reason, effect sizes should **generally be interpreted together** with their **uncertainty**, **measurement scale**, **study design**, and **substantive context**.

3.9.3 Statistical significance versus practical significance

Statistical significance and practical significance address different questions.

Statistical significance asks: **Are the observed data sufficiently inconsistent with the specified null model to provide evidence against it?**

Practical significance asks: **Is the magnitude of the observed effect large enough to matter in the real-world context of the problem?**

These questions need not have the same answer.

For example, suppose an online platform has millions of users and introduces a change that increases average user engagement by only 0.2 seconds per user. With a sufficiently large sample, the estimated effect could produce $p < 10^{-8}$.

[316] **Odds** are the **ratio of the probability** that an **event occurs** to the **probability that it does not occur**, $\text{odds} = p(1-p)$. For example, if $p = 0.20$, the odds are $0.20/0.80 = 0.25$, or $1:4$ (one occurrence for every four non-occurrences).

The result may therefore provide strong statistical evidence against a null model of exactly zero effect. However, the statistical evidence does not determine whether an increase of 0.2 seconds is operationally meaningful. The answer could depend on whether the change affects advertising revenue, server costs, user satisfaction, or other business outcomes.

Assessment of practical significance should consider factors such as:

- **Domain-specific thresholds for meaningful change:** A 1-point reduction in blood pressure may be statistically detectable but may be considered too small to justify a change in clinical practice.
- **Costs associated with implementing an intervention:** A new teaching program might increase students' test scores by 2%, but the improvement may need to be weighed against substantial training and staffing costs.
- **Benefits expected from the observed effect:** A manufacturing intervention that reduces defects by 0.5% could still be practically important if it saves millions of dollars annually.
- **Potential risks or harms:** A medication that produces a small improvement in symptoms may have limited practical value if it also causes frequent or serious side effects.
- **Implementation difficulty and feasibility:** An intervention may have a meaningful estimated effect but be difficult to implement because it requires specialized equipment, extensive training, or major organizational changes.
- **Size and characteristics of the affected population:** A small effect affecting millions of people may have substantial aggregate consequences, whereas the same effect in a very small population may have limited overall impact.
- **Available alternative interventions:** An intervention producing a modest benefit may be less practically significant if a simpler, cheaper intervention produces a similar or greater benefit.
- **Uncertainty in the estimated effect:** An estimated 5% improvement may appear meaningful, but if the confidence interval ranges from a 1% decrease to an 11% increase, the practical importance of the effect remains uncertain.

A useful interpretation therefore considers the **effect size and its uncertainty**, rather than treating the p-value as a measure of importance. A statistically detectable effect can be too small to matter in practice, while an important effect can remain statistically uncertain when the available data are limited.

Common Analytical Mistakes

Analytical mistakes can **arise at any stage of an AI study**, from **defining hypotheses** and **collecting data** to **selecting models**, **evaluating results**, and **interpreting associations**. Some errors produce **misleading statistical significance**, while others create **biased estimates**, **overstate** model performance, or lead to causal conclusions that the **evidence cannot support**. Many of these problems **do not require intentional misconduct**; they can emerge naturally from **repeated experimentation**, **undisclosed analytical choices**, **inappropriate evaluation procedures**, or **misunderstandings** about what the data can establish. Recognizing these **recurring failure modes** is therefore essential for producing results that are transparent, reproducible, and scientifically defensible.

3.10.1 p-Hacking

p-hacking refers to **exploiting undisclosed analytical flexibility** until a **desirable** or **statistically significant result** is obtained. The issue is not necessarily **intentional misconduct**; researchers may make **seemingly reasonable decisions** during analysis that, when combined, **increase the probability of obtaining a false-positive finding**. Common examples include:

- stopping data collection once $p < 0.05$;
- trying multiple definitions of the outcome;
- excluding observations after examining the results;
- testing numerous subgroups;
- selecting covariates based on statistical significance;
- repeatedly transforming variables;
- reporting only analyses that produce favorable results;
- switching from two-sided to one-sided tests after seeing the results;
- presenting an exploratory hypothesis as though it had been specified in advance.

Simmons, Nelson, and Simonsohn[317] demonstrated that **ordinary analytical flexibility** can **substantially increase** the probability of **false-positive findings**, even when researchers do not fabricate or manipulate data.

Prevention

Several practices can reduce the risk of p-hacking:

- preregister hypotheses and analysis plans;
- clearly distinguish exploratory from confirmatory analyses;
- report all outcomes and exclusions;
- disclose decisions that were made after examining the data;
- use holdout or independent validation data;
- conduct sensitivity analyses;
- report null findings as well as statistically significant findings;
- replicate results using new data;
- report effect sizes and confidence intervals alongside p-values.

3.10.2 Multiple Comparisons

Suppose m independent null hypotheses are tested at a significance level of $\alpha = 0.05$. The probability of obtaining **at least one false positive** is:

$$1 - (1 - \alpha)^m$$

For $m = 20$, $1 - (1 - 0.05)^{20} \approx 0.642$. Thus, even when every null hypothesis is true, **at least one apparently significant result will occur** in approximately 64% of such **families of independent tests**.

Bonferroni correction

The Bonferroni procedure controls the **family-wise error rate** (FWER) by **adjusting the significance level** for each individual test. If m hypotheses are being tested and the desired overall significance level is α, the adjusted significance level is:

$$\alpha_{\text{individual}} = \frac{\alpha}{m}$$

[317] J. P. Simmons, L. D. Nelson, and U. Simonsohn, *False-positive psychology: undisclosed flexibility in data collection and analysis allows presenting anything as significant*, Psychological science, 22 (11), pp 1359-1366, 2011.

Each hypothesis is then tested using this smaller significance level. For example, if $\alpha = 0.05$ and 5 hypotheses are tested, the significance level for each test becomes $0.05/5 = 0.01$.

The Bonferroni correction is **simple**, **easy to apply**, and **widely applicable** because it **does not require the tests to be independent**. However, it **can be conservative**, particularly **when many hypotheses are tested**. This may **reduce the statistical power** of the analysis and increase the likelihood of failing to detect genuinely significant effects (**Type II errors**). Hence, the basic tradeoff is:

$$\begin{cases} \text{Smaller } \alpha \Rightarrow \text{fewer false positives} \\ \text{Smaller } \alpha \Rightarrow \text{lower statistical power} \Rightarrow \text{more false negatives} \end{cases}$$

Example # 129: Conservativeness of Bonferroni correction
Suppose a researcher tests 20 different hypotheses using an overall significance level of $\alpha = 0.05$. **Without** any correction, each test would normally use $\alpha = 0.05$. **With** the Bonferroni correction, the significance level for each test becomes $\alpha_{\text{individual}} = \frac{0.05}{20} = 0.0025$.

Now imagine one of the tests produces a p-value of 0.01. **Without correction**, this result would be considered statistically significant because $0.01 < 0.05$. However, after the Bonferroni correction $0.01 > 0.05$ so the result is **not considered statistically significant**.

This illustrates why **Bonferroni can be conservative**: as the **number of hypotheses increases**, the **required significance level** becomes **increasingly smaller**. While this helps protect against false positives (Type I errors), it can also **make it harder to detect real effects**, increasing the risk of **Type II errors**.

False discovery rate

Benjamini and Hochberg[318] proposed controlling the **false discovery rate** (FDR) which is the **expected proportion of false discoveries** among the **hypotheses rejected as significant**.

First, sort the p-values: $p_{(1)} \leq p_{(2)} \leq p_{(3)} \leq \cdots \leq p_{(m)}$. For a desired FDR level q, find the largest k satisfying $p_{(k)} \leq \frac{k}{m} q$. Reject the hypotheses corresponding to $p_{(1)}, \cdots, p_{(k)}$. FDR control can provide **greater statistical power** than strict family-wise error control when many hypotheses are examined.

Example # 130: Applying the Benjamini–Hochberg Procedure to Control FDR

Suppose a researcher conducts **10 hypothesis tests** to identify genes that may be associated with a particular disease. The resulting p-values are:

$$0.001, 0.008, 0.012, 0.020, 0.035, 0.041, 0.070, 0.120, 0.300, 0.600$$

The researcher wants to control the false discovery rate (FDR) at $q = 0.05$.

Step 1: Sort the p-values: The values are already sorted i.e., $p_{(1)} \leq \cdots \leq p_{(10)}$.

Step 2: Calculate the Benjamini–Hochberg thresholds: For each ordered p-value, calculate $\frac{k}{m} q$ where $m = 10$ and $q = 0.05$ (**Table 44**).

Table 44: Benjamini-Hochberg step-up comparison for controlling the false discovery rate. Ordered p-values, $p_{(k)}$, are compared with the rank-dependent threshold $\frac{k}{m} q$, here with $m = 10$ hypotheses and $q = 0.05$. The inequality $p_{(k)} \leq \frac{k}{m} q$ is satisfied through $k = 4$, identifying the largest qualifying rank and therefore the first four ordered hypotheses as significant under the Benjamini-Hochberg procedure.

k	$p_{(k)}$	$\frac{k}{10}(0.05)$	Satisfies $p_{(k)} \leq \frac{k}{m} q$?
1	0.001	0.005	✓
2	0.008	0.010	✓
3	0.012	0.015	✓
4	0.020	0.020	✓
5	0.035	0.025	✗
6	0.041	0.030	✗
7	0.070	0.035	✗
8	0.120	0.040	✗
9	0.300	0.045	✗
10	0.600	0.050	✗

[318] Y. Benjamini and Y. Hochberg, *Controlling the false discovery rate: a practical and powerful approach to multiple testing*, Journal of the Royal Statistical Society, 1995.

The **largest** k satisfying the condition is $k = 4$.

Step 3: Reject the first four hypotheses: Therefore, the hypotheses corresponding to $p_{(1)}, p_{(2)}, p_{(3)}, p_{(4)}$ are rejected. This means that, under the chosen FDR procedure, these results provide sufficient statistical evidence to **reject their corresponding null hypotheses**. Consequently, we call the four findings with p-values $0.001, 0.008, 0.012, 0.020$ **statistically significant**.

Multiple comparisons in AI[319, 320, 321, 322]

Multiplicity can arise in AI development through repeated searches across:

- many features;
- many model architectures;
- many decision thresholds;
- many subgroups;
- multiple preprocessing procedures;
- different random seeds;
- repeated examinations of validation performance.

If the entire search process is ignored, the final reported result may be overly optimistic. In effect, the reported performance can reflect not only the quality of the model but also the extent of the search used to find it.

Example # 131: Predicting One-Year Disease Risk Using AI

Suppose a research team develops an AI model to predict whether a patient is at high risk of developing a particular disease within the next year. The team has a dataset containing 50,000 patients and initially divides it into training, validation, and test sets.

The researchers begin with a **reasonable hypothesis**: *a model using demographic characteristics, laboratory measurements, medical history, and medication information may be able to predict disease risk*.

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[319] D. D. Jensen and P. R. Cohen, *Multiple comparisons in induction algorithms*, Machine Learning, 2000.

[320] G. C. Cawley and N. L. C. Talbot, *On over-fitting in model selection and subsequent selection bias in performance evaluation*, The Journal of Machine Learning Research, 2010.

[321] J. Reunanen, *Overfitting in making comparisons between variable selection methods*, The Journal of Machine Learning Research, 2003.

[322] J. Pizarro, E. Guerrero, and P. L. Galindo, *Multiple comparison procedures applied to model selection*, Neurocomputing, 2002.

They **train several candidate models** and **evaluate their performance** on the **validation set**. However, the development process does not stop there. The researchers gradually **try many alternatives**:

- 120 different feature sets, including different combinations of laboratory measurements, demographic variables, and medical history;
- 15 model architectures, including logistic regression, random forests, gradient boosting, and several neural-network configurations;
- 40 different combinations of hyperparameters for each architecture;
- 10 probability thresholds for defining a patient as *high risk*;
- 8 preprocessing strategies, such as different approaches to missing data and variable scaling;
- 6 random seeds for stochastic algorithms;
- performance comparisons across 12 patient subgroups, such as age categories and sex;
- repeated examinations of validation performance during development.

Eventually, **one particular combination** produces an **impressive validation result** with an area under the ROC curve (AUROC) of 0.91. The researchers **might initially** describe this as **evidence** that the model performs extremely well.

But there is an important question: **How many opportunities did the researchers have to obtain an unusually high validation AUC?**

The answer is **not simply *one***. The final model was **selected after a large search** over features, architectures, preprocessing choices, hyperparameters, thresholds, random seeds, and subgroups. Even if none of these alternatives has a genuinely superior underlying performance, **extensive searching can make it increasingly likely** that at least one configuration will look unusually good on the validation data.

For example, imagine that the true predictive performance of all candidate approaches is roughly similar. Because validation performance contains random variation, some candidates will perform slightly better than their underlying ability, while others will perform slightly worse. If the researchers **examine only a few candidates**, an **unusually high result may be**

uncommon. If they examine thousands of candidate configurations, however, it becomes much easier to encounter one that benefits from favorable random variation.

This creates a subtle form of **selection bias**. The **validation set is no longer being used** merely to estimate how well a prespecified model performs. It has **also become part of the process** used to **search for and select the model**.

Consider a simplified version of the process:

- The team tries Model A and observes a validation AUROC of 0.78.
- Model B gives 0.81.
- Model C gives 0.79.
- After many additional experiments, Model 347 gives 0.91.
- The researchers **select Model 347** because it has the **highest observed validation performance**.
- They then **report the 0.91 AUROC** as though it **were an unbiased estimate** of the model's performance.

The problem is that Model 347 was **not selected independently** of the validation results. Its identity was determined partly because it performed **exceptionally well** on **that particular validation set**.

The same issue becomes even more apparent if the researchers **repeatedly inspect subgroups**. Suppose the overall model has an AUROC of 0.84, but its AUROC is 0.93 among one particular age group. If the researchers examined 12 subgroups and **highlighted the subgroup with the highest observed performance**, the 0.93 figure may **partly reflect the search itself**. The subgroup result was not simply discovered; it was **selected from many opportunities**.

A similar problem occurs with decision thresholds. Suppose the researchers examine thresholds ranging from 0.10 to 0.90 and select the threshold producing the highest F_1 score. If the **threshold was chosen after examining** the validation results, the resulting F_1 score is **subject to the same selection problem**. The threshold is effectively **another degree of freedom in the search**.

What happens to the final test set? Now suppose the researchers finally evaluate Model 347 once on a completely untouched test set and obtain an AUROC of 0.84. This result is lower than the validation AUROC of 0.91. That **difference does not necessarily mean that the model *failed***. Rather, it illustrates why the **distinction between model development and model evaluation matters**.

The **validation set was repeatedly used** during model development and therefore **helped determine** which model, preprocessing procedure, hyperparameters, and threshold **would ultimately be selected**. The test set, by contrast, was not used for those decisions.

The **test-set estimate** therefore provides a **more appropriate assessment** of how the selected model performs on new data, provided the **test set itself has remained genuinely untouched**. The key lesson is that the **relevant unit of analysis is not merely the final model**. It is the **entire search process that produced the final model**.

Why the final result can be misleading: Suppose two research teams both report an AUROC of 0.91.

Team 1 **specified** its model architecture, features, preprocessing procedure, and evaluation criterion **in advance** and performed a **single prespecified analysis**.

Team 2 evaluated thousands of combinations of features, architectures, preprocessing procedures, thresholds, random seeds, and subgroups and **reported the best validation result**.

Although both teams report the same numerical AUROC, the **evidential meaning of the two results is different**. In the second case, the reported number has been selected from a much larger collection of opportunities for favorable results.

Thus, the **apparent performance of an AI system** can depend not only on **how good the model is**, but also on **how extensively researchers searched for a favorable result**.

How can the problem be addressed? Researchers can reduce this problem by making the **search process explicit** and **separating model development from final evaluation**. For example, they can:

- reserve a test set that is not examined during model development;
- use cross-validation or nested cross-validation when appropriate;
- predefine important outcomes and evaluation procedures where feasible;
- account for multiplicity when making formal statistical inferences;
- avoid repeatedly using the same validation set as a source of final performance claims;
- report the extent of the model-search process rather than describing only the final selected model;
- evaluate the final model on genuinely independent data when possible.

The central principle is simple: **If many opportunities are used to find a favorable result, the uncertainty associated with that search should not be ignored**.

In AI development, therefore, a **reported performance** measure should be **interpreted in the context of the entire sequence of choices** and **comparisons** that led to the final model, not merely as a property of the final model considered in isolation.

3.10.3 Correlation mistaken for causation

An **observed association** $P(Y|X)$ **does not by itself** identify the **causal effect of intervening** on X: $P\big(Y|\mathrm{do}(X)\big)$. **Several mechanisms** can produce an association **without establishing** a causal relationship.

Confounding

A third variable Z influences both X and Y: $Z \to X, Z \to Y$.

Reverse causation

The direction of causality may be reversed i.e., $Y \to X$.

Selection bias

Inclusion in the observed dataset may depend on X, Y, or variables that cause them, creating an association that does not represent the target population.

Common temporal trend

Two variables may change over time without either causing the other. For example, unrelated quantities can both increase simply because of a shared time trend.

Measurement artifact

The same measurement procedure, instrument, coding rule, or source of error may affect both variables and create an apparent association.

Feedback

Causality may operate in both directions. Earlier values of Y may influence later values of X, while earlier values of X influence later values of Y.

Therefore, **causal conclusions** require an **appropriate experimental design** or **explicit causal assumptions** supported by domain knowledge and evidence.

3.10.4 Other recurring mistakes

Treating missing values as zeros

A missing laboratory result is not equivalent to a laboratory measurement of zero. Missingness and zero are different states and should generally be represented separately.

Reporting only averages

Average performance can conceal substantial differences between subgroups. An AI system with a strong overall average may still perform poorly for particular populations or cases.

Examining only correlations

Correlation matrices primarily capture pairwise linear associations. They can miss nonlinear relationships, interactions, threshold effects, and higher-order structure.

Ignoring units

Incorrect or inconsistent units can completely distort an analysis. A unit-conversion error can dominate all other observed patterns.

Removing every outlier

Not every outlier is an error. Rare but valid observations may represent precisely the unusual cases that an AI system must be able to handle.

Using the Test Set for Exploratory Data Analysis

Repeatedly inspecting test-set performance turns the test set into part of the development process. Once the test set influences model or analysis decisions, it no longer provides an unbiased final evaluation.

Using an inappropriate baseline

A sophisticated model should be compared with meaningful alternatives, including existing practice and simple baseline models. Comparing only against random prediction can make a model appear more impressive than it actually is.

Equating prediction with explanation

A feature can improve predictive performance without explaining the mechanism that generated the outcome. Predictive usefulness and causal or mechanistic explanation are distinct objectives.

Recommended Integrated Workflow

A rigorous AI analysis should follow a **structured workflow** (Figure 63) in which **decisions** about data, modeling, evaluation, and interpretation are **made deliberately** and **documented** throughout the study. The goal is not merely to obtain a strong-performing model, but to ensure that the **resulting estimates** and **conclusions** are **reproducible**, **appropriately evaluated**, and **supported by the available evidence**.

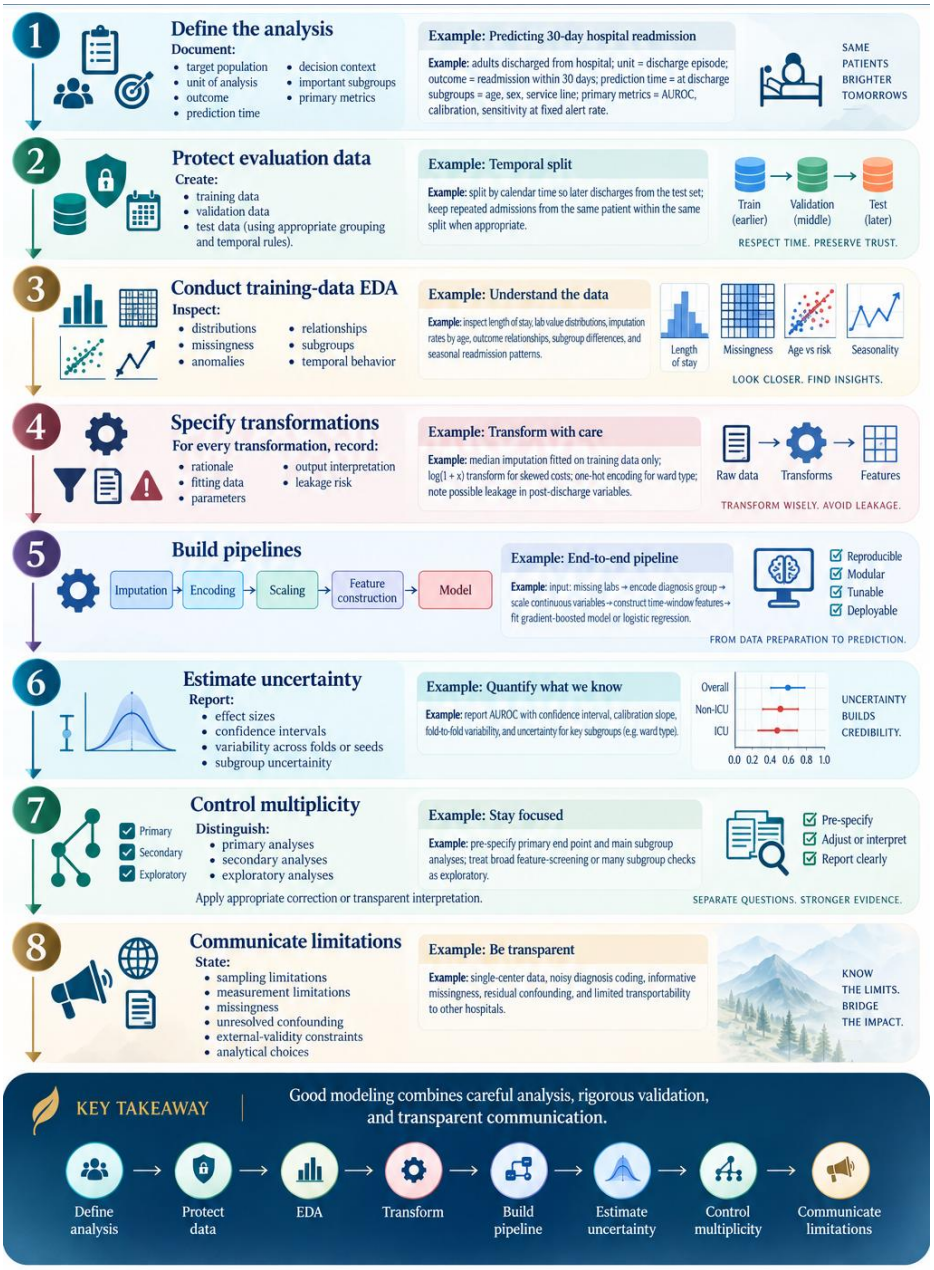

Figure 63: Recommended integrated workflow for EDA and predictive modeling.

Cohen's κ and Fleiss' κ

This section provides an instructive example of computing **Cohen's** κ[323,324] and **Fleiss'** κ[325,326] to assess **inter-rater reliability**. Cohen's κ is used to measure the **agreement between two raters** who **independently classify** the **same set of items** into categorical outcomes, whereas Fleiss' κ extends this approach to assess **agreement among multiple raters**. Both statistics account for the **agreement expected by chance**, providing a more informative assessment of inter-rater agreement than simple percentage agreement alone. The following examples illustrate the calculation of each statistic, including the construction of the **contingency table**[327], the computation of **observed** and **expected agreement**, and the interpretation of the resulting κ coefficients.

3.11.1 Cohen's κ – An Instructive Example

Cohen's κ is a statistical measure used to assess the level of agreement between **two raters** who classify the same items into categories. It accounts for the agreement that could occur by chance.

For example, two teachers independently grade the **same 50 students** as either **Pass** or **Fail**. We want to determine how well their judgments agree. Suppose the teachers classify the 50 students as tabulated in **Table 45**.

Step 1: Calculate observed agreement (P_o)

Observed agreement is the proportion of classifications on which the two raters agree i.e.,

$$P_o = \frac{\text{Number of agreements}}{\text{Total observations}}$$

[323] **Cohen's κ (Cohen's kappa):** A statistical measure of **inter-annotator agreement** between two annotators that **adjusts for agreement expected by chance**. Example: Two radiologists independently classify 100 X-rays as normal or abnormal and achieve a Cohen's κ of 0.82, indicating strong agreement beyond chance.

[324] Jacob Cohen, A coefficient of agreement for nominal scales, Educational and Psychological Measurement, 20(1), pp 37-46, 1960.

[325] **Fleiss' κ (Fleiss' kappa):** A statistical measure of **inter-annotator agreement among three or more annotators** that **adjusts for chance agreement**. Example: Five clinicians independently label 200 skin lesion images as benign, malignant, or uncertain. A Fleiss' κ of 0.76 indicates substantial agreement across the group.

[326] Joseph L. Fleiss, *Measuring nominal scale agreement among many raters*, Psychological Bulletin, 76(5), 378-382., 1971.

[327] A **contingency table** was introduced by **Karl Pearson** in **1904** and is used to display the frequency distribution and relationship between two or more categorical variables.

Table 45: Paired pass–fail outcomes for evaluations by Teacher A and Teacher B. The contingency table cross-classifies the outcomes assigned by the two teachers for 50 students. Both teachers recorded a pass for 30 students and a fail for 12 students, while their assessments disagreed for 8 students: 5 were passed by Teacher A but failed by Teacher B, and 3 were failed by Teacher A but passed by Teacher B. The marginal totals show 35 passes and 15 fails for Teacher A, compared with 33 passes and 17 fails for Teacher B.

	Teacher B: Pass	Teacher B: Fail	Total
Teacher A: Pass	30	5	**35**
Teacher A: Fail	3	12	**15**
Total	**33**	**17**	**50**

Substituting the values: $P_o = \frac{30+12}{50} = 0.84$. Therefore, the observed agreement is 84%. **However, some agreement could occur simply by chance.** Cohen's κ adjusts for this.

Step 2: Calculate expected chance agreement (P_e)

First, calculate the **marginal proportions** for each category (**Table 46**).

Table 46: Marginal pass and fail proportions for Teacher A and Teacher B. The table summarizes each teacher's overall classification rates across 50 students. Teacher A assigned a pass to 70% of students and a fail to 30%, whereas Teacher B assigned a pass to 66% and a fail to 34%. These marginal proportions provide a concise comparison of the overall assessment tendencies of the two teachers.

Category	Teacher A	Teacher B
Pass	$35/50 = 0.70$	$33/50 = 0.66$
Fail	$15/50 = 0.30$	$17/50 = 0.34$

The expected agreement by chance is the sum of the products of the corresponding category proportions $P_e = \sum_{i=1}^{k} P_{Ai} \times P_{Bi}$ where k is the number of categories, P_{Ai} is the proportion of cases assigned to category i by Rater A, and P_{Bi} is the proportion of cases assigned to category i by Rater B. In the current example:

$$P_e = (P_{A,\text{Pass}} \times P_{B,\text{Pass}}) + (P_{A,\text{Fail}} \times P_{B,\text{Fail}}) = (0.70 \times 0.66) + (0.30 \times 0.34) = 0.564$$

Thus, the expected chance agreement is 56.4%.

Step 3: Calculate Cohen's κ

The formula is $\kappa = \frac{P_o - P_e}{1 - P_e}$. For the example under study:

$$\kappa = \frac{P_o - P_e}{1 - P_e} = \frac{0.84 - 0.564}{1 - 0.564} \approx 0.633$$

Step 4: Interpret the result

One commonly used interpretation scale is the following, proposed by Landis and Koch[328] (Table 47).

Table 47: Conventional interpretation of Cohen's κ values. The table summarizes commonly used qualitative categories for interpreting the magnitude of Cohen's κ, ranging from poor agreement for $\kappa \leq 0$ to almost perfect agreement for $\kappa \in [0{,}81, 1.00]$. These categories provide a convenient descriptive guide for assessing agreement beyond chance, although the practical importance of a given κ value should also be interpreted in the context of the application, prevalence of categories, and consequences of disagreement.

Kappa value	Interpretation
≤ 0	Poor agreement
0.01 – 0.20	Slight agreement
0.21 – 0.40	Fair agreement
0.41 – 0.60	Moderate agreement
0.61 – 0.80	Substantial agreement
0.81 – 1.00	Almost perfect agreement

Our result, $\kappa = 0.633$, indicates **substantial agreement according to this scale**. Note that these ranges are guidelines, not universal thresholds. The teachers agree on 84% of students, but Cohen's κ **adjusts that figure for the agreement expected by chance**, resulting in a value of approximately 0.633.

3.11.2 Fleiss' κ – An Instructive Example

[328] J. R. Landis and G. G. Koch, *The measurement of observer agreement for categorical data*. Biometrics, 33(1), 159-174, 1977.

Fleiss' κ is a statistical measure used to **assess inter-rater reliability** when **three or more raters independently classify** the same set of items into categorical outcomes. Like Cohen's κ, it accounts for the agreement expected by chance.

Suppose three experts independently classify 5 research abstracts into two categories, i.e., **Relevant** and **Not Relevant**. Each expert evaluates every abstract. The ratings are summarized in **Table 48**.

Table 48: Distribution of relevance ratings across five abstracts. The table summarizes ratings from three evaluators for each abstract, showing the number classified as Relevant and Not Relevant. Across the five abstracts, 15 ratings were recorded in total, comprising 8 Relevant and 7 Not Relevant judgments.

Abstract	Relevant	Not Relevant	Total raters
1	3	0	3
2	2	1	3
3	2	1	3
4	1	2	3
5	0	3	3
Total ratings	**8**	**7**	**15**

Step 1: Calculate observed agreement ($\bar{P}$)

The observed agreement for subject i is computed as follows:

$$P_i = \frac{1}{n(n-1)}\left(\sum_{j=1}^{k} n_{ij}^2 - n\right)$$

where n is the number of raters per subject, n_{ij} is the number of raters who assign subject i to category j, and k is the number of categories.

In this example, $n = 3$ raters and $k = 2$ categories. Therefore, $n(n-1) = 3 \times 2 = 6$. For Abstract 1:

$$P_1 = \frac{1}{6}\left(\sum_{j=1}^{2} n_{1j}^2 - n\right) = \frac{1}{6}(n_{11}^2 + n_{12}^2 - 3) = \frac{1}{6}(3^2 + 0^2 - 3) = 1$$

For Abstract 2:

$$P_2 = \frac{1}{6}\left(\sum_{j=1}^{2} n_{2j}^2 - n\right) = \frac{1}{6}(n_{21}^2 + n_{22}^2 - 3) = \frac{1}{6}(2^2 + 1^2 - 3) = \frac{1}{3}$$

Applying the same calculation to all five abstracts yields **Table 49**.

Table 49: Item-level agreement among three raters across five abstracts. The table reports the number of Relevant and Not Relevant ratings for each abstract together with the corresponding item-specific agreement measure, P_i. Complete agreement occurs for Abstracts 1 and 5 ($P_i = 1$), whereas Abstracts 2-4 show partial agreement ($P_i = 1/3$), reflecting a two-to-one split among the three raters.

Abstract	Relevant	Not Relevant	P_i
1	3	0	1.000
2	2	1	0.333
3	2	1	0.333
4	1	2	0.333
5	0	3	1.000

The **overall observed agreement** is the average of the item-specific agreements:

$$\bar{P} = \frac{1}{N}\sum_{i=1}^{N} P_i$$

where N is the number of items. In the example under study: $\bar{P} = \frac{1}{5}\sum_{i=1}^{5} P_i = 0.60$. Thus, the observed agreement among the raters is 60%.

Step 2: Calculate expected chance agreement (P_e)

First, calculate the **proportion of all ratings** assigned to each category. For **Relevant** we have $p_1 = \frac{8}{15}$ and for **Not Relevant** we have $p_2 = \frac{7}{15}$.

The **expected chance agreement** is the sum of the squared category proportions i.e.,

$$\bar{P}_e = \sum_{j=1}^{k} p_j^2$$

where k is the number of categories and p_j is the proportion of all ratings assigned to category j.

In the example under study, we have $\bar{P}_e = \left(\frac{8}{15}\right)^2 + \left(\frac{7}{15}\right)^2 = 0.5022$. The expected agreement by chance is approximately 50.22%.

Step 3: Calculate Fleiss' κ

The formula for is:

$$\kappa = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$$

Substituting the calculated values yields: $\kappa = \frac{0.600 - 0.5022}{1 - 0.5022} \approx 0.196$. Therefore, Fleiss' kappa is approximately 0.20.

Step 4: Interpret the result

Using the commonly cited interpretation guidelines of Landis and Koch[329] (**Table 47**), a Fleiss' $\kappa \approx 0.196$ indicates **slight agreement beyond chance** according to these guidelines. Although the observed agreement is 60%, the expected chance agreement is already about 50.22%. After adjusting for chance, the resulting kappa indicates relatively low inter-rater reliability.

These interpretation ranges are guidelines **rather than universal standards**, and the κ value should be considered alongside the study context and category distribution.

[329] J. R. Landis and G. G. Koch, *The measurement of observer agreement for categorical data*. Biometrics, 33(1), 159-174, 1977.

Three Illustrative Case Studies

We now discuss three case studies utilizing the concepts, principles, and practices studied in this module.

3.12.1 Anscombe's Quartet and the Datasaurus Dozen

Background

Francis Anscombe constructed four small datasets to demonstrate that numerical summaries alone may be seriously misleading. The four datasets have approximately the same:

- mean of X;
- mean of Y;
- variance of X;
- variance of Y;
- Pearson correlation;
- linear regression equation

Their common fitted regression line is approximately $\hat{y} = 3 + 0.5x$ and their correlation is approximately $r \approx 0.816$. Yet their scatterplots show fundamentally different structures.

The four structures

Dataset I: approximately linear relationship

The first dataset displays a reasonably linear association with ordinary residual variation. A linear regression is broadly appropriate.

Dataset II: nonlinear relationship

The second dataset follows a curved pattern. A linear model summarizes the overall direction but fails to represent the systematic nonlinearity.

Dataset III: influential vertical outlier

Most observations follow a linear pattern, but one observation has an unusually large Y-value. The mean, variance, correlation, and regression coefficients are strongly affected by that point.

Dataset IV: high-leverage observation

Most X-values are nearly identical. A single observation with an unusual X-value largely determines the regression line and correlation. Without the high-leverage observation, there is almost no information for estimating the relationship.

Why identical statistics do not imply identical data

A statistic is a many-to-one mapping: $S: \mathcal{D} \to \mathbb{R}^k$. Many different datasets can produce the same summary vector: $S(\mathcal{D}_1) = S(\mathcal{D}_2)$ even though $\mathcal{D}_1 \neq \mathcal{D}_2$. The summary therefore cannot preserve every property of the original data. Mean and variance do not describe:

- spatial arrangement;
- multimodality;
- curvature;
- leverage;
- clustering;
- local density;
- discontinuity.

The Datasaurus extension

Matejka and Fitzmaurice[330] developed a procedure for creating datasets with nearly identical summary statistics but radically different graphical appearances, including a dataset shaped like a dinosaur. The Datasaurus Dozen datasets share approximately equal:

- x-means;
- y-means;
- standard deviations;
- correlations,

yet appear as:

- circles;
- lines;
- stars;

[330] J. Matejka and G. Fitzmaurice, *Same Stats, Different Graphs: Generating Datasets with Varied Appearance and Identical Statistics through Simulated Annealing*, Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems, 2017.

- clusters;
- curves;
- a dinosaur.

This demonstrates that the problem is not limited to four artificially simple datasets. Vastly different structures can satisfy similar low-dimensional numerical constraints.

Implications of AI

Suppose two training datasets have the same:

- number of observations;
- class prevalence;
- feature means;
- feature variances;
- correlation matrix.

They may still differ in:

- subgroup separation;
- nonlinear structure;
- outliers;
- density;
- support;
- rare cases;
- decision boundaries.

A model trained on one may behave very differently from a model trained on the other. For example, two hospitals may have equal average patient age and disease prevalence. However:

- one may contain two distinct age clusters;
- another may have a continuous age distribution;
- one may lack older high-risk patients;
- another may contain measurement errors.

Summary-table similarity does not establish transportability.

Recommended analytical response

For each numerical relationship:

- calculate descriptive statistics;
- create a scatterplot;
- examine residuals;
- identify influential observations;
- evaluate plausible nonlinear forms;
- compare subgroups;
- repeat the analysis with and without questionable points;
- document whether conclusions are stable.

The case demonstrates why visualization is an unavoidable stage of serious data analysis: **Summary statistics describe selected properties; plots reveal structure**.

3.12.2 UC Berkeley Graduate Admissions and Simpson's Paradox

Background

Aggregated University of California, Berkeley graduate-admissions data from 1973 appeared to show a substantial difference between admission rates for male and female applicants. Approximately: 44% of male applicants and 35% of female applicants were admitted.

At the aggregate level, the data appeared consistent with bias against female applicants. However, analysis by department produced a very different pattern. The principal explanation was that female applicants applied disproportionately to departments with lower overall admission rates, while male applicants applied more frequently to less selective departments[331].

Aggregate analysis

Let $G \in \{M, F\}$ represent applicant gender and Y represent admission. The aggregate comparison is $P(Y = 1|G = M)$ versus $P(Y = 1|G = F)$. This comparison combines all departments. However, departments differ substantially in:

- applicant volume;
- selectivity;

[331] P. J. Bickel, E. A. Hammel, and J. W. O'Connell, *Sex Bias in Graduate Admissions: Data from Berkeley: Measuring bias is harder than is usually assumed, and the evidence is sometimes contrary to expectation*, Science, 1975.

- admission capacity;
- applicant composition.

Department-specific analysis

Let D denote the department. The stratified comparison examines $P(Y = 1|G, D = d)$ for each department d. The aggregate admission rate is a weighted average:

$$P(Y = 1|G) = \sum_{d} P(Y = 1|G = g, D = d)P(D = d|G = g)$$

Even if department-specific rates are similar, aggregate rates can differ when $P(D = d|G = M) \neq P(D = d|G = F)$. This is the mathematical foundation of the reversal[332].

Simpson's Paradox

Simpson's paradox occurs when a relationship observed in aggregated data changes direction or substantially changes after conditioning on another variable. For example:

$$P(Y = 1|G = M) > P(Y = 1|G = F)$$

may hold overall, while within many departments:

$$P(Y = 1|G = M, D = d) \leq P(Y = 1|G = F, D = d)$$

The paradox is not a mathematical contradiction. The aggregate and conditional quantities answer different questions.

EDA lessons

- **Categorical summaries:** The analysis requires more than a two-by-two table. Admission proportions should be displayed:
  - overall;
  - by department;

[332] The **mathematical foundation of the reversal** is the **law of total probability**: an overall admission rate is determined not only by the admission rate within each department, but also by how applicants from each group are distributed across departments. If two groups have different proportions of applicants applying to departments with different levels of selectivity, their overall admission rates can differ even when their admission rates within individual departments are similar or identical.

In the stronger form known as **Simpson's paradox**, the difference can even reverse direction: one group may have a higher admission rate within every department, yet a lower admission rate overall because that group is disproportionately represented in more selective departments. Thus, the reversal arises from the interplay between department-specific admission rates and differences in the composition of the groups across departments.

- by gender within department;
- with department applicant counts.

- **Subgroup behavior:** The overall association was partly produced by different application patterns across departments.
- **Weighting:** Large departments contribute more heavily to aggregate rates.
- **Visualization:** Useful displays include:
  - grouped bar charts;
  - mosaic plots[333];
  - department-level dot plots;
  - plots of admission rate against department selectivity.

The causal question is more difficult

It is tempting to conclude that department-adjusted comparisons automatically establish the absence of discrimination. That conclusion is too strong without a causal model. Department choice may itself be influenced by:

- previous educational opportunities;
- expectations;
- career advice;
- social norms;
- perceived acceptance;
- structural barriers.

Thus, whether one should condition on department depends on the causal question.

Question # 1: Were men and women treated differently within the same department?

Question # 2: Did the complete admissions system produce unequal opportunities?

Causal interpretation requires deciding whether department is:

- a confounder;

[333] **Mosaic plot:** A graphical display in which the area of each tile represents the frequency or proportion of observations for a combination of categorical variables, making differences in group composition visually apparent. Example: A mosaic plot of $\text{gender} \times \text{department} \times \text{admission outcome}$ could show how the proportions of male and female applicants are distributed across departments and how admissions differ within each department.

- a mediator;
- a selection variable;
- part of the outcome-generating mechanism.

Association alone cannot resolve this.

Modern AI analogy

Suppose an admissions model predicts acceptance using:

- prior grades;
- university;
- program choice;
- recommendation strength.

Aggregate evaluation may show equal accuracy across genders. However, subgroup analysis may reveal:

- lower accuracy within highly selective programs;
- different calibration;
- different missingness;
- unequal false-negative rates;
- differential access to recommendation letters.

Similarly, a model can appear fair overall while failing within important intersections

Recommended workflow

- Define the unit of analysis: applicant or application.
- Construct overall contingency tables.
- Stratify by department.
- Examine department size and selectivity.
- Calculate effect sizes and confidence intervals.
- Draw a causal diagram.
- Distinguish descriptive from causal conclusions.
- Examine additional relevant variables.
- Communicate both aggregate and conditional results.
- Avoid presenting one adjustment as the uniquely correct answer without causal justification.

The case demonstrates: **Aggregation can conceal, exaggerate, or reverse subgroup relationships.**

3.12.3 Many Analysts, One Dataset

Background

Silberzahn and colleagues[334] gave the same dataset and research question to 29 analytical teams comprising 61 analysts. The question was whether soccer referees were more likely to give red cards to darker-skin-toned players than to lighter-skin-toned players. All teams used the same underlying data, yet they made different choices about:

- data exclusions;
- variable definitions;
- model family;
- covariates;
- aggregation;
- repeated observations;
- transformation;
- uncertainty estimation.

The estimated odds ratios ranged from approximately 0.89 to 2.93. Twenty teams found a statistically significant positive relationship, while nine did not.

Why results differed

The dataset contained decisions that were not determined automatically by the research question. Analysts had to choose:

- whether to aggregate observations by player;
- whether to model individual referee-player interactions;
- which leagues to include;
- how to code skin tone;
- which player characteristics to adjust for;
- whether to use logistic, count, or multilevel models;
- how to handle repeated observations;
- which observations to treat as influential.

[334] R. Silberzahn et al., *Many Analysts, One Data Set: Making Transparent How Variations in Analytic Choices Affect Results*, Advances in Methods and Practices in Psychological Science, 1(3), 337-356, 2018.

Each decision changed the effective statistical estimand.

Feature engineering as an analytical choice

Suppose the raw data contain:

- number of games;
- number of red cards;
- player position;
- club;
- league;
- referee;
- skin-tone rating.

Possible engineered outcomes include:

$$Y_i = \mathbf{1}\{\text{player ever received a red card}\}, Y_i = \frac{\text{red cards}}{\text{games}}$$

or a match-level binary indicator. These are not equivalent. Possible exposure measures include:

- total games;
- total referee encounters;
- playing time;
- matches started.

Different representations answer different questions.

Model specification

One analyst might fit $\text{logit}\, P(Y = 1) = \beta_0 + \beta_1 \text{skin tone}$. Another analyst may fit $\text{logit}\, P(Y = 1) = \beta_0 + \beta_1 \text{skin tone} + \beta_2 \text{position} + \beta_3 \text{league} + \text{u}_{\text{free}}$. The coefficient β_1 does not have the same interpretation in the two models.

The second model conditions on additional variables and includes referee-specific random effects. Neither specification is automatically correct without a clear estimand and causal rationale.

Analyst degrees of freedom

The study demonstrates that analytical uncertainty has at least two components:

$$\text{total uncertainty} \Rightarrow \text{sampling uncertainty} + \text{specification uncertainty}$$

A conventional confidence interval from one selected model usually reflects only uncertainty conditional on that model. It does not account for:

- plausible alternative preprocessing;
- feature definitions;
- model structures;
- exclusion rules;
- outcome codings.

This explains why two competent analysts can obtain different conclusions without either making a simple calculation error.

Connection to p-hacking

The many-analysts study was designed transparently and does not imply misconduct. However, it reveals why selective reporting can be dangerous. If a single analyst tries many specifications and reports only the one with $p < 0.05$, the reader cannot see the broader distribution of plausible results. The relevant evidence may instead be $\{\hat{\theta}_1, \hat{\theta}_2, \cdots, \hat{\theta}_K\}$ representing a range of defensible analytical choices.

Multiverse and specification-curve analysis

A multiverse analysis evaluates many defensible specifications. For each specification k, calculate $\hat{\theta}_k$, $\mathrm{SE}(\hat{\theta}_k)$, CI_k. The results can be ordered and plotted. Questions include:

- Is the direction stable?
- Does significance depend on one arbitrary decision?
- Which choices create the largest changes?
- Are some specifications scientifically more defensible?
- Does the estimated effect remain practically meaningful?

Recommended analytical safeguards

- **Pre-analysis planning**

  Specify:

  - primary outcome;
  - unit of analysis;
  - exclusion criteria;
  - feature definitions;

- statistical model;
- primary subgroup analysis

- **Data dictionary**
  Ensure consistent interpretation of:
  - skin-tone rating;
  - red-card event;
  - match exposure;
  - player identity;
  - referee identity.

- **Sensitivity analysis**
  Vary:
  - model family;
  - adjustment variables;
  - coding choices;
  - influential observations;
  - clustering assumptions.

- **Report effect sizes**
  Present:
  - odds ratios;
  - absolute risk differences where possible;
  - confidence intervals;
  - base rates.

- **Preserve exploratory findings**
  Exploratory analyses may be valuable, but they should be identified as exploratory rather than portrayed as uniquely prespecified results.

- **Replication**
  Evaluate the question using:
  - another time period;
  - another league;
  - independently coded data;
  - a preregistered protocol.

Implications for AI development

AI teams also make many analytical choices:

- missing-value handling;
- categorical encoding;
- feature windows;
- model architecture;
- class weighting;
- threshold selection;
- data partitions;
- random seeds;
- subgroup definitions.

If only the best-performing pipeline is reported, uncertainty arising from pipeline selection is hidden. A credible AI evaluation should report:

- the search space;
- selection procedure;
- validation protocol;
- final untouched test result;
- variability across repetitions;
- sensitivity to plausible design choices.

The central lesson is: **Data do not determine one inevitable analysis; analytical choices are part of the evidence**.

Reading List

To support students in furthering their understanding of the topics covered in this module, here's a curated reading list organized by topic. This list includes books and research papers that delve deeper into each of the key areas discussed in this module.

Exploratory Data Analysis: Foundations, Philosophy, and Practice

- J. W. Tukey, *Exploratory Data Analysis*, Addison-Wesley, 1977.
- J. T. Behrens, *Principles and Procedures of Exploratory Data Analysis*, Psychological Methods, 2(2), pp. 131-160, 1997.
- N. A. Heckert and J. J. Filliben, *NIST/SEMATECH e-Handbook of Statistical Methods: Chapter 1, Exploratory Data Analysis*, National Institute of Standards and Technology, 2003.
- F. J. Anscombe, *Graphs in Statistical Analysis*, The American Statistician, 27(1), pp. 17-21, 1973.
- D. C. Hoaglin, F. Mosteller, and J. W. Tukey, *Understanding Robust and Exploratory Data Analysis*, Wiley, 1983.

Exploration versus Confirmation, Units of Analysis, and Independent Evidence

- B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, *The Preregistration Revolution*, Proceedings of the National Academy of Sciences, 115(11), pp. 2600-2606, 2018.
- S. E. Lazic, *The Problem of Pseudoreplication in Neuroscientific Studies: Is It Affecting Your Analysis?*, BMC Neuroscience, 11, Article 5, 2010.
- D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, C. F. Dormann, *Cross-validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure*, Ecography, 40(8), pp. 913-929, 2017.
- Open Science Collaboration, *Estimating the Reproducibility of Psychological Science*, Science, 349(6251), aac4716, 2015.

Distributions, Robust Statistics, and Unusual Observations

- P. J. Huber and E. M. Ronchetti, *Robust Statistics*, 2nd ed., Wiley, 2009.
- G. E. P. Box and D. R. Cox, *An Analysis of Transformations*, Journal of the Royal Statistical Society: Series B, 26(2), pp. 211-243, 1964.
- V. Chandola, A. Banerjee, and V. Kumar, *Anomaly Detection: A Survey*, ACM Computing Surveys, 41(3), Article 15, 2009.

- J. L. Hintze and R. D. Nelson, *Violin Plots: A Box Plot-Density Trace Synergism*, The American Statistician, 52(2), pp. 181-184, 1998.
- R. J. Hyndman and Y. Fan, *Sample Quantiles in Statistical Packages*, The American Statistician, 50(4), pp. 361-365, 1996.

Missing Data, Sampling Bias, Representativeness, and the Big Data Paradox

- R. J. A. Little and D. B. Rubin, *Statistical Analysis with Missing Data*, 3rd ed., Wiley, 2019.
- X.-L. Meng, *Statistical Paradises and Paradoxes in Big Data (I): Law of Large Populations, Big Data Paradox, and the 2016 US Presidential Election*, The Annals of Applied Statistics, 12(2), pp. 685-726, 2018.
- J. Buolamwini and T. Gebru, *Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification*, Proceedings of the 1st Conference on Fairness, Accountability and Transparency, PMLR 81, pp. 77-91, 2018.
- Z. Obermeyer, B. Powers, C. Vogeli, and S. Mullainathan, *Dissecting Racial Bias in an Algorithm Used to Manage the Health of Populations*, Science, 366(6464), pp. 447-453, 2019.
- S. E. Lazic, *The Problem of Pseudoreplication in Neuroscientific Studies: Is It Affecting Your Analysis?*, BMC Neuroscience, 11, Article 5, 2010.

Data Visualization and Statistical Graphics

- E. R. Tufte, *The Visual Display of Quantitative Information*, 2nd ed., Graphics Press, 2001.
- L. Wilkinson, *The Grammar of Graphics*, 2nd ed., Springer, 2005.
- C. O. Wilke, *Fundamentals of Data Visualization*, O'Reilly Media, 2019.
- F. J. Anscombe, *Graphs in Statistical Analysis*, The American Statistician, 27(1), pp. 17-21, 1973.
- J. L. Hintze and R. D. Nelson, *Violin Plots: A Box Plot-Density Trace Synergism*, The American Statistician, 52(2), pp. 181-184, 1998.

Descriptive Statistics, Quantiles, Skewness, and Transformations

- R. J. Hyndman and Y. Fan, *Sample Quantiles in Statistical Packages*, The American Statistician, 50(4), pp. 361-365, 1996.
- P. J. Huber and E. M. Ronchetti, *Robust Statistics*, 2nd ed., Wiley, 2009.
- G. E. P. Box and D. R. Cox, *An Analysis of Transformations*, Journal of the Royal Statistical Society: Series B, 26(2), pp. 211-243, 1964.
- D. C. Hoaglin, F. Mosteller, and J. W. Tukey, *Understanding Robust and Exploratory Data Analysis*, Wiley, 1983.
- J. W. Tukey, *Exploratory Data Analysis*, Addison-Wesley, 1977.

Covariance, Correlation, Association, and Causal Caution

- C. Spearman, *The Proof and Measurement of Association between Two Things*, The American Journal of Psychology, 15(1), pp. 72-101, 1904.
- J. Pearl, *Causal Inference in Statistics: An Overview*, Statistics Surveys, 3, pp. 96-146, 2009.
- J. Pearl, *Causal Diagrams for Empirical Research*, Biometrika, 82(4), pp. 669-688, 1995.
- M. A. Hernan and J. M. Robins, *Causal Inference: What If*, Chapman & Hall/CRC, 2020.
- A. Gelman and H. Stern, *The Difference Between Significant and Not Significant Is Not Itself Statistically Significant*, The American Statistician, 60(4), pp. 328-331, 2006.

Feature Engineering: Representation, Scaling, Transformation, and Feature Construction

- M. Kuhn and K. Johnson, *Feature Engineering and Selection: A Practical Approach for Predictive Models*, Chapman & Hall/CRC, 2019.
- T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009.
- G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning: With Applications in Python*, Springer, 2023.
- G. E. P. Box and D. R. Cox, *An Analysis of Transformations*, Journal of the Royal Statistical Society: Series B, 26(2), pp. 211-243, 1964.

Categorical Encoding, High-Cardinality Features, and Leakage-Safe Preprocessing

- D. Micci-Barreca, *A Preprocessing Scheme for High-Cardinality Categorical Attributes in Classification and Prediction Problems*, ACM SIGKDD Explorations Newsletter, 3(1), pp. 27-32, 2001.
- S. Kaufman, S. Rosset, C. Perlich, and O. Stitelman, *Leakage in Data Mining: Formulation, Detection, and Avoidance*, ACM Transactions on Knowledge Discovery from Data, 6(4), Article 15, 2012.
- M. Kuhn and K. Johnson, *Feature Engineering and Selection: A Practical Approach for Predictive Models*, Chapman & Hall/CRC, 2019.
- D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller, D. I. Warton, B. A. Wintle, F. Hartig, C. F. Dormann, *Cross-validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure*, Ecography, 40(8), pp. 913-929, 2017.

Temporal, Sequential, and Representation-Based Features

- R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed., OTexts, 2021.
- A. Vaswani et al., *Attention Is All You Need*, Advances in Neural Information Processing Systems 30, 2017.
- D. Sculley et al., *Hidden Technical Debt in Machine Learning Systems*, Advances in Neural Information Processing Systems 28, pp. 2503-2511, 2015.
- D. R. Roberts et al., *Cross-validation Strategies for Data with Temporal, Spatial, Hierarchical, or Phylogenetic Structure*, Ecography, 40(8), pp. 913-929, 2017.

Population, Samples, Sampling Variability, and Statistical Foundations

- X.-L. Meng, *Statistical Paradises and Paradoxes in Big Data (I)*, The Annals of Applied Statistics, 12(2), pp. 685-726, 2018.
- J. Neyman, *Outline of a Theory of Statistical Estimation Based on the Classical Theory of Probability*, Philosophical Transactions of the Royal Society of London, Series A, 236(767), pp. 333-380, 1937.
- S. E. Lazic, *The Problem of Pseudoreplication in Neuroscientific Studies*, BMC Neuroscience, 11, Article 5, 2010.
- G. James et al., *An Introduction to Statistical Learning: With Applications in Python*, Springer, 2023.

Confidence Intervals, Estimation, and Bootstrap Inference

- J. Neyman, *Outline of a Theory of Statistical Estimation Based on the Classical Theory of Probability*, Philosophical Transactions of the Royal Society of London, Series A, 236(767), pp. 333-380, 1937.
- B. Efron, *Bootstrap Methods: Another Look at the Jackknife*, The Annals of Statistics, 7(1), pp. 1-26, 1979.
- G. Cumming, *The New Statistics: Why and How*, Psychological Science, 25(1), pp. 7-29, 2014.
- S. Greenland et al., *Statistical Tests, p-Values, Confidence Intervals, and Power: A Guide to Misinterpretations*, European Journal of Epidemiology, 31, pp. 337-350, 2016.
- D. Lakens, *Calculating and Reporting Effect Sizes to Facilitate Cumulative Science*, Frontiers in Psychology, 4, Article 863, 2013.

Hypothesis Testing, p-Values, Type I/II Errors, and Statistical Power

- R. L. Wasserstein and N. A. Lazar, *The ASA Statement on p-Values: Context, Process, and Purpose*, The American Statistician, 70(2), pp. 129-133, 2016.

- S. Greenland et al., *Statistical Tests, p-Values, Confidence Intervals, and Power: A Guide to Misinterpretations*, European Journal of Epidemiology, 31, pp. 337-350, 2016.
- J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Lawrence Erlbaum Associates, 1988.
- G. Cumming, *The New Statistics: Why and How*, Psychological Science, 25(1), pp. 7-29, 2014.
- R. L. Wasserstein, A. L. Schirm, and N. A. Lazar, *Moving to a World Beyond* $p < 0.05$, The American Statistician, 73(sup1), pp. 1-19, 2019.

Effect Sizes, Uncertainty, and Practical Significance

- D. Lakens, *Calculating and Reporting Effect Sizes to Facilitate Cumulative Science: A Practical Primer for t-Tests and ANOVAs*, Frontiers in Psychology, 4, Article 863, 2013.
- J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed., Lawrence Erlbaum Associates, 1988.
- G. Cumming, *The New Statistics: Why and How*, Psychological Science, 25(1), pp. 7-29, 2014.
- A. Gelman and H. Stern, *The Difference Between Significant and Not Significant Is Not Itself Statistically Significant*, The American Statistician, 60(4), pp. 328-331, 2006.
- R. L. Wasserstein, A. L. Schirm, and N. A. Lazar, *Moving to a World Beyond* $p < 0.05$, The American Statistician, 73(sup1), pp. 1-19, 2019.

Multiple Comparisons, p-Hacking, and Analytical Flexibility

- J. P. Simmons, L. D. Nelson, and U. Simonsohn, *False-Positive Psychology: Undisclosed Flexibility in Data Collection and Analysis Allows Presenting Anything as Significant*, Psychological Science, 22(11), pp. 1359-1366, 2011.
- Y. Benjamini and Y. Hochberg, *Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing*, Journal of the Royal Statistical Society: Series B, 57(1), pp. 289-300, 1995.
- B. A. Nosek et al., *The Preregistration Revolution*, Proceedings of the National Academy of Sciences, 115(11), pp. 2600-2606, 2018.
- Open Science Collaboration, *Estimating the Reproducibility of Psychological Science*, Science, 349(6251), aac4716, 2015.
- R. L. Wasserstein, A. L. Schirm, and N. A. Lazar, *Moving to a World Beyond* $p < 0.05$, The American Statistician, 73(sup1), pp. 1-19, 2019.

Integrative Reading: Statistical Thinking for AI and Machine Learning

- M. Kuhn and K. Johnson, *Feature Engineering and Selection: A Practical Approach for Predictive Models*, Chapman & Hall/CRC, 2019.
- G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, *An Introduction to Statistical Learning: With Applications in Python*, Springer, 2023.
- T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed., Springer, 2009.
- J. Pearl, *Causal Inference in Statistics: An Overview*, Statistics Surveys, 3, pp. 96-146, 2009.
- M. A. Hernan and J. M. Robins, *Causal Inference: What If*, Chapman & Hall/CRC, 2020.
- D. Sculley et al., *Hidden Technical Debt in Machine Learning Systems*, Advances in Neural Information Processing Systems 28, pp. 2503-2511, 2015.

Practice Questions for Quizzes and/or Exams

Please find below a list of **practice questions** to help you review **key concepts** and prepare for quizzes and/or exams.

Foundations of Exploratory Data Analysis

1. Define exploratory data analysis (EDA) and explain its role in the end-to-end AI and machine learning pipeline.
2. What are the principal objectives of EDA, and how do numerical summaries and graphical methods complement one another?
3. Why should EDA be treated as an iterative process rather than a fixed sequence of statistical operations?
4. Distinguish between exploratory analysis and confirmatory analysis in terms of their objectives, analytical procedures, and interpretation of findings.
5. Why should a hypothesis discovered through EDA not automatically be treated as independently confirmed evidence when tested using the same dataset?
6. Explain how separate exploratory and confirmatory datasets, preregistration, replication, and transparent reporting can improve the credibility of statistical findings.
7. Distinguish among the unit of observation, unit of prediction, unit of sampling, and unit of evaluation. Illustrate the differences using a medical imaging application.
8. A medical imaging dataset contains $50{,}000$ images from $2{,}000$ patients. Why would treating the dataset as $50{,}000$ independent patient-level observations potentially produce misleading statistical conclusions?

Understanding Distributions and Detecting Anomalies

9. Explain the five complementary characteristics of a numerical distribution: location, spread, shape, tails, and unusual observations.
10. Distinguish among symmetric, positively skewed, negatively skewed, unimodal, bimodal, and multimodal distributions.
11. What are heavy-tailed distributions, and why can they create difficulties for statistical models that assume approximately normal observations?
12. Explain the meaning and practical significance of zero inflation, truncation, and heterogeneous mixtures in real-world datasets.
13. How can inappropriate histogram bin widths and bin boundaries conceal genuine structure or create misleading distributional patterns?
14. Why can excessive smoothing or insufficient smoothing in kernel density estimation produce incorrect interpretations of distributional shape?

15. Define an anomaly and distinguish among a legitimate extreme observation, measurement error, data-entry error, and evidence of distribution shift.
16. Why should observations beyond the conventional boxplot whiskers not automatically be removed from a dataset?
17. Describe a structured anomaly-investigation workflow covering flagging, verification, explanation, correction or retention, and documentation.

Missingness, Bias, Representativeness, and Subgroup Behavior

18. Distinguish among variable-level, record-level, pattern-level, and subgroup-specific missingness.
19. A dataset contains $2{,}000$ patient records, and income is missing for 450 patients. Calculate the variable-level missingness rate. What additional information would be needed to determine whether missingness is concentrated within particular patient groups?
20. How can missingness itself contain useful information about the underlying data-generating process? Explain using an example involving laboratory tests ordered selectively by clinicians.
21. Explain the consequences of subgroup-specific missingness for effective sample size, imputation uncertainty, feature distributions, and model reliability.
22. Distinguish among sampling bias, measurement bias, label bias, selection bias, survivorship bias, and historical bias.
23. How can a dataset be internally consistent and accurately recorded while still reproducing historical inequalities?
24. Explain the Big Data Paradox and why increasing the number of observations does not necessarily eliminate systematic bias.
25. Interpret the three components of the following decomposition of estimation error:

$$\bar{X}_S - \bar{X}_U = \rho_{R,X} \sqrt{\left(\frac{N-n}{n}\right)} \sigma_X$$

What does each component represent, and what does the equation reveal about the relationship between data quality, data quantity, and problem difficulty?

26. Why should subgroup analysis be guided by domain knowledge, substantive relevance, and adequate sample sizes rather than by repeatedly searching for statistically significant findings?

Data Visualization and Responsible Statistical Graphics

27. Why is visualization necessary even when a dataset has already been summarized using means, variances, correlations, and regression coefficients?

28. What is the central lesson of Anscombe quartet, and how does it demonstrate the limitations of relying exclusively on numerical summaries?
29. Explain how histograms, kernel density plots, boxplots, empirical cumulative distribution functions, and Q-Q plots provide complementary perspectives on numerical distributions.
30. What information can be obtained from an empirical cumulative distribution function that may be difficult to identify from a histogram?
31. How can a Q-Q plot be used to identify heavy tails, light tails, skewness, heterogeneous mixtures, and unusual observations?
32. What does heteroscedasticity mean, and how might it appear in a scatterplot?
33. What is overplotting, and how can transparency, jittering, hexagonal binning, density contours, and aggregation help address it?
34. What is a correlation heatmap? Explain its uses and limitations when relationships are nonlinear, outliers are influential, or correlations are estimated using different effective sample sizes.
35. What is a pair plot, and why does its usefulness diminish as the number of variables becomes large?
36. Explain how chart selection, axis scaling, category ordering, color, aggregation, uncertainty display, and disclosure of exclusions contribute to responsible data visualization.

Descriptive Statistics and Distributional Summaries

37. Define the arithmetic mean, weighted mean, and median. Under what distributional characteristics might each provide a useful summary?
38. Consider the observations: $\{2,3,3,4,100\}$. Calculate the mean and median. Explain why the two statistics provide substantially different descriptions of the same dataset.
39. Define sample variance and sample standard deviation. Why is the denominator $(n-1)$ conventionally used when estimating population variance from independent sample observations?
40. Distinguish among variance, standard deviation, interquartile range, and median absolute deviation. Which measures are more resistant to extreme observations? .
41. Explain the relationship among quantiles, quartiles, and percentiles. What information do the 25^{th}, 50^{th}, 90^{th}, and 95^{th} percentiles provide about a distribution?
42. A cloud infrastructure company records the following network latencies in milliseconds: $[120,135,142,150,165]$. Calculate the 80^{th} percentile using Hyndman and Fan Type 7 and Type 6 interpolation. If the SLA requires the 80^{th} percentile to remain below 155 milliseconds, determine the outcome under each method.

43. Define skewness mathematically and explain the interpretation of positive, negative, and approximately zero skewness.
44. Why can logarithmic transformations be useful for strongly right-skewed variables? Discuss their effects on distributional shape, interpretation, and subsequent statistical modeling.
45. Define covariance and explain how its sign and magnitude should be interpreted. Why are covariance values not directly comparable across variables measured in different units?
46. Define Pearson correlation and explain how standardizing covariance produces a dimensionless measure of linear association.

Relationships, Correlation, and Causal Interpretation

47. Distinguish among positive linear, negative linear, nonlinear, monotonic, threshold-based, interaction-driven, and outlier-driven relationships.
48. Why can Pearson correlation be close to zero even when two variables are strongly related? Illustrate using the relationship $Y = X^2$ for a distribution of X symmetric around zero.
49. Compare Pearson correlation and Spearman rank correlation in terms of the relationships they measure and their sensitivity to extreme observations.
50. Why might calculating Pearson correlation for ordinal categories encoded as integers produce a misleading interpretation?
51. How can the correlation between two time-dependent variables be influenced by shared trends, seasonality, or temporal dependence?
52. Explain why an aggregate association may become weaker, stronger, or reversed after stratifying the data into relevant subgroups.
53. Distinguish among confounding, reverse causation, selection effects, and chance as possible explanations for an observed association.
54. Why does correlation not establish causation? Explain how randomized interventions, causal models, and explicit causal assumptions can strengthen causal conclusions.

Feature Engineering, Scaling, Normalization, and Categorical Encoding

55. Define feature engineering and explain the meaning of the transformation $\phi(x_{\text{raw}}) = x_{\text{model}}$. How can feature engineering improve the representation of information without necessarily introducing new information?
56. Why must transformations whose parameters are learned from data be fitted using the training partition rather than the complete dataset?

57. Explain standardization and derive the transformation used to convert a numerical feature into a standardized representation using the training mean and standard deviation.
58. Compare standardization, robust scaling, min-max normalization, and unit-vector normalization. Identify an appropriate application for each.
59. Why are distance-based algorithms, regularized regression, and gradient-based optimization generally more sensitive to feature scaling than many tree-based models?
60. Distinguish among logarithmic transformations, square-root transformations, and nonlinear feature construction. How can each influence model behavior and interpretability?
61. Define binning and compare equal-width, equal-frequency, domain-defined, and supervised binning.
62. What information can be lost when a continuous variable is converted into categories? Explain how artificial discontinuities arise near bin boundaries.
63. Compare ordinal encoding, one-hot encoding, frequency encoding, and target encoding. Discuss their suitability for nominal, ordinal, and high-cardinality variables.
64. Why is target encoding especially vulnerable to target leakage? Explain how cross-fitting, out-of-fold estimates, smoothing, and explicit handling of unseen categories can reduce this risk.

Advanced Feature Construction and Feature-Engineering Risks

65. Define an interaction feature and explain the interpretation of the interaction coefficient in the model $\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$.
66. What are polynomial features, and how can they allow a model that is linear in its parameters to represent nonlinear relationships?
67. Why can increasing the degree of polynomial features create problems involving multicollinearity, excessive dimensionality, unstable extrapolation, and overfitting?
68. Define lag features, rolling means, rolling standard deviations, differences, percentage changes, event counts, and time-since-event features.
69. A fraud detection system generates a prediction at time t. Explain why a rolling average constructed using observations from $(t + 1)$ or later would introduce temporal leakage.
70. Why should cyclical variables such as hour of day, day of week, and month of year not always be represented using ordinary integer encoding?
71. Explain how sine and cosine transformations preserve circular proximity when encoding cyclical variables. Write the formulas and identify the appropriate period for hour-of-day encoding.

72. Identify and explain eight major feature-engineering risks: target leakage, temporal leakage, identity leakage, proxy discrimination, unstable high-cardinality features, train-serving inconsistency, excessive dimensionality, and spurious interactions.

Statistical Foundations, Sampling, and Uncertainty

73. Distinguish between a population and a sample. Why must the target population be explicitly defined before making statistical claims?
74. Compare simple random sampling, stratified sampling, cluster sampling, systematic sampling, and convenience sampling.
75. How can sampling bias arise when the probability of inclusion in a dataset is related to the characteristic being estimated?
76. Define sampling variability and explain why two independently selected samples from the same population may produce different estimates.
77. Distinguish between a population parameter, sample statistic, estimator, and point estimate.
78. Define bias and variance of an estimator. Interpret the following expressions: $\mathrm{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$ and $\mathrm{Var}(\hat{\theta}) = \mathbb{E}\left[(\hat{\theta} - \mathbb{E}[\hat{\theta}])^2\right]$.
79. Explain the mean squared error decomposition $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) + \mathrm{Bias}(\hat{\theta})^2$. Why does this relationship matter when comparing statistical estimators?
80. Identify the major sources of uncertainty in AI and statistical analysis, including sampling, measurement, missingness, labels, model specification, parameter estimation, feature selection, distribution shift, and analyst decisions.

Confidence Intervals and Bootstrap Inference

81. Define a confidence interval and distinguish it from a point estimate.
82. What is the correct frequentist interpretation of a 95% confidence interval? Why should it not ordinarily be interpreted as a 95% probability that the fixed population parameter lies within the particular observed interval?
83. Write the conventional confidence interval for a population mean when the population standard deviation is unknown. Explain the roles of the sample mean, sample standard deviation, sample size, and critical value.
84. A researcher records examination scores from 100 students and obtains a sample mean of 72 and a sample standard deviation of 15. Calculate an approximate 95% confidence interval for the population mean using the normal critical value 1.96.
85. Explain how sample size, variability, confidence level, and dependence among observations can influence the width of a confidence interval.

86. Define the bootstrap method and describe the procedure for constructing a percentile bootstrap confidence interval for a sample median.
87. Why should bootstrap resampling respect the original sampling or dependence structure? Explain using a dataset containing repeated measurements from individual patients.

Hypothesis Testing, Effect Sizes, and Statistical Significance

88. Define the null hypothesis and alternative hypothesis. Explain their roles in statistical hypothesis testing.
89. What is a test statistic, and how does it connect the observed data with the null hypothesis?
90. Define a p-value and explain why it is not the probability that the null hypothesis is true.
91. Distinguish between Type I and Type II errors. Explain the roles of significance level and statistical power.
92. What is an effect size, and why should effect sizes be reported alongside p-values and confidence intervals?
93. Compare mean differences, standardized mean differences, risk differences, risk ratios, odds ratios, and correlation coefficients as measures of effect size.
94. Explain how a very small effect can become statistically significant in a sufficiently large dataset while remaining practically unimportant.
95. What is the difference between statistical significance and practical significance? Illustrate using an AI system that produces a statistically detectable but operationally negligible improvement over an existing baseline.

Common Analytical Mistakes, Reproducibility, and Integrative Case Studies

96. Define p-hacking and explain how selective reporting, repeated testing, outcome switching, and data-dependent analytical decisions can produce misleading statistical findings.
97. Why do multiple comparisons increase the risk of false-positive findings? Compare the objectives of Bonferroni correction and false discovery rate control.
98. Explain the central lesson of the UC Berkeley graduate admissions case study. How can Simpson paradox arise when overall admission rates are interpreted without considering differences across departments?
99. What does the Many Analysts, One Dataset study demonstrate about analytical flexibility, model specification, and reproducibility? Explain why different defensible analytical choices can produce substantially different results from the same dataset.

100. Develop a comprehensive EDA and modeling protocol for a new AI project that incorporates the following elements:
    - Defining the target population, unit of analysis, outcome, prediction time, and decision context.
    - Protecting training, validation, and test partitions.
    - Conducting EDA using training data.
    - Examining distributions, missingness, anomalies, relationships, temporal behavior, and subgroups.
    - Specifying feature transformations and documenting their fitting data, parameters, interpretation, and leakage risks.
    - Selecting appropriate statistical models and evaluation measures.
    - Reporting uncertainty, effect sizes, subgroup behavior, and sensitivity analyses.
    - Documenting analytical decisions and reproducibility requirements.
    - Preserving the final test set for independent evaluation.
    - Communicating the limitations of the resulting statistical and predictive claims.

Below is a list of **scenario-based questions** designed to help students apply the concepts from this module to real-world AI systems.

S1: University Student Performance Prediction

A university develops an AI system to predict examination performance using student attendance, study hours, sleep duration, previous grades, and online learning activity. The dataset contains $12{,}000$ student-course records collected from $2{,}000$ students over three academic years.

During EDA, researchers discover that students sleeping between seven and eight hours generally obtain higher examination scores. However, the relationship becomes weaker when students are divided into undergraduate and postgraduate groups.

The researchers also identify several students reporting more than 18 hours of daily study and notice that students with low attendance have disproportionately missing sleep records.

After experimenting with multiple predictors and regression models, the team selects the model showing the strongest relationship between sleep duration and examination performance. It then reports a statistically significant association using the same dataset.

Questions

- Identify the unit of observation, unit of prediction, and appropriate unit of evaluation in this project.
- Which graphical techniques would help examine the relationship between sleep duration and examination performance?
- Why might the aggregate relationship differ from the relationships observed within undergraduate and postgraduate groups?
- How should the unusually high reported study hours and subgroup-specific missingness be investigated?
- What is problematic about discovering a relationship, selecting a model, and confirming the resulting hypothesis using the same dataset?
- Propose an appropriate exploratory and confirmatory analysis strategy for this project.

S2: Hospital Readmission Prediction and Data Quality

A hospital network develops an AI system to predict whether patients will be readmitted within 30 days of discharge. The dataset contains $50{,}000$ hospital admissions from five hospitals and includes age, blood pressure, laboratory measurements, previous admissions, length of stay, and diagnosis codes.

The following problems emerge during EDA:

- Hospital A records blood pressure using calibrated digital devices, while Hospital B uses older devices that consistently produce higher readings.
- Laboratory measurements are missing for 60% of low-risk patients because clinicians do not routinely order the tests.
- Hospital length of stay is strongly right-skewed, with a small number of patients remaining hospitalized for several months.
- Several blood pressure values are unusually high, although clinical experts confirm that some represent genuine medical emergencies.
- Some patients contribute multiple hospital admissions.

The team proposes replacing missing laboratory values with the overall mean, deleting observations beyond the conventional boxplot whiskers, and randomly dividing individual admission records into training and test partitions.

Questions

- Identify the measurement bias and missingness problems in the dataset.
- Why might the missing laboratory values contain clinically meaningful information?
- Which descriptive statistics and visualizations would be appropriate for examining hospital length of stay?
- Why should observations beyond the boxplot whiskers not automatically be deleted?

- What statistical and modeling problems may arise from randomly partitioning admissions belonging to the same patient?
- Propose a more reliable EDA, preprocessing, and evaluation workflow for this project.

S3: Credit Default Prediction and Feature Engineering

A financial technology company develops a machine learning model to predict whether a customer will default on a loan. The dataset contains income, debt ratio, occupation, city, credit history, employment duration, previous defaults, and loan application dates.

Annual income ranges from $20{,}000$ to several million monetary units, whereas debt ratio ranges from 0 to 1. The team uses a distance-based classifier without scaling the numerical features.

The occupation variable contains hundreds of categories, including several occupations represented by only one customer. To encode occupation, the team replaces each category with its observed default rate calculated from the complete dataset.

The company also constructs a feature called total defaults during the next 12 months and includes it when predicting default at the time of loan application.

During deployment, a new occupation category appears that was absent during training. Furthermore, the production system supplies raw income values even though a revised version of the model was trained using logarithmically transformed income.

Questions

- Why can the difference in numerical scale between income and debt ratio affect a distance-based classifier?
- Compare standardization, robust scaling, and logarithmic transformation as possible treatments for the income variable.
- What are the risks of using ordinary target encoding for occupations represented by very few customers?
- Explain how out-of-fold target encoding and smoothing can reduce leakage and overfitting.
- Identify the temporal leakage and train-serving inconsistency problems in this project.
- How should the system handle occupation categories that were not observed during training?

S4: Smart City Air Quality Monitoring and Statistical Visualization

A city deploys an AI-assisted air quality monitoring system using $\mathrm{PM}_{2.5}$ sensors installed across residential, commercial, and industrial areas. The objective is to characterize pollution patterns and predict potentially hazardous air quality conditions.

During EDA, the data team discovers that most daily $\mathrm{PM}_{2.5}$ concentrations are moderate, but occasional severe pollution episodes produce extremely high observations. Consequently, the mean concentration is substantially higher than the median.

The team also identifies the following patterns:

- Some sensors report identical values for several consecutive hours.
- Measurements are missing during electricity outages.
- Pollution concentrations increase during morning and evening traffic periods.
- Several monitoring stations show abrupt changes after a sensor firmware update.
- Industrial areas exhibit greater variability and more extreme concentrations than residential areas.

The city dashboard displays average pollution using a three-dimensional bar chart with a truncated vertical axis. The dashboard excludes missing measurements without disclosure and reports only the citywide mean.

Questions

- What distributional characteristics would explain the difference between mean and median $\mathrm{PM}_{2.5}$ concentrations?
- Which graphical methods would help investigate temporal trends, unusual observations, and differences across monitoring stations?
- How can sensor firmware changes introduce measurement bias or apparent distribution shifts?
- Why might reporting only the citywide mean conceal important information about pollution exposure?
- How could the median, interquartile range, and upper percentiles complement the mean in describing pollution?
- Identify the problems with the proposed dashboard and recommend appropriate principles of responsible visualization.

S5: E-Commerce A/B Testing and Statistical Inference

An online retailer develops a new AI recommendation system intended to increase customer purchases. To assess its effectiveness, the company conducts an A/B test involving $40{,}000$ randomly assigned customers.

Group A receives the existing recommendation system, while Group B receives the new AI system.

The experiment produces the results in **Table 45**.

A statistical test comparing conversion rates produces a reported p-value below 0.05.

However, the research team also conducts 30 additional tests across customer segments, product categories, average spending, browsing time, and engagement measures. It reports only the statistically significant findings.

Table 45: Illustrative comparison of customer purchasing behavior between two groups.

Measure	Group A	Group B
Number of customers	20,000	20,000
Customers making a purchase	2,000	2,400
Conversion rate	10%	12%
Average purchase value	$80	$82

Furthermore, the new recommendation system requires substantially greater computational resources, and management argues that statistical significance alone establishes that the system should replace the existing model.

Questions

- Formulate the null and alternative hypotheses for comparing the conversion rates of Groups A and B.
- Calculate the absolute and relative improvement in conversion rate achieved by the new recommendation system.
- What does a p-value below 0.05 indicate in the context of the prespecified conversion-rate test?
- Why can conducting 30 additional statistical tests increase the risk of false-positive findings?
- Explain how the Bonferroni correction or false discovery rate control could address the multiple-comparison problem.
- Why should practical significance, effect size, uncertainty, computational cost, and operational requirements be considered alongside statistical significance when deciding whether to deploy the new AI system?

These are **take-home reflection questions** intended to encourage **deep thinking and planning**. There are no single correct answers. Each question can't be answered in a couple sentences. It will push you to think deeply about the concepts, principles, and practices in this module.

R1: Smart City Air Quality Monitoring and Public Health

A metropolitan government plans to develop an AI system that predicts hazardous air pollution levels and issues early warnings to citizens. The proposed system combines $PM_{2.5}$

measurements from 150 monitoring stations, weather observations, traffic activity, industrial emissions, hospital admissions, and historical pollution records.

During preliminary analysis, the data team discovers several unexpected characteristics:

- $\mathrm{PM}_{2.5}$ concentrations are strongly right-skewed, with occasional extreme pollution episodes.
- Measurements from industrial districts exhibit greater variability than measurements from residential areas.
- Several sensors consistently report higher concentrations after a firmware update.
- Missing observations are concentrated during electricity outages and severe weather.
- Pollution concentrations increase during morning and evening traffic periods.
- Hospital admissions and $\mathrm{PM}_{2.5}$ concentrations both increase during winter.
- A few stations have unusually high readings that the data team initially proposes removing.

The city government wants to publish a dashboard showing average pollution levels and use the AI model to determine when public health warnings should be issued.

However, the proposed dashboard displays only citywide averages, while the model is evaluated using a random division of individual hourly observations.

Questions

- **Design a comprehensive EDA strategy:** How would you investigate the distributions, temporal behavior, missingness, anomalies, and geographical differences in the pollution data? Identify the numerical summaries and visualizations you would use and explain their complementary roles.
- **Investigate the unusual observations:** How would you distinguish legitimate extreme pollution episodes from measurement errors, equipment failures, and changes caused by sensor firmware updates? Explain why automatically removing all extreme observations could undermine the intended application.
- **Select appropriate descriptive statistics:** How should the mean, median, standard deviation, interquartile range, and upper percentiles be used to communicate pollution exposure? Explain why different summaries may be appropriate for different operational questions.
- **Examine the relationship between pollution and hospital admissions:** Why does their observed wintertime association not establish that changes in $\mathrm{PM}_{2.5}$ directly caused the observed changes in admissions? Discuss alternative explanations and additional evidence required for a causal claim.
- **Develop a temporal feature-engineering strategy:** Propose features derived from pollution history, traffic patterns, weather, and seasonal behavior. Explain how lag features, rolling averages, interaction features, and cyclical encoding could be used without introducing temporal leakage.

- **Critique the proposed evaluation strategy:** Why might randomly distributing hourly observations across training and test partitions produce misleading performance estimates? Design an evaluation strategy that accounts for temporal dependence, geographical variation, and deployment at previously unseen monitoring stations.
- **Redesign the public dashboard:** Propose visualizations that communicate citywide conditions, neighborhood-level differences, temporal trends, uncertainty, missing measurements, and extreme events without misleading the intended audience.
- **Define an appropriate decision criterion:** Explain how predictive performance, uncertainty, false alarms, missed hazardous events, and the consequences of public warnings should influence the decision to issue a pollution alert.

R2: Multihospital AI for Predicting Clinical Deterioration

A healthcare network collects electronic medical records from eight hospitals to develop an AI system that predicts clinical deterioration within the next 12 hours.

The dataset contains $250{,}000$ patient records, including age, laboratory measurements, vital signs, medication history, previous admissions, diagnoses, and intensive care transfers.

During EDA, the researchers find that the hospitals differ considerably in their patient populations and measurement practices. Some hospitals treat mostly older patients with multiple chronic conditions, whereas others serve younger populations. Laboratory tests are ordered more frequently for patients who already appear seriously ill.

The researchers also discover that:

- The overall relationship between age and deterioration risk is relatively weak, but stronger relationships appear within certain clinical subgroups.
- Several biomarkers have highly skewed distributions.
- Some patients contribute multiple hospital admissions.
- Missing laboratory measurements are concentrated in particular hospitals and patient groups.
- Medication orders placed after clinical deterioration are present in the historical records.
- A model with excellent overall performance has substantially poorer recall among patients from rural hospitals.

The team proposes imputing missing values using the complete dataset, scaling all numerical variables before partitioning, and reporting a single pooled performance metric.

Questions

- **Define the statistical units:** Distinguish among the unit of observation, unit of prediction, unit of sampling, and unit of evaluation. Explain how repeated admissions

from the same patient affect statistical independence, uncertainty estimation, and model evaluation.

- **Develop a structured subgroup analysis:** How would you examine distributions, missingness, outcome prevalence, and predictive performance across hospitals, age groups, clinical categories, and relevant intersections? Explain how subgroup selection should be justified.
- **Investigate the weak aggregate association:** Why might the overall age–deterioration relationship differ substantially from the relationships observed within particular clinical groups? Discuss possible aggregation effects, confounding, interactions, and heterogeneous populations.
- **Design an appropriate preprocessing strategy:** Compare standardization, robust scaling, logarithmic transformation, and imputation for the available clinical variables. Explain how the selected transformations should be fitted and applied within the modeling pipeline.
- **Identify possible leakage:** Examine the medication orders, preprocessing procedures, patient records, and prediction timeline. Explain which forms of leakage may arise and how they could inflate the reported performance.
- **Develop an evaluation and uncertainty framework:** Propose suitable patient-level and hospital-level validation procedures, subgroup metrics, confidence intervals, and bootstrap resampling strategies. Explain how dependence among records should influence the resampling unit.
- **Interpret unequal model performance:** What additional evidence would be needed to determine whether poorer recall in rural hospitals results from population differences, data quality, measurement practices, missingness, or inadequate model generalization?
- **Assess deployment readiness:** Develop a plan for deciding whether the system should be deployed across all hospitals, restricted to selected settings, or subjected to further data collection and evaluation. Explain how uncertainty and subgroup-specific limitations should be communicated to clinicians.

R3: Credit Card Fraud Detection and the Hidden Risks of Feature Engineering

A financial institution develops an AI system to identify potentially fraudulent credit card transactions in real time. The dataset contains 20 million transactions collected over three years.

Each transaction includes the amount, merchant category, geographical location, device identifier, transaction timestamp, customer history, and a label indicating whether fraud was subsequently confirmed. Fraudulent transactions constitute only 0.4% of the dataset.

The data science team performs EDA and discovers that transaction amounts are strongly right-skewed, transaction activity varies by hour and day, and some merchant categories contain very few observations.

The team proposes constructing the following features:

- Total transactions during the previous 24 hours.
- Average transaction amount over the previous seven days.
- Number of geographical locations visited during the previous month.
- Time elapsed since the previous transaction.
- Average fraud rate associated with each merchant.
- Hour of day and day of week.
- Difference between the current transaction amount and the historical customer average.

During development, several problems emerge. The merchant fraud rate is calculated using the complete dataset. Some rolling features include transactions recorded after the transaction being evaluated. The dataset is randomly partitioned without considering customer identity or chronological order.

The final model achieves 99.6% accuracy, and management argues that the system is ready for autonomous deployment.

Questions

- **Critique the reported accuracy:** Explain why 99.6% accuracy may provide little evidence of useful fraud detection when the positive class constitutes only 0.4% of the dataset. Identify more informative evaluation measures.
- **Develop an EDA protocol:** Which graphical and numerical methods would you use to investigate transaction amounts, fraud prevalence, merchant categories, customer behavior, temporal patterns, and unusual transactions?
- **Design a leakage-safe feature-engineering pipeline:** For each proposed feature, explain what information may legitimately be used, how the feature should be calculated, and what information must be excluded at prediction time.
- **Investigate categorical encoding:** Compare one-hot encoding, frequency encoding, and target encoding for merchant categories. Explain the challenges associated with rare merchants, unseen categories, and unstable fraud-rate estimates.
- **Analyze cyclical and temporal representations:** Explain why ordinary integer encoding of hour of day may be inadequate. Propose an alternative representation and discuss how it could be combined with lag features and customer-specific transaction histories.
- **Redesign the evaluation strategy:** How would you partition the dataset to evaluate future transactions while accounting for customer history, temporal dependence, and the intended deployment population? Distinguish evaluation of future transactions from evaluation on entirely unseen customers.
- **Account for asymmetric errors:** How should the financial and operational consequences of missed fraud, unnecessary transaction blocking, customer dissatisfaction, and manual investigation affect model evaluation and threshold selection?

- **Design a production monitoring strategy:** What should be monitored when fraud patterns, merchant categories, transaction volumes, and customer behavior change? Explain how the system should respond when important features become unavailable or unreliable.

R4: A University-Wide AI Tutoring Experiment

A university introduces an AI-based tutoring platform intended to improve examination performance. The system provides personalized practice questions, study recommendations, explanations, and feedback.

The university conducts a semester-long experiment involving $4{,}000$ students. Half are randomly assigned to receive access to the new platform, while the remaining students continue using existing academic-support resources.

The initial analysis shows that students assigned to the AI tutoring platform obtain examination scores averaging 2.5% higher than those assigned to the control group. A statistical test produces $p = 0.03$. However, the researchers discover several complications:

- Students assigned access vary substantially in their actual platform usage.
- Students with strong previous academic performance use the platform more frequently.
- Students belong to different departments with different examinations and grading practices.
- Some students share AI-generated explanations with classmates in the control group.
- The researchers test 25 additional outcomes and subgroup comparisons.
- Several additional analyses produce statistically significant results.
- The platform increases educational expenditure and requires substantial technical support.

University leadership proposes making the platform compulsory because the experiment has produced a statistically significant result.

Questions

- **Formulate the statistical question precisely:** Define the target population, experimental unit, primary outcome, null hypothesis, alternative hypothesis, and effect size of interest. Distinguish assignment to the platform from actual platform usage.
- **Design the exploratory analysis:** Which descriptive statistics and visualizations would help examine score distributions, prior achievement, platform usage, department-level differences, and unusual observations?

- **Examine the experimental design:** How do random assignment, unequal platform usage, sharing of materials, and differences in academic departments influence the interpretation of the observed results?
- **Interpret the statistical evidence:** What does $p = 0.03$ establish under the specified statistical test, and what does it not establish? Explain why statistical significance alone is insufficient for deciding whether the platform should be adopted.
- **Address analytical flexibility:** How does conducting 25 additional comparisons affect the credibility of statistically significant findings? Compare the roles of preregistration, Bonferroni correction, false discovery rate control, and independent replication.
- **Evaluate practical significance:** How would you assess whether an average improvement of 2.5% justifies the financial, computational, administrative, and educational costs of the system?
- **Investigate subgroup behavior:** How would you determine whether the platform provides similar benefits across departments, prior achievement levels, and other educationally meaningful groups without repeatedly searching for favorable subgroup results?
- **Develop a responsible reporting framework:** Explain how the final report should communicate effect sizes, confidence intervals, subgroup results, statistical uncertainty, multiple comparisons, implementation costs, and limitations of the experiment.

R5: Predicting Agricultural Yield and the Problem of Analytical Flexibility

A national agricultural research institute wants to predict wheat yield using satellite imagery, weather records, soil measurements, irrigation history, fertilizer usage, and farm-management information.

The available dataset contains $100{,}000$ field-season observations collected from $12{,}000$ farms over ten years.

The institute invites five independent data science teams to analyze the dataset and identify the most important predictors of wheat yield. Although every team receives the same data, their conclusions differ substantially.

- Team A removes fields with extremely low yields, treating them as outliers.
- Team B retains these fields because they may represent genuine crop failures.
- Team C applies logarithmic transformations to yield and fertilizer usage.
- Team D creates nonlinear and interaction features involving rainfall, temperature, and soil properties.
- Team E uses a different missing-data strategy and partitions the dataset by farm and growing season.

Some teams conclude that fertilizer usage is the dominant predictor of yield. Others find rainfall, irrigation, soil characteristics, or nonlinear interactions to be more important.

The institute also discovers that most observations come from large commercial farms, while smallholder farms are underrepresented. Certain farms contribute many more field-season observations than others. Management wants to select the analysis reporting the highest predictive accuracy and use it to formulate national agricultural recommendations.

Questions

- **Define the analytical problem and statistical units:** Distinguish among the field-season observation, farm, growing season, unit of prediction, and unit of evaluation. Explain how repeated observations from the same farm affect statistical independence.
- **Develop a comprehensive EDA strategy:** How would you investigate yield distributions, missingness, unusual observations, geographical variation, temporal trends, and relationships among weather, soil, irrigation, and fertilizer variables?
- **Examine the treatment of extreme observations:** How would you distinguish erroneous yield records from genuine crop failures? Explain how retaining or excluding these observations could change the interpretation and operational usefulness of the resulting model.
- **Evaluate the competing feature-engineering strategies:** Compare the implications of logarithmic transformations, polynomial features, interaction terms, temporal aggregation, categorical encoding, and alternative missing-data treatments.
- **Investigate representativeness:** Why might 100,000 observations provide misleadingly precise estimates for the intended national farming population? Explain how the Big Data Paradox and unequal contributions from individual farms are relevant.
- **Design a defensible validation strategy:** How would you evaluate generalization across farms, regions, growing seasons, and previously unseen agricultural conditions? Explain the respective roles of training, validation, and final test data.
- **Assess the different analytical conclusions:** Does disagreement among the five teams necessarily indicate that some have made errors? Explain how defensible analytical choices can generate different results and how a specification-curve or multiverse analysis could expose this variation.
- **Distinguish prediction from causation:** Why does identifying fertilizer usage as an important predictive feature not establish that increasing fertilizer will increase yield? What additional experimental design or causal assumptions would be required to support agricultural intervention recommendations?
- **Develop a statistical reporting framework:** How should the institute communicate predictive performance, effect sizes, confidence intervals, sensitivity analyses, subgroup behavior, and uncertainty arising from model and feature-selection decisions?

- **Recommend a reproducible analytical protocol:** Design a process through which the five teams could reconcile their results without simply selecting the analysis with the most favorable performance. Include pre-analysis planning, a common data dictionary, agreed evaluation procedures, documented analytical choices, sensitivity analysis, and independent confirmation.